CHAPTER 61
PROPELLER

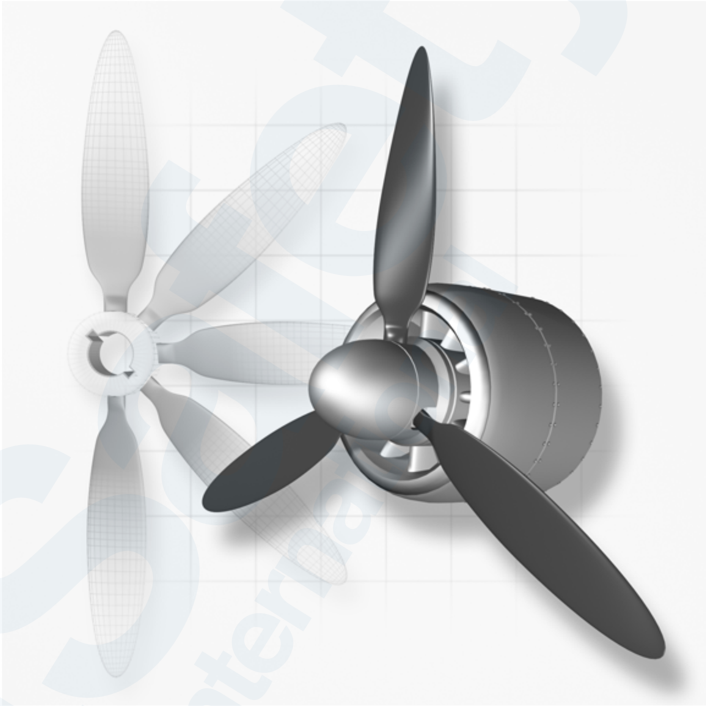
CONTENTS

Page

61 PROPELLER

ILLUSTRATIONS

CHAPTER 61
PROPELLER



61-00-00 GENERAL

Refer to Figure 61-1. Propeller Assembly.

The aircraft is equipped with two propeller systems. Each system has:

- Dowty CR 408/6-123 - F/17 propeller assembly
- PEC
- Pitch Control Unit (PCU)
- Overspeed Governor (OSG) and pump
- Auxiliary Feather pump.

The propeller used on the Pratt and Whitney PW 150A engine for the deHavilland Dash 8 Series 400 aircraft can be described by its model number as follows:

- **C** - Civil
- **R** - Dowty Aerospace Propellers
- **408** - Aircraft type identification
- **6** - Number of blades
- **123** - Blade root end size in mm
- **F** - Flange mounted
- **17** - Function/Installation characteristics.

The propeller system has the following subsystems:

- Propeller Assembly (61-10-00)
- Propeller Controlling (61-20-00).

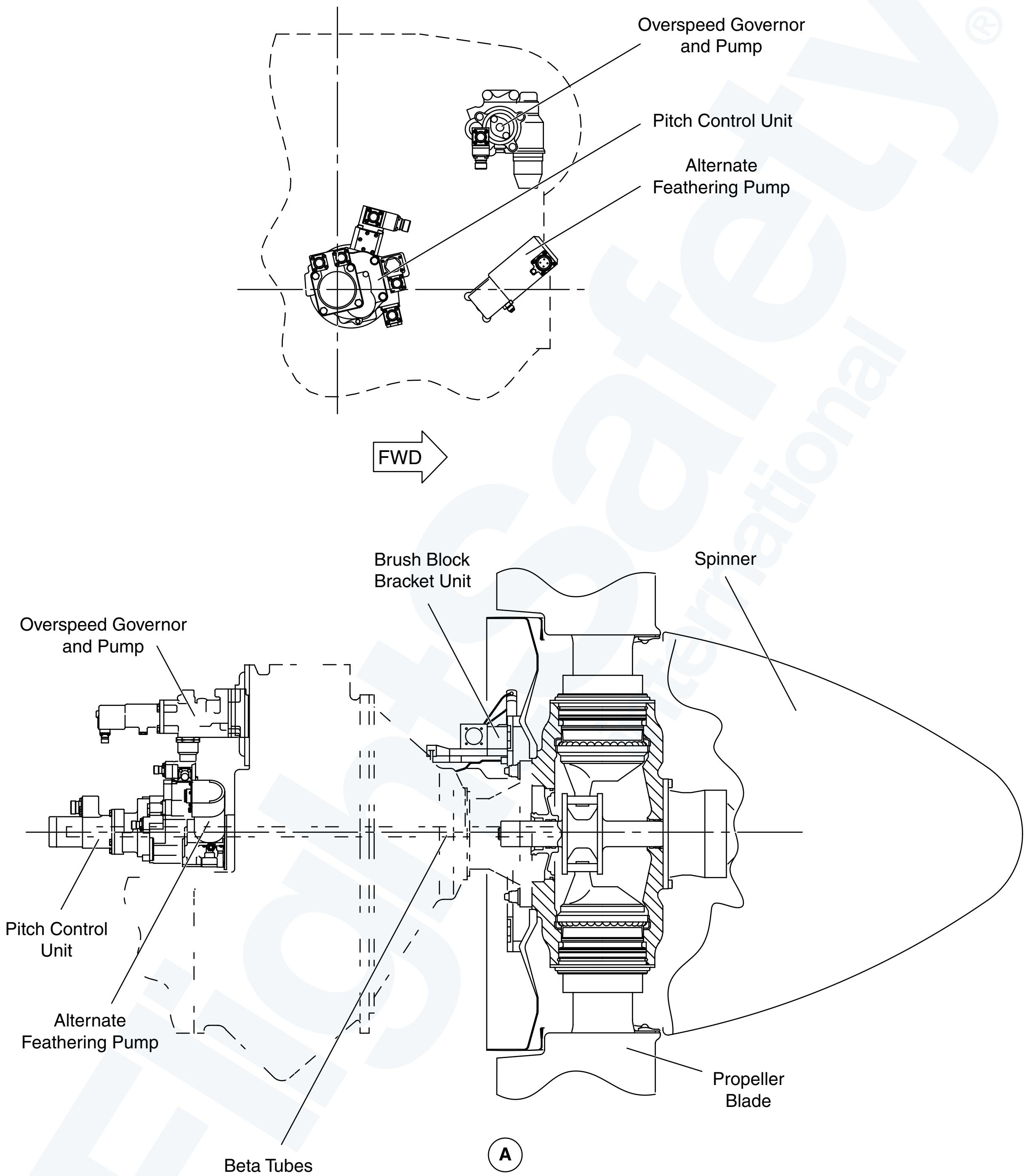


Figure 61-1. Propeller Assembly

SYSTEM DESCRIPTION

The propeller system is a variable pitch six bladed propeller assembly. The blades are counter weighted to move towards coarse pitch in the event of oil pressure failure.

Mode and function control is by the PEC. The oil pressure originates from the related engine and is supplied to the OSG and pump.

The pump increases the oil pressure and ports the metered oil to a servo valve. The servo valve receives electrical signals from the PEC and directs oil pressure to the coarse/fine sides of the propeller PCU. The PCU has a double acting pitch change mechanism to control blade pitch.

NOTES

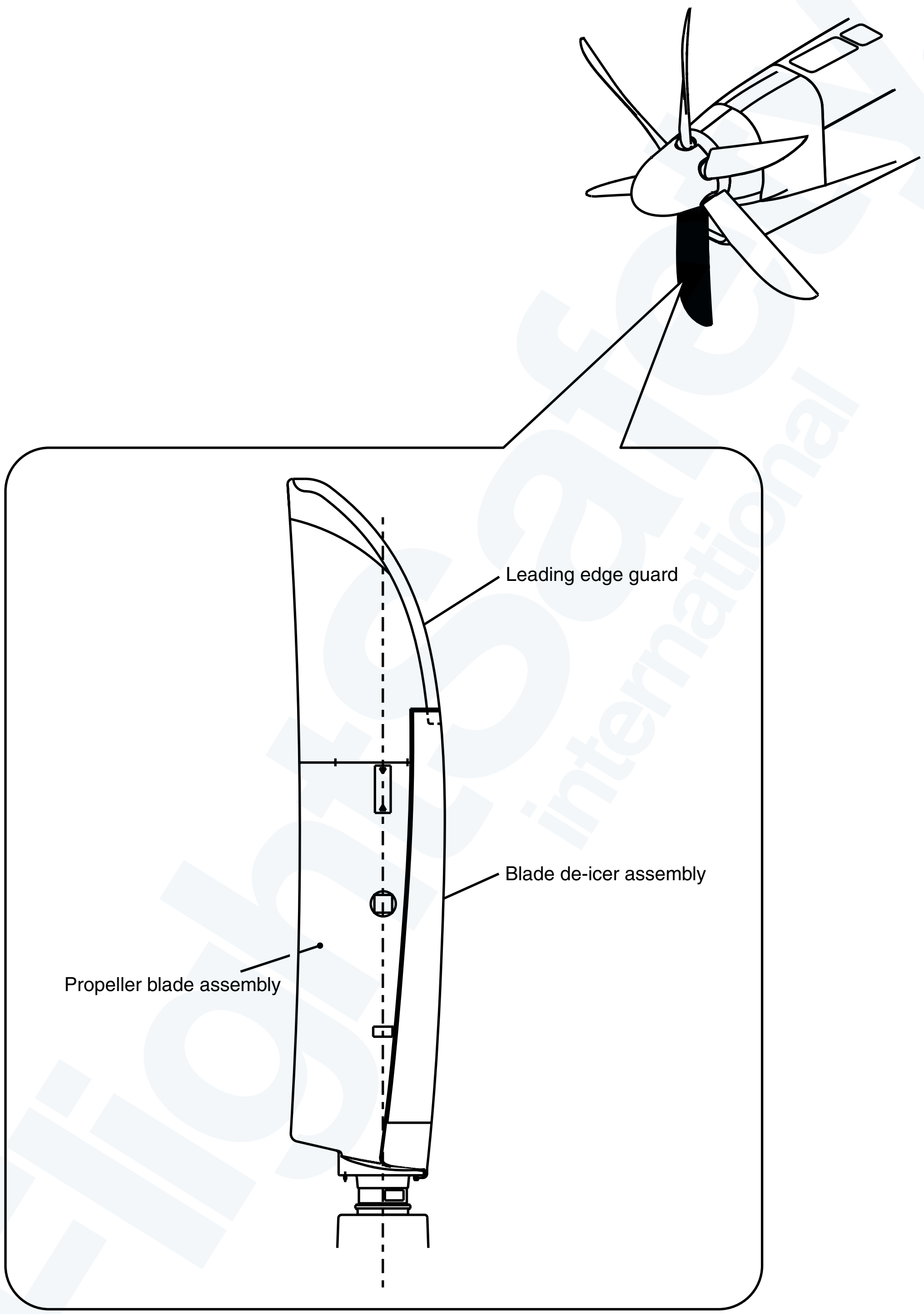


Figure 61-2. Propeller Blade Assembly

61-10-00 PROPELLER ASSEMBLY

INTRODUCTION

The propeller assembly has the following components:

- Propeller Blade Assembly
- Propeller Hub Assembly
- Spinner Assembly.

COMPONENT DESCRIPTION

Propeller Blade Assembly

Refer to:

- Figure 61-2. Propeller Blade Assembly.
- Figure 61-3. Blade-to-Hub Attachment.

Blade Assembly

The six blade assemblies are installed in a flange mounted hub.

Each blade is an all composite aerofoil construction with a steel outer root sleeve supported by a two bearings. The aerofoil has a foam core and twin carbon fiber spars with an overall braided carbon/glass fiber shell. The Blades provide forward and reverse thrust to the aircraft.

Lightning Braid

The lightning conductor braid is fitted between the preload nut and the hub. It is a braided chord impregnated with conductive material. The composite liner and washer effectively insulate the blade from the hub. The function of the lightning braid is to ensure electrical continuity from the blade to the hub.

Counter Weights

The blade counterweights are at the blade root.

They are heavy masses consisting of a tungsten carbide weight, bolted to an aluminum bracket that is clamped to each blade root. There is a key on the blade root and the counterweight to ensure proper installation. The counterweights are sized and phased so that when there is no oil pressure, they will produce a turning moment to the coarse direction, this overcomes the blades natural drive fine (centrifugal) turning moment.

Ball Bearing Roller

29 ball bearings between root the of the blade and the hub, form the inner blade bearings. The function of the bearing is to provide smooth pitch operation.

Taper Roller Bearings

There is a taper roller bearing between root of the blade and the hub. It forms the outer blade bearing. It provides smooth pitch operation.

Secondary Retention Cable

The secondary retention cable is fitted in the hub around the blade root. It is a steel wire cable with a locking plate at the end. The cable will retain the blade in the hub in the event that the ball bearings fail.

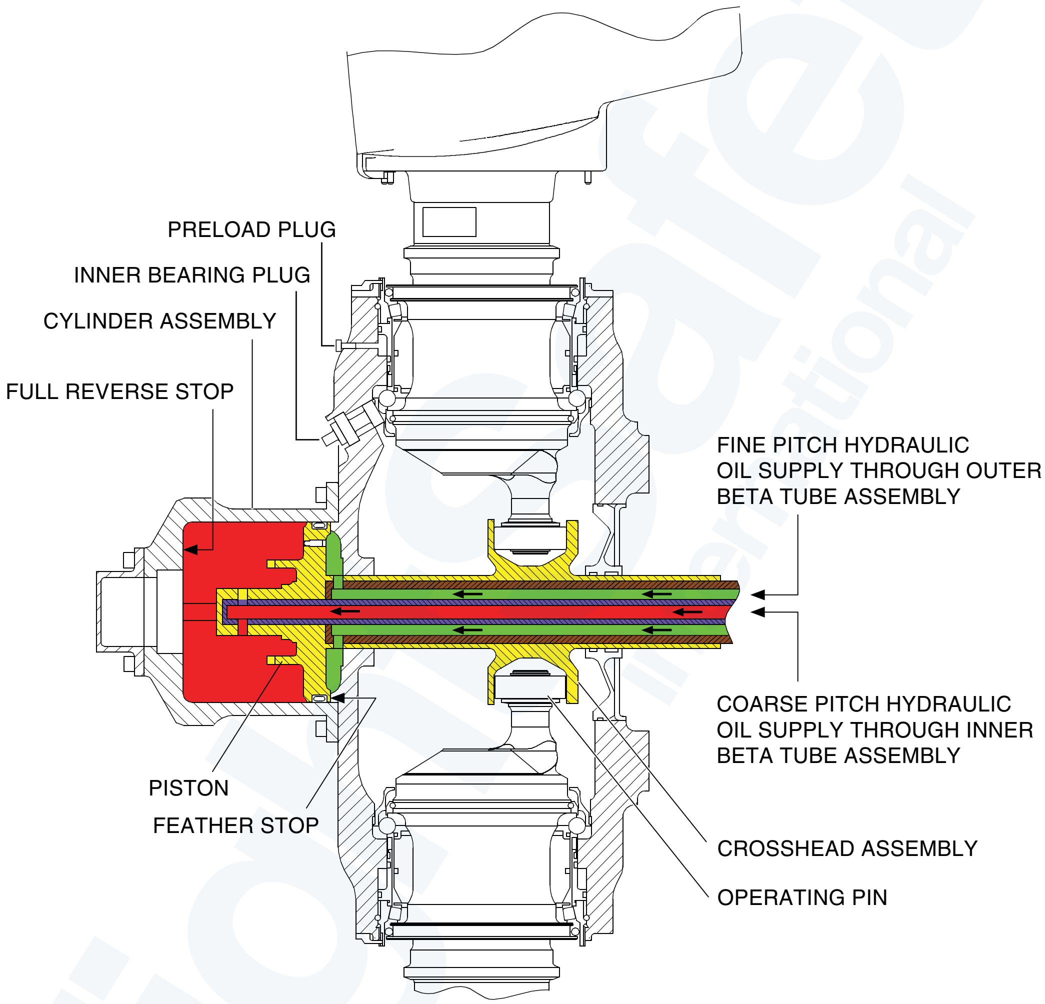


Figure 61-3. Blade-to-Hub Attachment

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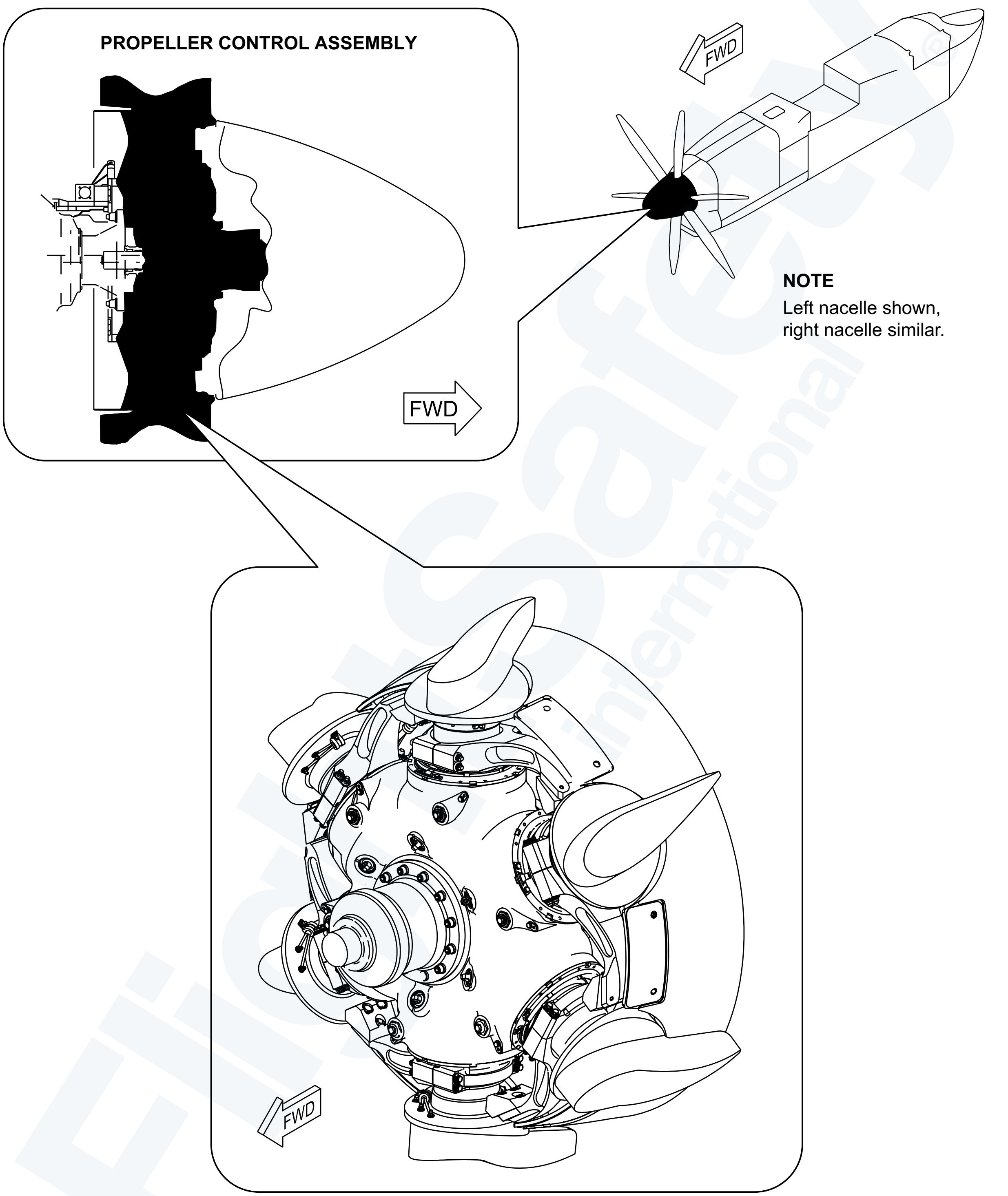


Figure 61-4. Propeller Hub Assembly

Actuator and Backplate Hub Assembly

Refer to:

- Figure 61-4. Propeller Hub Assembly.
- Figure 61-5. Propeller Backplate.
- Figure 61-6. Propeller Attachment and Backplate Assembly.

Actuator

The pitch change actuator is housed in a cylinder assembly secured to the hub.

The cylinder is made of aluminum alloy and has a removable cap at the front end.

The piston is inside the cylinder forward of the crosshead assembly and held in position by a key and keyway. The piston is fitted with a “LEE” jet which allows a small continuous flow of hot oil to flow from one side to the other to keep the assembly warm in flight.

The actuator controls the pitch of the blade.

Hub Assembly

The hub assembly is on the Propeller flange of the RGB.

The hub is manufactured from a single piece aluminum alloy. The forward face has 12 equally spaced holes for attaching the cylinder. The aft face of the hub has 15 integral steel mounting studs and three location/drive dowels. The nuts on the studs have a minimum running torque of 50 lbs/in.

The hub supports six blades and has six pairs of blade root bearings races. A front-mounted aluminum alloy piston and cylinder and steel crosshead/shaft change blade pitch.

To prevent fretting at the shaft to the propeller interface, a composite shim is installed on the flange

Backplate Assembly

The backplate assembly is installed on the rear of the hub assembly.

The backplate is constructed of carbon fiber composite. The slip ring is installed directly onto the backplate and has an aluminum alloy housing with three bronze rings in a plastic molding. The target screws that supply propeller speed and phase angle feedback are on the external diameter of the slipring.

The backplate forms the aerodynamic interface between the spinner and engine nacelle.

The slipring is used to transfer electrical power for blade de-icing.

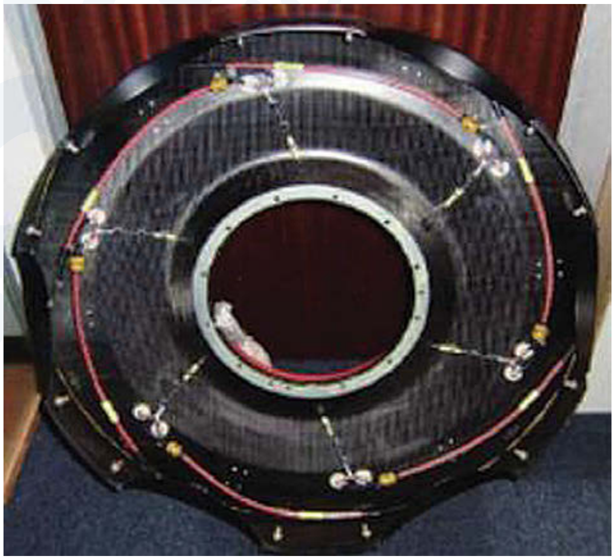
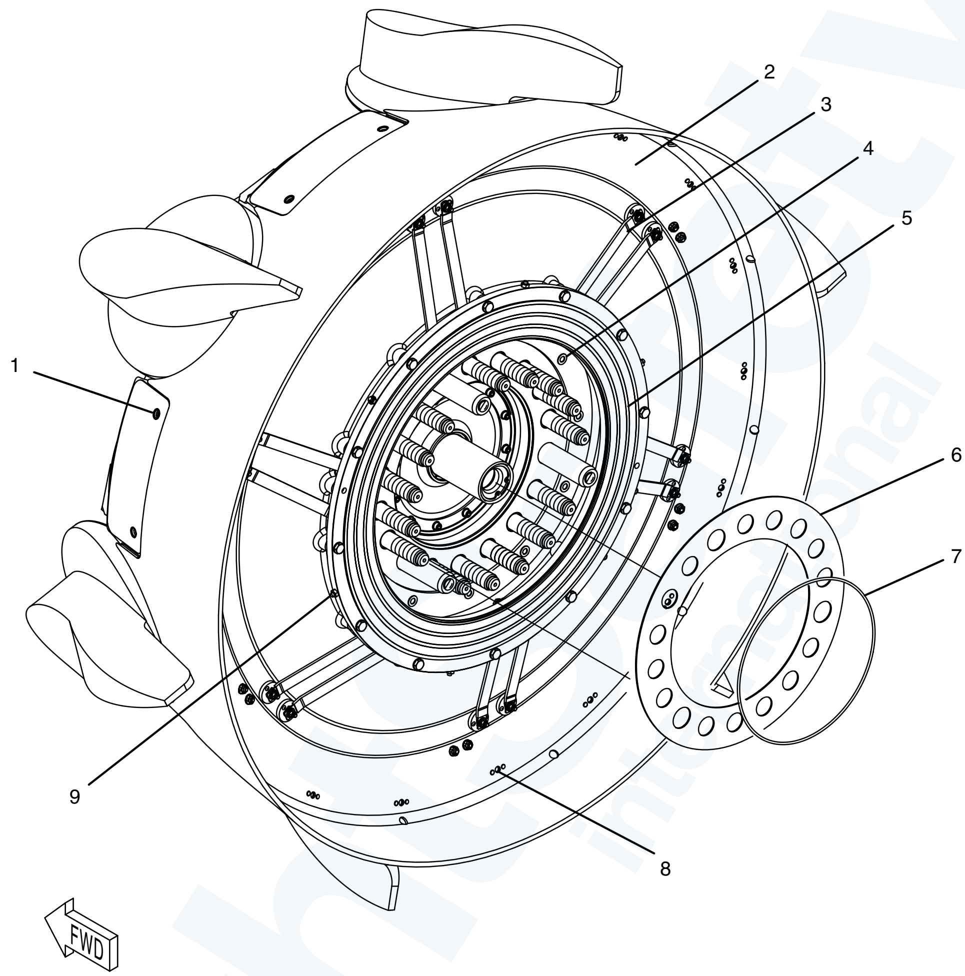


Figure 61-5. Propeller Backplate

61 PROPELLER



LEGEND

1. Spinner retention nut location (typical)
2. Backplate
3. Deice bus bars (typical)
4. Backplate attachment bolt (typical)
5. Deice slip ring
6. Gasket
7. O-ring seal
8. Static/dynamic balance weight location (typical)
9. MPU target screw (typical)

Figure 61-6. Propeller Attachment and Backplate Assembly

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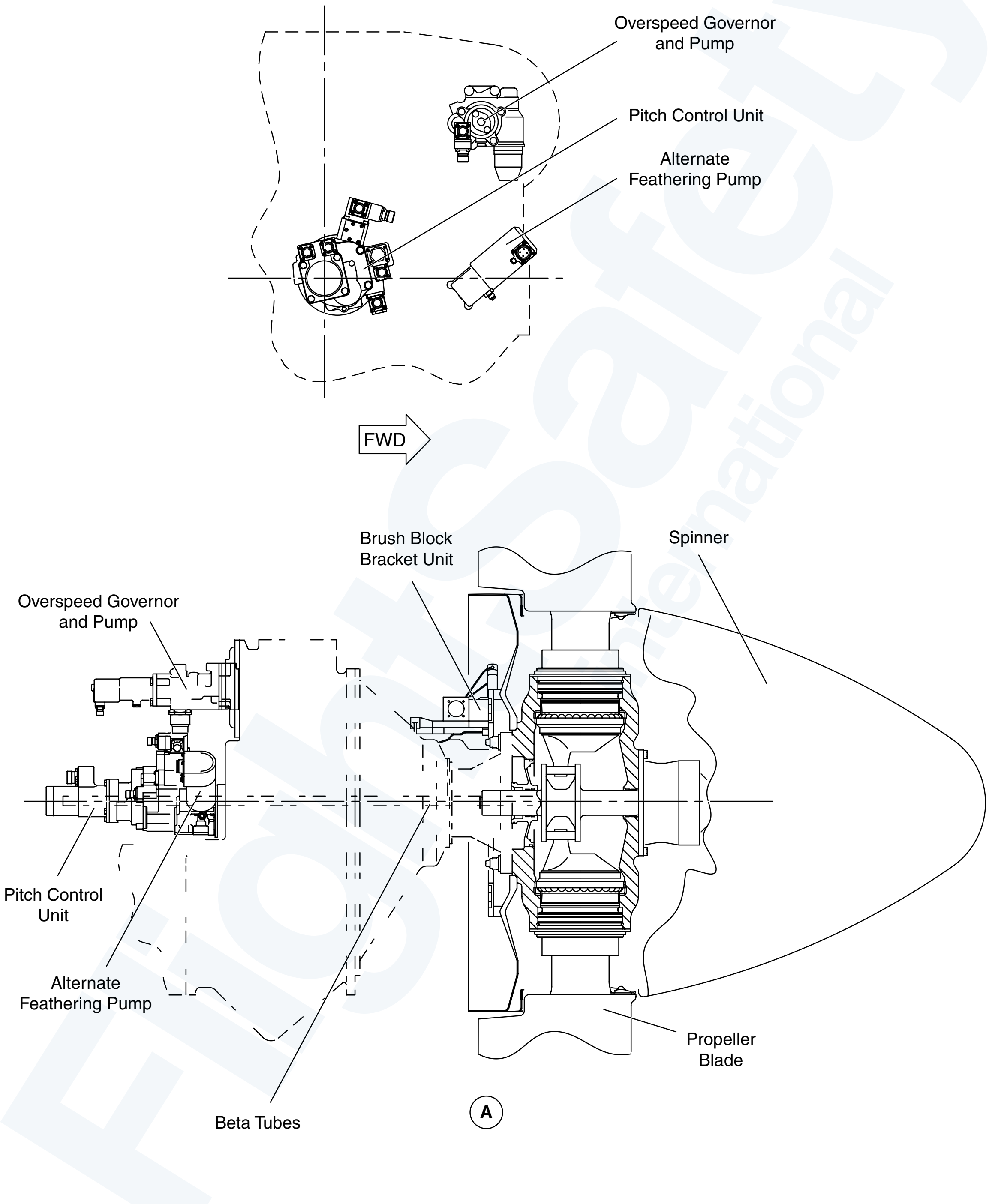


Figure 61-7. Propeller Component Locations

Spinner Assembly

Refer to:

- Figure 61-7. Propeller Component Locations.
- Figure 61-8. Propeller Spinner.

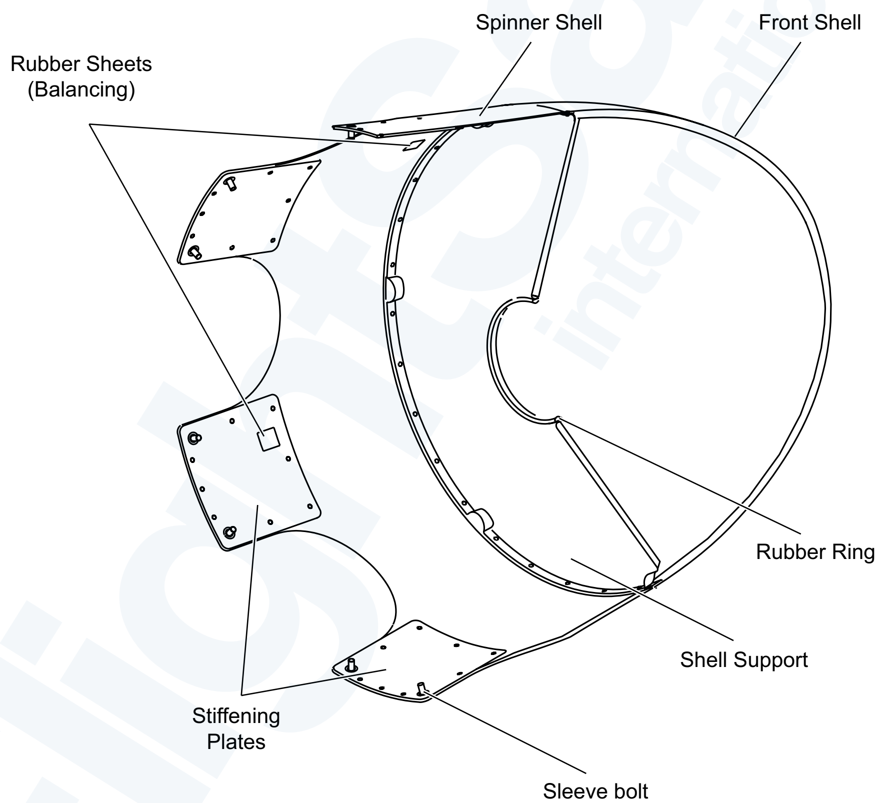
The spinner is installed on the propeller backplate assembly with 12 quick release fasteners.

It is constructed of composite material made in three pieces.

The spinner gives an aerodynamic fairing over the front end of the propeller.

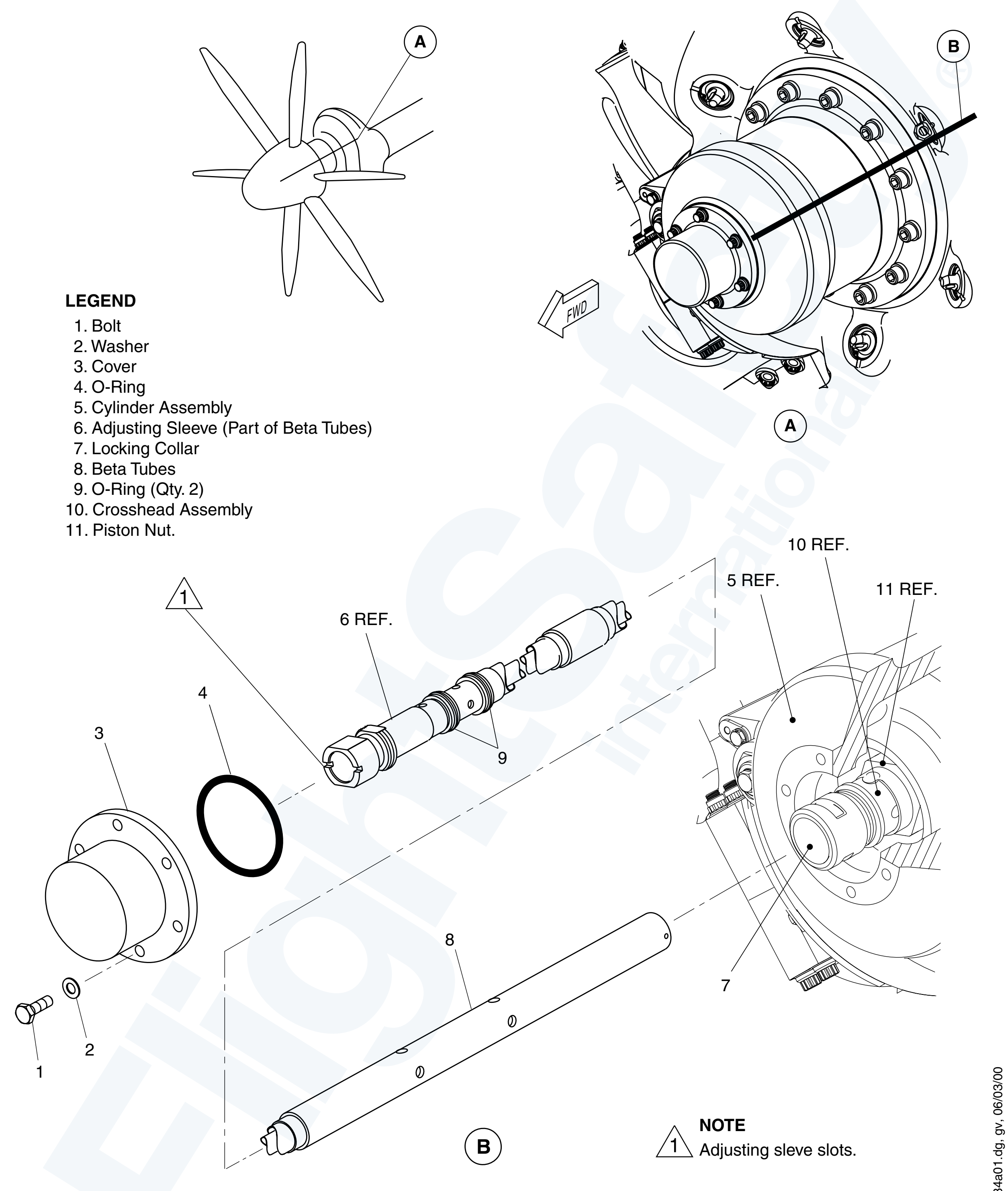
A centralizing/support diaphragm on the pitch change cylinder makes sure that the spinner runs correctly, when rotating with the propeller.

It is balanced by the addition of rubber pads stuck to the inner surface. Although it is a separately balanced component, it is marked by the manufacturer with a blade #1 position.



NOTE
Portion of spinner removed for clarity.

Figure 61-8. Propeller Spinner



Beta Tubes

Figure 61-9. Beta Tubes

REMOVAL OF THE BETA TUBES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to Figure 61-9. Beta Tubes.

The maintenance procedure that follows is for the removal of the beta tubes.

Obey all the electrical/electronic safety precautions.

WARNING

USE GOGGLES AND GLOVES WHEN YOU DO WORK WITH ENGINE OIL.

WARNING

DO NOT TOUCH THE COMPONENTS OF THE PROPELLER OR THE PROPELLER CONTROL EQUIPMENT UNTIL THEY ARE COOL.

Remove the beta tubes as follows:

NOTE

If the beta tubes are to be removed from and installed to the same hub LRU, record the beta tube calibration depth at removal. Set the beta tubes to the same value at installation.

1. Remove the spinner.
2. Place a container under the beta cap to collect the oil.
3. Remove the bolts with the washers which attach the cover to the cylinder assembly.
4. Remove the cover with the o-ring from the cylinder assembly.

CAUTION

INSTALL THE BLADE BATS TO TWO BLADES WHICH ARE OPPOSITE TO EACH OTHER, BETWEEN THE ARROW HEADS ON THE LABEL ATTACHED TO THE CAMBER FACE OF THE BLADE.

5. To turn the propeller blades to the full reverse mechanical stop use the blade bats. This will move the crosshead assembly forward and help you to remove the beta tube.
6. Through the front of the cylinder assembly put beta tube wrench.
7. Engage the wrench with the slots in the adjusting sleeve.
8. Push the outer sleeve of beta tube spanner against the locking collar.
9. Compress the spring to disengage the locking collar from the crosshead spline.
10. Turn the wrench in a counterclockwise direction to loosen and remove the beta tube from the propeller.
11. Remove the O-ring seals from the beta tube
12. Clean the beta tube.
13. Put the beta tube in a clean polyethylene bag. Keep the beta tube in its wooden packing case for storage and transport.
14. Do a visual inspection of the bore in the propeller crosshead assembly Make sure that there are no particles of rubber in the bore. Remove any rubber particles if necessary.
15. Install a protective cover to the cylinder assembly.
16. Use the bolts and the washers to loosely attach the cover.

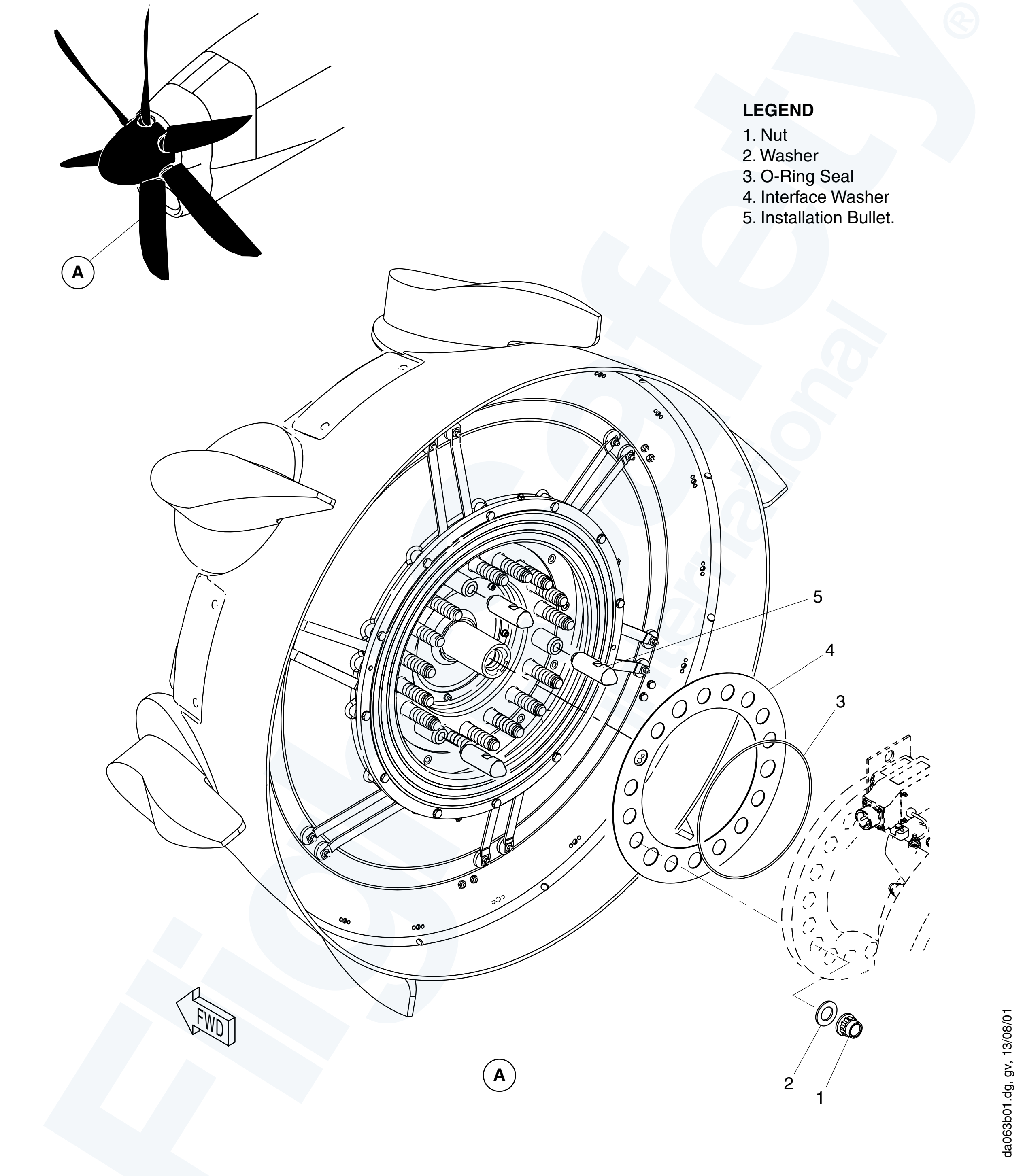


Figure 61-10. Propeller

REMOVAL OF THE PROPELLER

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to:

- Figure 61-10. Propeller.
- Figure 61-11. Propeller - Criss Cross Pattern.

The maintenance procedure that follows is for the removal of the propeller.

Obey all the electrical/electronic safety precautions.

WARNING

USE GOGGLES AND GLOVES WHEN YOU DO WORK WITH ENGINE OIL.

Remove the propeller as follows:

1. Remove the spinner.
2. Remove the beta tubes.
3. Remove the brush block unit.
4. Remove the dual pulse probe assembly.
5. Break the torque on fifteen nuts.
6. Remove the top and bottom three nuts and washers.
7. Attach the lifting equipment. Use it to hold the weight of the propeller assembly.
8. Install the three installation bullets.
9. Put a container below the propeller to collect unwanted used oil.
10. Get access to the propeller attachment studs. Remove the fifteen nuts and washers with the torque adapter.
11. Remove the propeller assembly from the driveshaft flange.
12. Remove the O-ring from the driveshaft flange.
13. Remove the interface washer from the driveshaft flange or propeller hub.
14. Install the transport cover to the slip ring assembly.
15. Install the transport cover to the shaft of the crosshead.

INSTALLATION OF THE PROPELLER

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to:

- Figure 61-10. Propeller.
- Figure 61-11. Propeller - Criss Cross Pattern.

The maintenance procedure that follows is for the installation of the propeller.

The installed blade assembly and bearing must be re-torqued after a ground run with a propeller forward constant speed check, after a maximum of 10 flight hours, or at the next overnight stop.

Do a visual inspection of the propeller shaft bearing sleeves.

WARNING

USE GOGGLES AND GLOVES WHEN YOU WORK WITH ENGINE OILS.

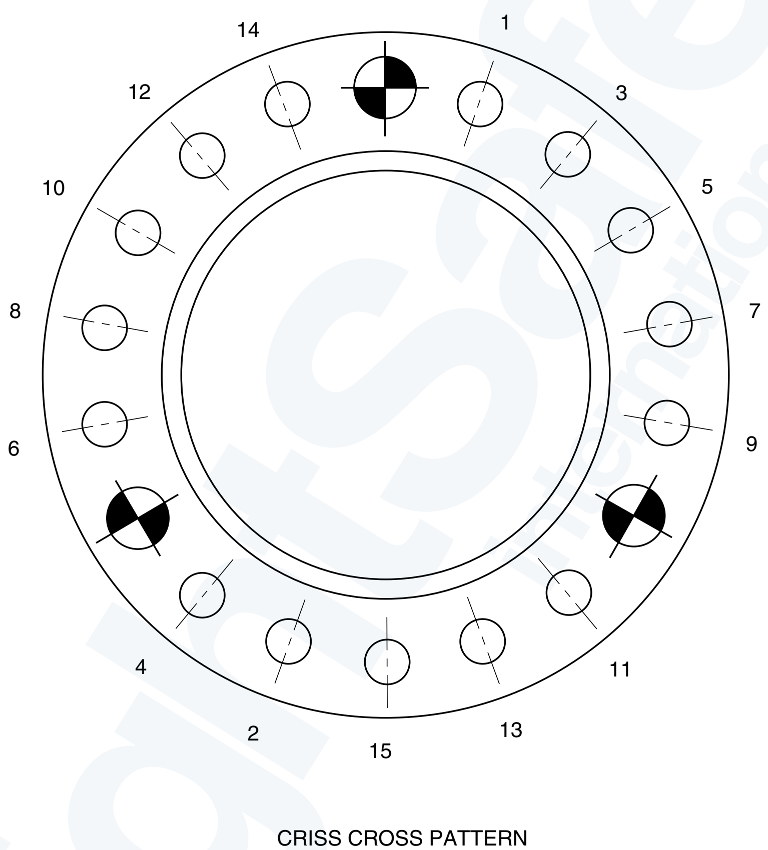


Figure 61-11. Propeller - Criss Cross Pattern

Install the propeller as follows:

1. Remove the brush block unit and the dual pulse probe assembly.
2. Attach the lifting equipment to the propeller.
3. Remove the transport covers from the slip ring assembly and the cross-head shaft.
4. Use synthetic lubricating oil to lubricate the O-ring and the threads and joint faces of the propeller mounting studs, nuts and washers.
5. Install the O-ring on the spigot of the engine drive shaft flange.
6. Attach the interface washer to the propeller hub.
7. Install the three installation bullets to the three dowels of the propeller.
8. Carefully install the propeller to the drive shaft flange.

WARNING

REPLACE THE PROPELLER ATTACHMENT NUTS IF THE RUNNING TORQUE IS LESS THAN 50 LBF IN (5.65 NM).

9. Install the fifteen nuts and the washers to attach the propeller.
10. Remove the three installation bullets from the three dowels.
11. Remove the load of the propeller from the lifting equipment.
12. Do a check of the running torque of the fifteen nuts.
13. Remove the lifting equipment.
14. Torque the fifteen nuts to 315 to 320 lbf ft (427 and 434 N m) with the torque adapter.

WARNING

YOU MUST USE THE CORRECT TORQUE FORMULA WHEN YOU USE THE TORQUE ADAPTER TO TORQUE THE PROPELLER ATTACHMENT NUTS.

NOTE

Torque the fifteen nuts in a sequence that tightens them at opposite sides of the diameter.

15. Loosen and re-torque all the nuts one at a time in the correct sequence, This must be done three times for each attachment nut.
16. Install a blade bat to one blade and another blade bat to the opposite blade and turn the blades smoothly through the full pitch range.
17. Remove the blade bats.
18. Install and adjust the brush block unit.
19. Install and adjust the dual pulse probe assembly.
20. Remove the brush retaining assembly from the brush block unit.
21. Install the beta tube assembly.
22. Install the spinner.

Do the calibration of timer monitor control unit TMCU.

Do the functional test and calibration of the propeller de-icing system.

Download the ANVS propeller balance data to erase the data from the ANCU This step is not necessary if the same propeller is re-installed.

Do the necessary operational tests.

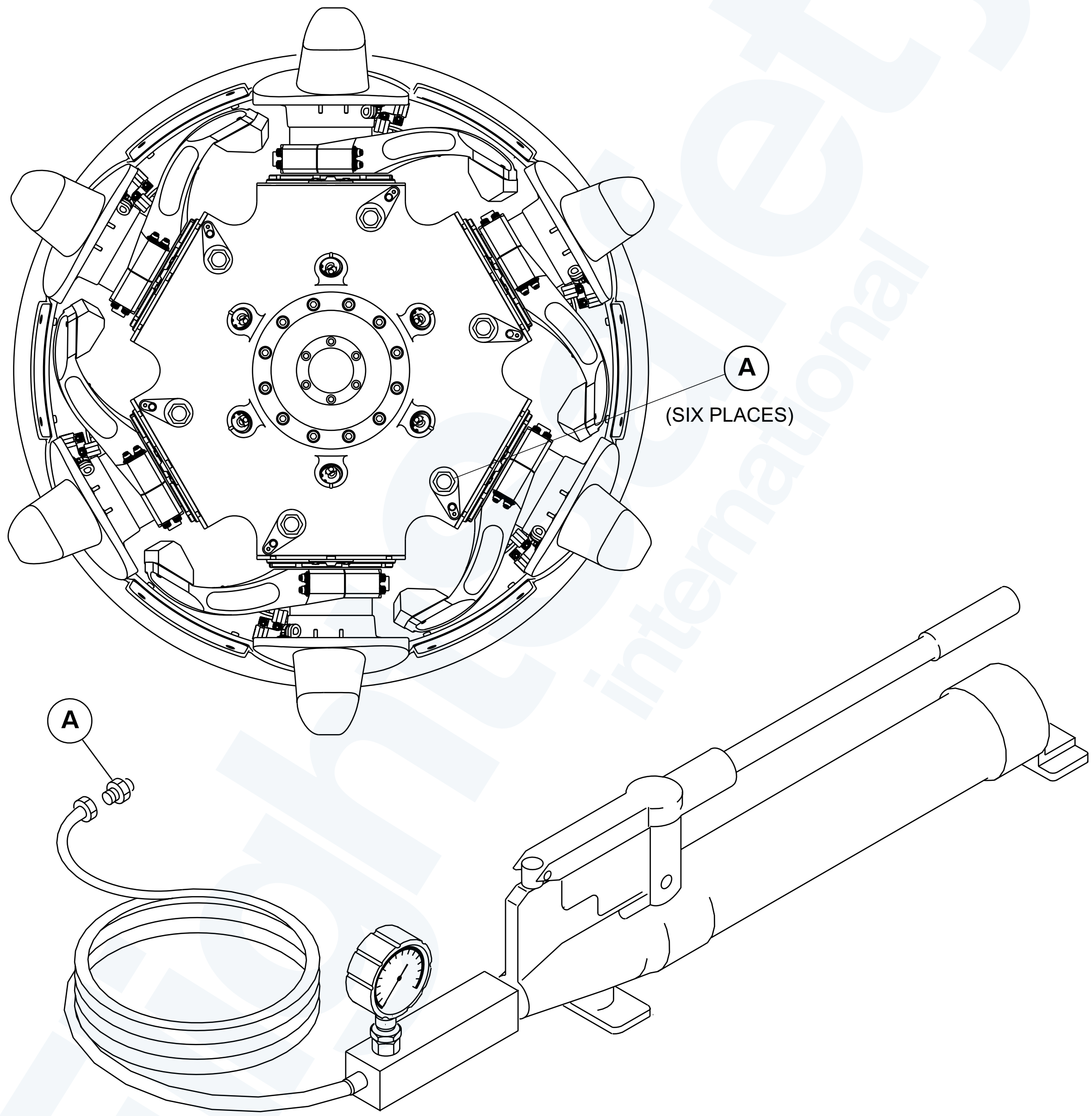


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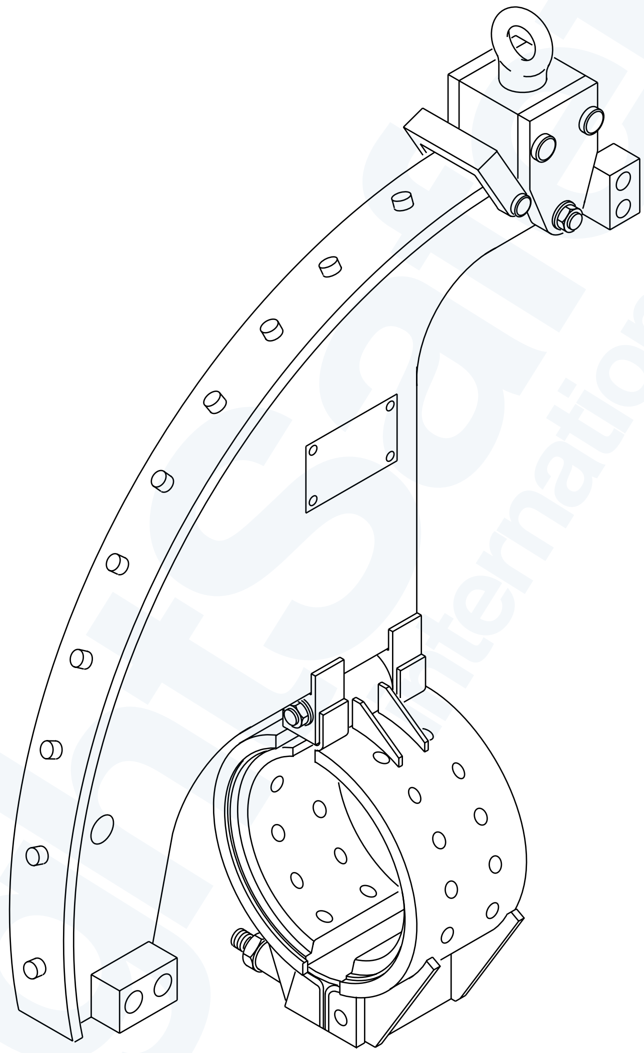
Figure 61-12. Pump - Hydraulic Hand (Series 400)



PROPELLER LIFTING EQUIPMENT

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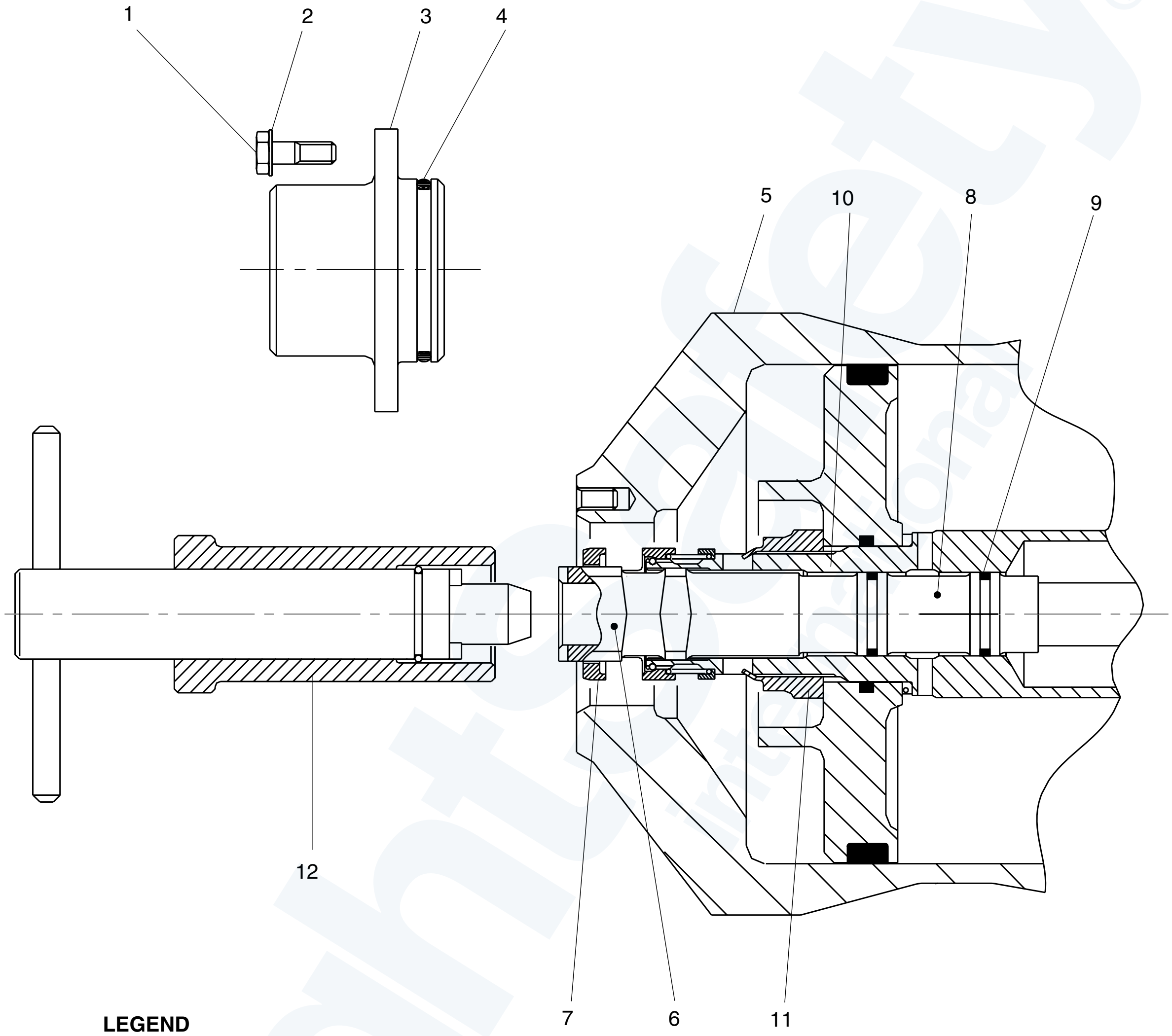
Figure 61-13. Propeller Lifting Equipment



LEGEND

1. Bolt
2. Washer
3. Cover
4. O-Ring
5. Cylinder Assembly
6. Adjusting Sleeve
7. Locking Collar
8. Beta Tubes
9. O-Ring (2 Places)
10. Cosshead Assembly
11. Piston Nut
12. Beta Tube Spanner.

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Figure 61-14. Beta Tubes - Cross Section View

INSTALLATION OF THE BETA TUBES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to:

- Figure 61-14. Beta Tubes - Cross Section View.
- Figure 61-15. Beta Tubes - Beta Calibration and Initial Standout Dimension.

The maintenance procedure that follows is for the installation of the beta tubes.

Make sure that the aircraft is in the same configuration as in the removal task.

Install the beta tubes as follows:

NOTE

If the beta tubes are to be removed from and installed to the same hub LRU, record the beta tube calibration depth at removal. Set the beta tubes to the same value at installation.

CAUTION

Install the blade bats to two blades which are opposite to each other between the arrow heads on the label attached to the camber face of the blade.

Beta Tube Rigging

If there is oil in the piston and gear box reservoir use the Aux feather pump to move the propeller pitch to max reverse then to flight fine for final measurement check, be sure to prime the piston with oil by dry motoring the engine before angle changes.

If the beta cap has been removed and no oil is in the hub cylinder then the blade bats must be used for blade angle changes, use two bats on opposing blades.

If removing the beta tube for propeller removal make sure to check the depth of the beta tube before removal as this measurement can be used to save time when installing the unit.

WARNING

BEFORE REMOVAL OF THE BETA CAP HANG A SUITABLE CONTAINER TO CATCH THE RESIDUAL OIL FROM CYLINDER HOUSING.

BE CAREFUL WHEN REMOVING THE BETA TUBE AND STORE IT IN A CLEAN DRY AREA, PROPERLY PROTECTED WITH THE ENDS CLOSED OFF.

WARNING

ALWAYS MAKE SURE THE SPRING LOCK IS DISENGAGED WHEN ROTATING THE BETA TUBE AND REENGAGED WHEN ROTATION IS FINISHED.

BE SURE TO MEASURE THE DEPTH OF THE PISTON FROM THE CYLINDER FACE BEFORE REMOVAL OF THE BETA TUBE.

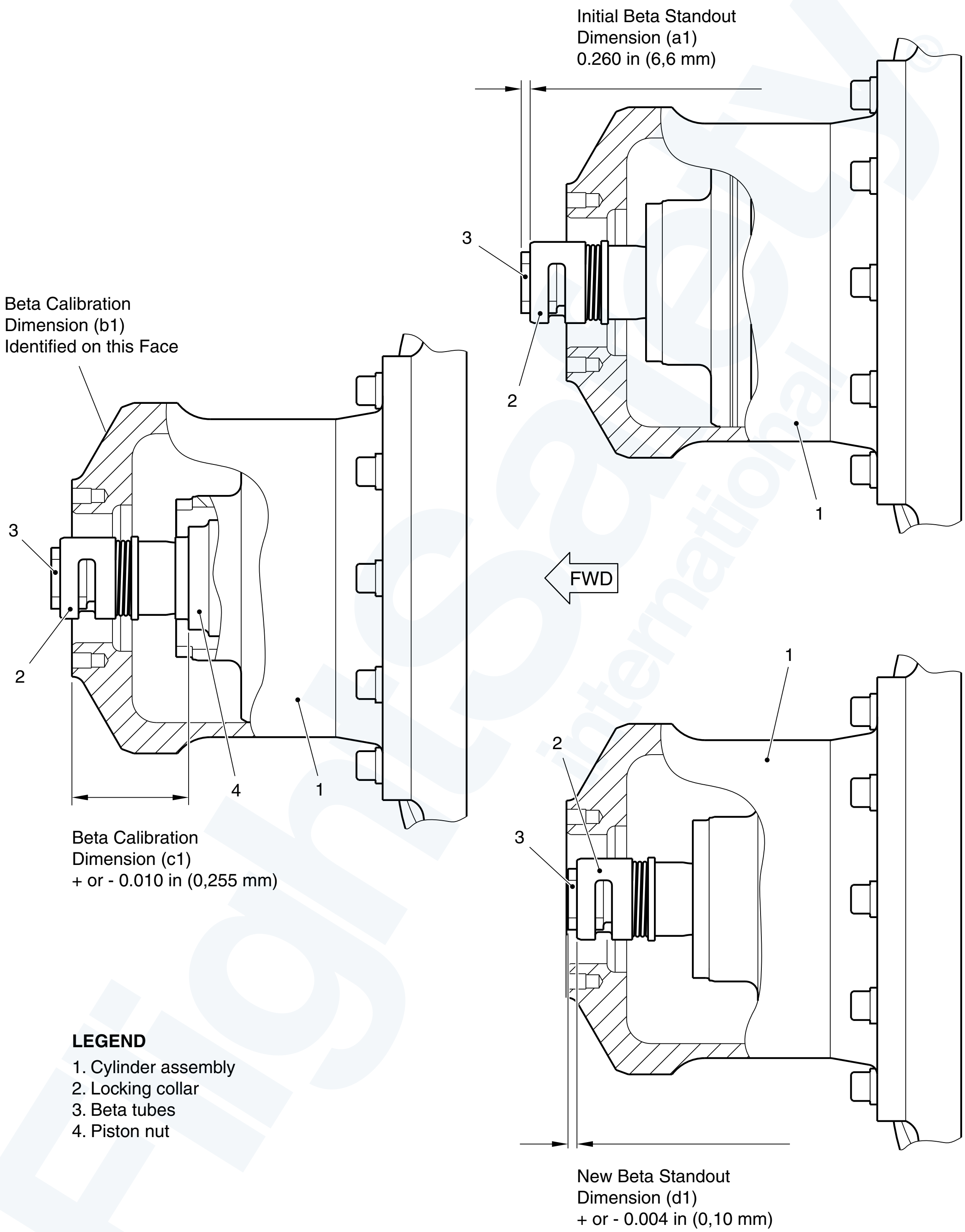

Figure 61-15. Beta Tubes - Beta Calibration and Initial Standout Dimension

Rigging of the Beta Tubes

Rigging of the beta tubes is essential to ensure that the propeller system is providing the correct blade angles, and therefore the required torque.

Refitting the same propeller:

- Set the propeller onto the flight fine stop by operating the unfeather switch with PLA @ RATING and CLA @ FUEL OFF
- Before removing the beta tubes measure the Beta Calibration Dimension between the front face of the cylinder assembly and the front face of the piston nut. Then if you are re-fitting the same propeller after an engine change, it should only be necessary to ensure that the Beta Calibration Dimension is the same +/– 0.010 in. (0.254 mm).

Fitting a new propeller and/or new beta tubes:

- When a new propeller and beta tubes are fitted iaw AMM TASK 61-20-01-400-801 then it is necessary to measure and adjust certain dimensions as follows:
- Set the propeller onto the flight fine stop by operating the unfeather switch with PLA @ RATING and CLA @ FUEL OFF
- Measure the Beta Calibration Dimension (C1) between the front face of the cylinder assembly and the front face of the piston nut.
- Measure the Initial Standout Dimension (A1) from the front face of the beta tubes to the front face of the locking collar.
- Read the Marked Beta Calibration Dimension (B1) found on the front face of the cylinder.
- By using the following formula D1 = A1 + B1 – C1 the new Beta Standout Dimension can be calculated.
- Adjust the beta tubes to the new Beta Standout Dimension.
- Reset the propeller onto the flight fine stop.
- Check the measured Beta Calibration Dimension is the same as the marked dimension +/– 0.010 in (0.254 mm).

NOTE:

If you are changing the propeller or the beta tubes, AMM Task 61-20-01-400-801 must be carried out in its entirety.

FUNCTIONAL TEST FOR THE PROPELLER ELECTRONIC CONTROLLER - CALIBRATION PROCEDURE

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Functional Test for the Propeller Electronic Controller - PEC Calibration Procedure

CAUTION

DO NOT OPERATE THE FEATHERING PUMP FOR MORE THAN 3 MINUTES WITHOUT A STOP OF 30 MINUTES. IF YOU DO THIS, YOU CAN CAUSE DAMAGE TO THE ELECTRICAL MOTOR.

Dry motor to make sure that the auxiliary oil tank is full of oil.

Un-feather the propeller onto the hydraulic flight fine stop as follows:

- Set the PLA to RATING.
- Set the CLA to FUEL OFF.
- Set the maintenance discrete to ON.
- Set and hold the maintenance Un-feather switch until the blades have un-feathered and stopped moving.

NOTE

The blades may initially oscillate about the flight fine stop.

- Hold the un-feather switch for an extended time period till the blade oscillation stops.
- Set the maintenance discrete to OFF.

Do the steps that follow:

- Set CDS GND MAINT.
- Set MAINT. On the ARCDU.
- Select OTHER SYSTEMS.
- Select EMU.
- Select POWERPLANT INTERFACE.
- Select TRIM DATA.
- Select PROP#1 BLADE PITCH TRIM or PROP#2 BLADE PITCH TRIM as necessary.

Untrimmed blade angle should read 16.0º + 3º or − 9º. If not then:

- Then refer to the PEC fault codes in the FIM.
- Make sure that the propeller has moved onto the hydraulic flight fine stop.
- Make sure that the beta tubes are at the correct set dimension.

In the Flight Compartment:

- Set the CLA to MAX (1020) and the PLA to RATING for the engine to be trimmed.
- Set the CLA to FUEL OFF and the PLA to RATING for the other engine.
- Set the maintenance discrete to ON.
- Press RIGGING TRIM ON for 20 seconds.

On the ARCDU:

- Make sure the STATUS message changes from IN PROG to COMPLETE.

If ABORTED and/or FAILED and/or the blade angle is not within the untrimmed blade angle limits:

- Set the maintenance discrete to OFF
- Start the engine.
- Operate the engine at ground idle power with the propeller un-feathered for 1 min.
- Shutdown the engine.
- Repeat the adjustment procedure.

If the STATUS message changes from IN PROG to ABORTED and/or FAILED and/or the blade angle is not within the un-trimmed blade angle limits:

- Refer to the PEC fault codes in the FIM.

If access to the propeller fault codes is necessary:

Set CDS GND MAINT.

POWERPLANT MAIN MENU.

- Set MAINT.
- OTHER SYSTEMS.
- EMU.
- POWERPLANT FAULTS.
- Select PROP#1 or PROP#2 to see the fault codes.

Make sure that the trimmed blade angle is 16.0º + or − 0.5º for both channels. If not then:

- Refer to the Propeller Electronic Controller (PEC) fault codes in the FIM.
- Make that the beta tubes are at the correct set dimension.

Do the steps that follow to feather the propeller with the feather pump:

- Set the maintenance discrete to ON.
- Feather the propeller using the ALT FTHR switch.
- Set the maintenance discrete to OFF.
- Operational test of the propeller fault code indication.

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61-20-00 PROPELLER CONTROLLING SYSTEM

INTRODUCTION

The propeller control system modulates blade angle or pitch, to achieve the necessary propeller RPM (Np) and blade pitch (Beta) control.

GENERAL

The power levers send signals to FADEC which processes the signals and forwards them to the related PEC to control the propeller pitch.

The condition levers send signals to the PEC directly to set Np and to feather the propellers.

The PEC supplies the current to the servo-valve drive, through a dual line hydro-mechanical actuation system, to change the blade pitch.

SYSTEM DESCRIPTION

Refer to Figure 61-16. Propeller Control Block Diagram.

The propeller control system includes:

- Beta tube assembly
- Beta feedback transducer
- Dual pulse probe assembly
- PCU
- Pitch control unit Adapter
- OSG
- Feathering pump
- PEC
- PROPELLER CONTROL panel.

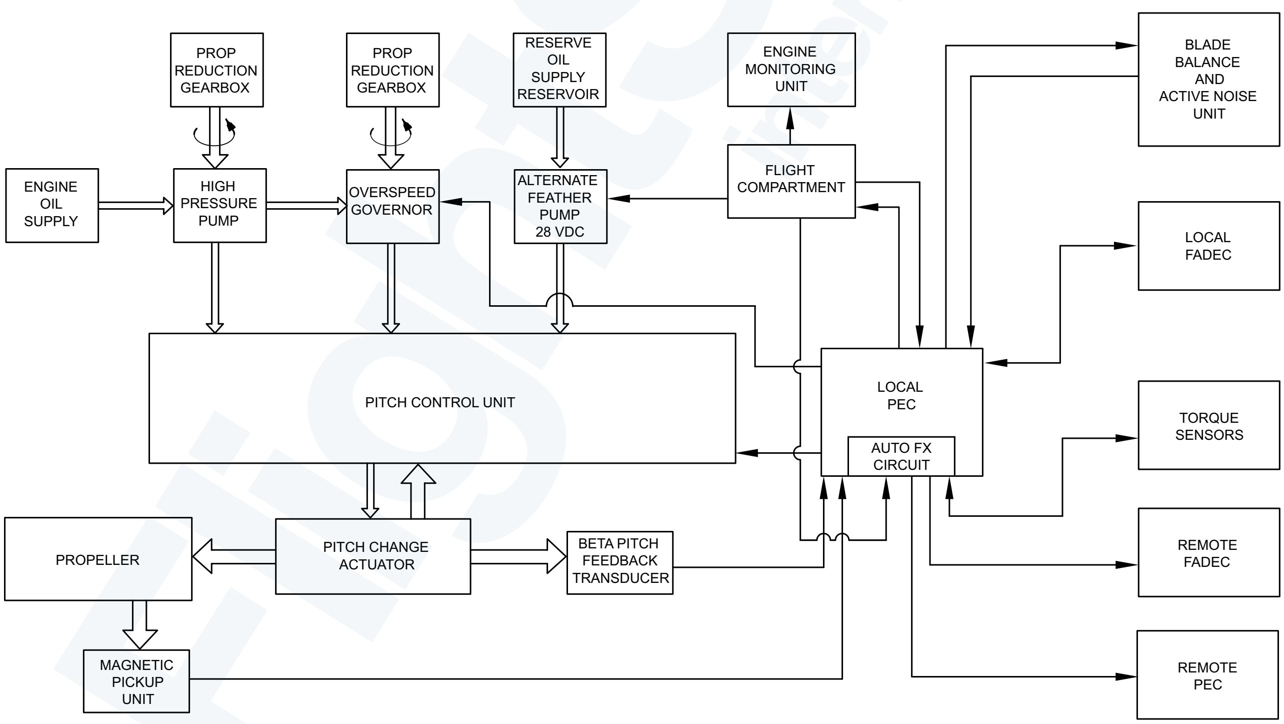

Figure 61-16. Propeller Control Block Diagram

Refer to Figure 61-17. Propeller Control Loop Logic.

The main modes of propeller operation for control during flight and on the ground are:

- Constant Speed Mode (PLA between above Flight Idle and Rated Power Detent)
- Beta Control Mode (PLA between above Flight Idle to below Disc)
- Reverse speed control mode (PLA between below Disc and Max Reverse)
- Synchro-phase control mode (slave propeller)
- Feather Mode.

The propeller control system has these units:

- Propeller Electronic Control Unit (PECU)
- Beta Tube Assembly
- Magnetic Pick-up Unit
- PCU
- OSG - High pressure Pump
- Feathering Pump
- PROPELLER CONTROL panel.

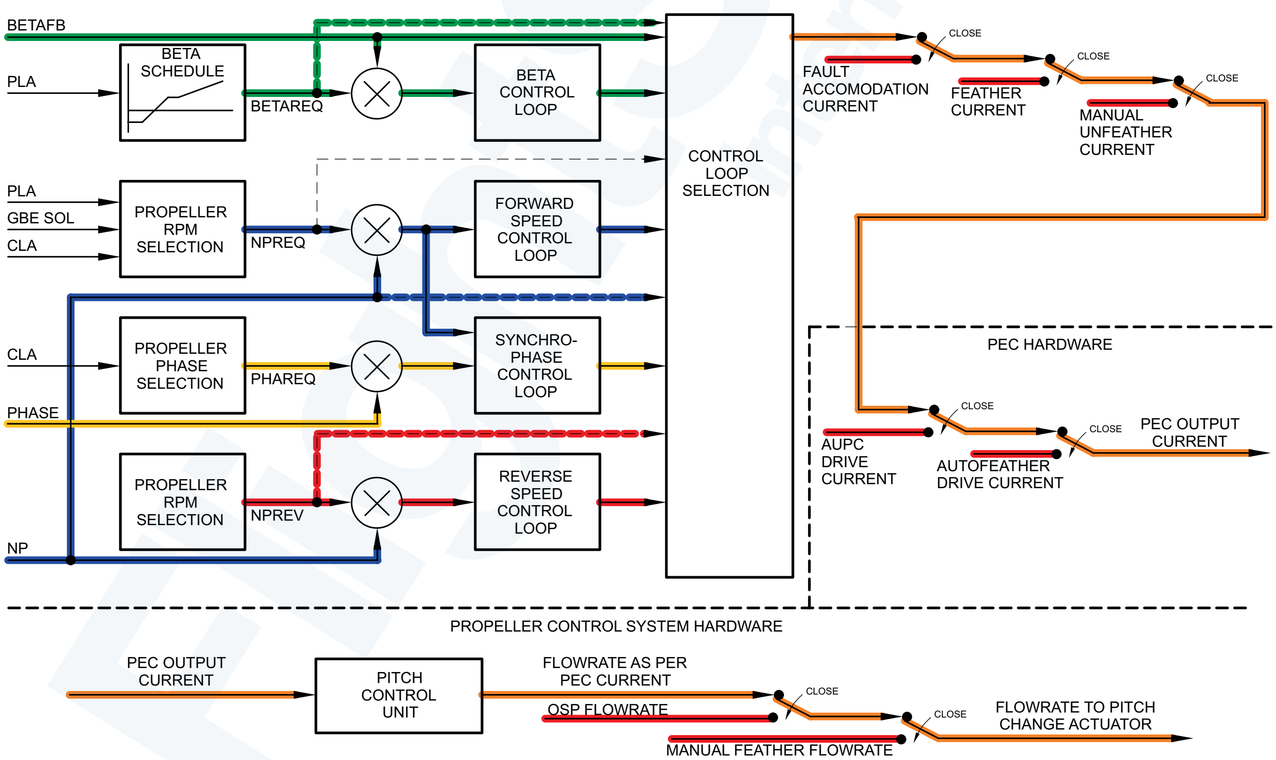

Figure 61-17. Propeller Control Loop Logic

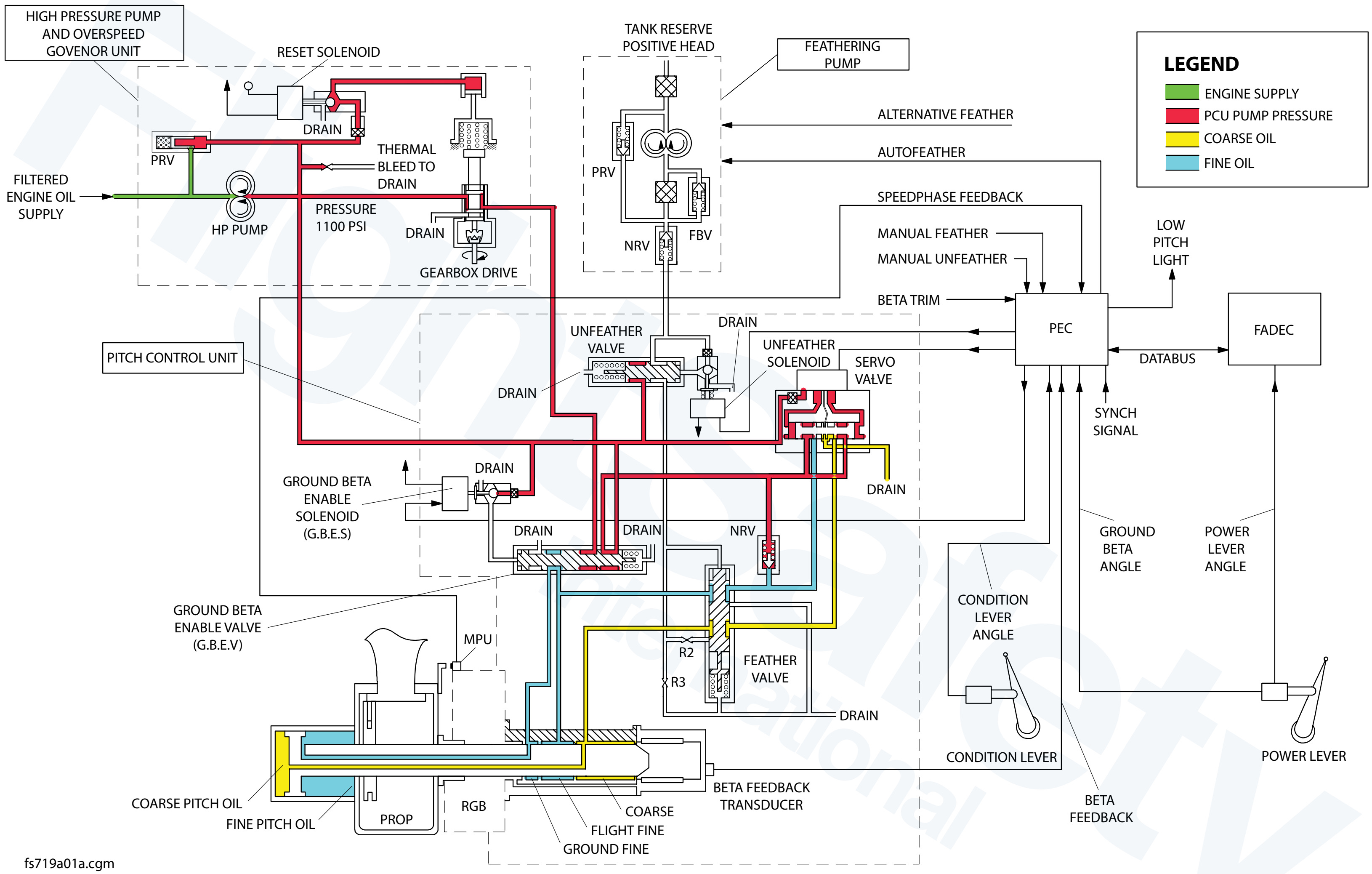

Figure 61-18. Ground Beta Mode

Beta Control

Refer to:

- Figure 61-18. Ground Beta Mode.
- Figure 61-19. Propeller Function Diagram.

The beta control system controls propeller pitch in the beta mode where propeller pitch is a function of PLA.

In this mode the system operates in closed loop blade angle control.

The blade angle is set by the dual PLA RVDT. The PEC receives PLA signals from FADEC.

During beta control, the PEC directs the servo-valve to meter oil into the fine or coarse pitch chamber to achieve the required blade angle.

The measured blade pitch is determined from the Beta Feedback Transducer (BFT) output signal and the required blade angle, is determined from a schedule against PLA. The system limits the pitch change rate in order to match the rate of change of torque absorption of the propeller to the rate of change of engine torque.

This reduces propeller overspeeds and underspeeds during power lever transients. The rate limit is based on blade pitch and direction of blade pitch change.

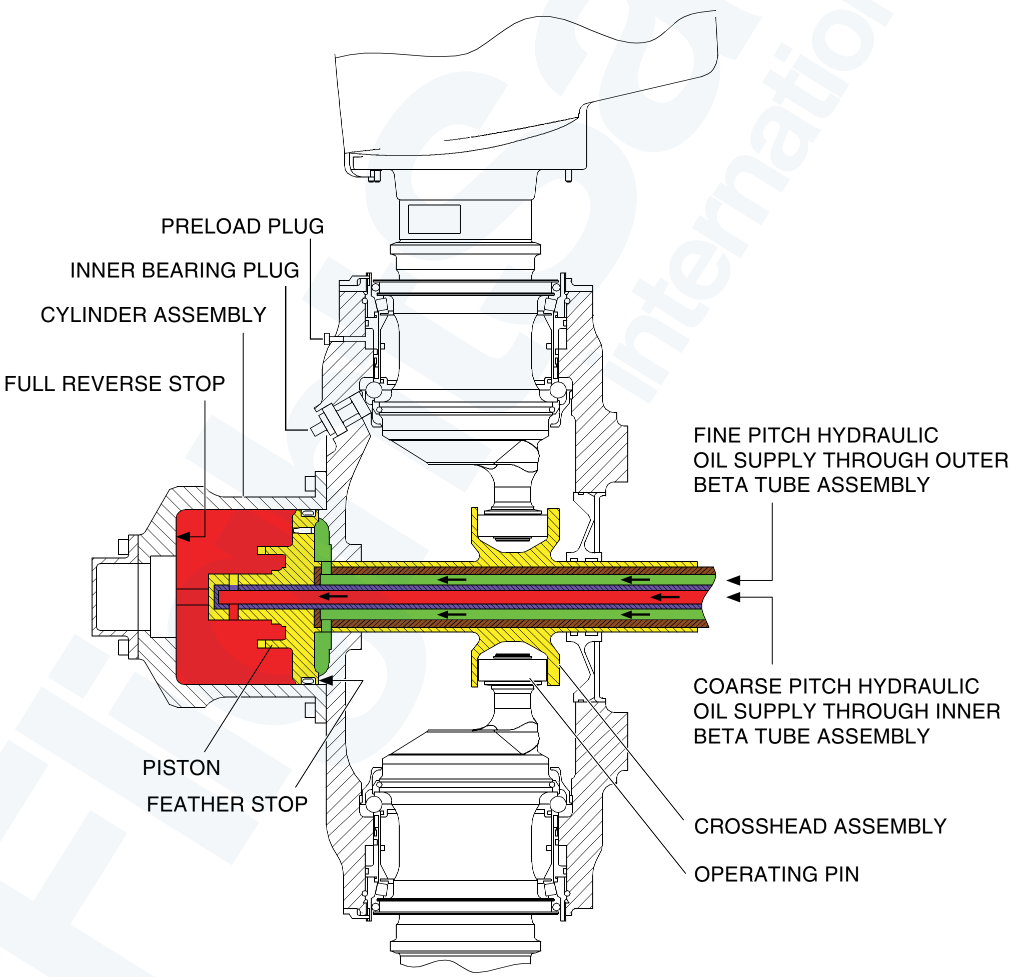

Figure 61-19. Propeller Function Diagram

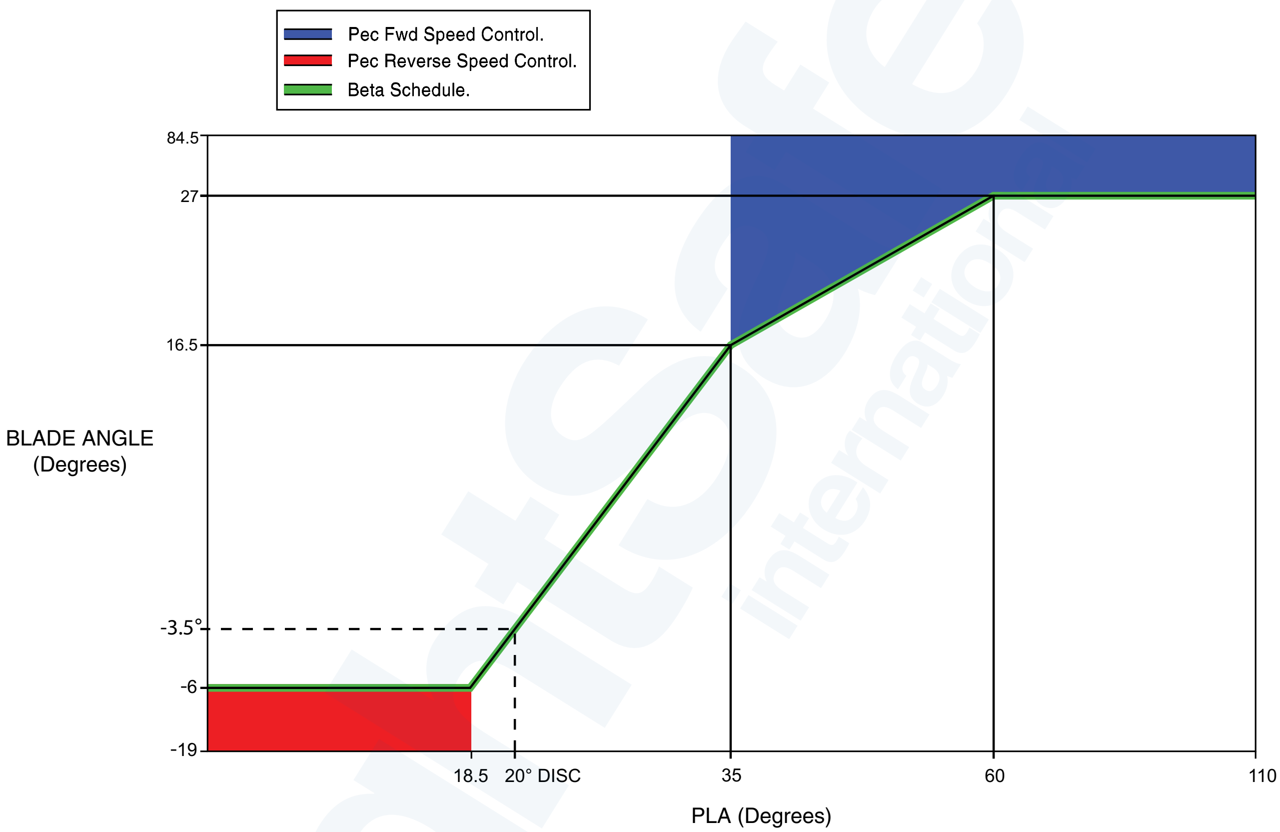

Figure 61-20. Beta Schedule

Refer to Figure 61-20. Beta Schedule.

The Beta schedule imposes the minimum blade angle that is allowed for any PLA. The position of the ports in the PCU/Beta tubes make sure that fine pitch in the in-flight mode is limited to a minimum blade angle of 16°.

This hydraulic cut-off of oil pressure is specified as the “hydraulic flight fine stop” interlock. The flight fine stop keeps a minimum pitch consistent with positive counterweight effort driving towards coarse pitch, to make sure the OSG is effective throughout the in-flight pitch range. A “soft” flight fine stop at approximately 16.5° is programmed into the PEC.

This stop makes sure the blade angle does not drop below 16.5° while the power lever is at, or above flight idle in flight. This is specified as a “software flight fine stop”.

A detent on the power lever quadrant stops unintentional movement of the lever below flight idle during flight. A power lever switch which closes below a PLA of 33°, energizes the Ground Beta Enable Solenoid (GBES), if the aircraft is on the ground or RA is less than 20 feet. When the GBES is energized, a pilot valve vents the chamber at the end of the Ground Beta Enable Valve spool (GBEV) to drain. The spring at the other end moves to its ground position.

Fine pitch oil pressure then enters another chamber in the PCU, which allows ground beta blade angles down to reverse.

With the GBEV in the ground position, the HP oil supply from the OSG is isolated and the second stage of the servo valve is supplied directly by the HP pump; this removes the OSG from the circuit.

Failure of the GBEV spool to move to its in-flight position will be sensed during an OSG test.

When the power lever is in the beta range, propeller speed is governed by the FADEC controlling the engine fuel system to produce 660 N_P.

NOTE

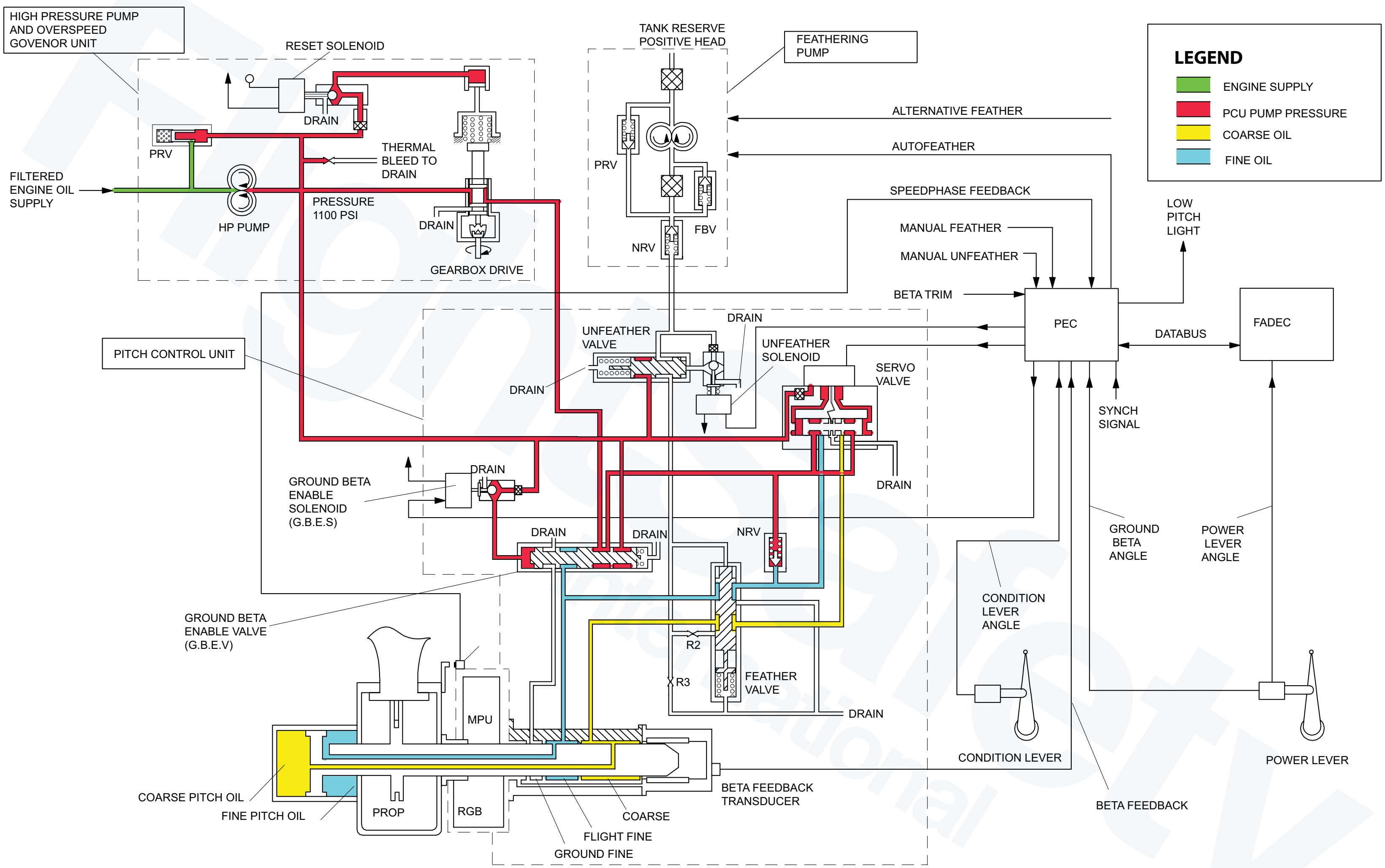

Figure 61-21. Steady State Constant Speed Control

Forward Speed Control

Refer to Figure 61-21. Steady State Constant Speed Control.

The purpose of the forward speed control system is to control the propeller in the constant speed mode.

In this mode the system operates in closed loop propeller RPM control. Propeller rpm (N_p) is calculated from the propeller dual pulse probe assembly output by timing.

Three discrete speeds 850, 900 and 1020 N_p can be selected from CLA. In the event of the PLA being moved into the over-travel position, NPREQ is set to 1020 N_p. This speed is latched until the CLA is moved to the 1020 N_p position, or the start/feather position.

During in-flight "constant speed" operation, the PEC directs the servo-valve to meter high pressure oil into the propeller fine pitch chamber. This is to balance the coarse seeking moment applied to the blades, so that the propeller stays at the selected speed (N_p). If there is a loss of HP oil supply, the blades will "autocoarsen" to a safe pitch condition, to give low windmilling drag.

If the N_p is greater than the demanded speed, the servo-valve will send the oil pressure to the coarse pitch chamber to reduce propeller speed.

Constant speed mode is entered when propeller speed reaches 850, 900 or 1020 rpm, depending on the condition lever selection.

HP oil for constant speeding flows through the OSG before it reaches the servo-valve.

If the servo valve sticks at the fine pitch selection, N_p will increase to approximately 1060 RPM. The OSG spool will then isolate the propeller control system from the HP oil supply. N_p will decrease due to propeller counterweight action. The OSG will then reconnect the HP oil supply and a stable governing condition at 1060 RPM will be achieved.

The OSG spool spring is in a cylinder, on a piston, that is connected to the HP oil supply, through a solenoid operated pilot valve. The solenoid is energized by the PROP O/SPEED GOVERNOR TEST switch on the pilot's side console. A WOW input to the PEC prevents test operation during flight.

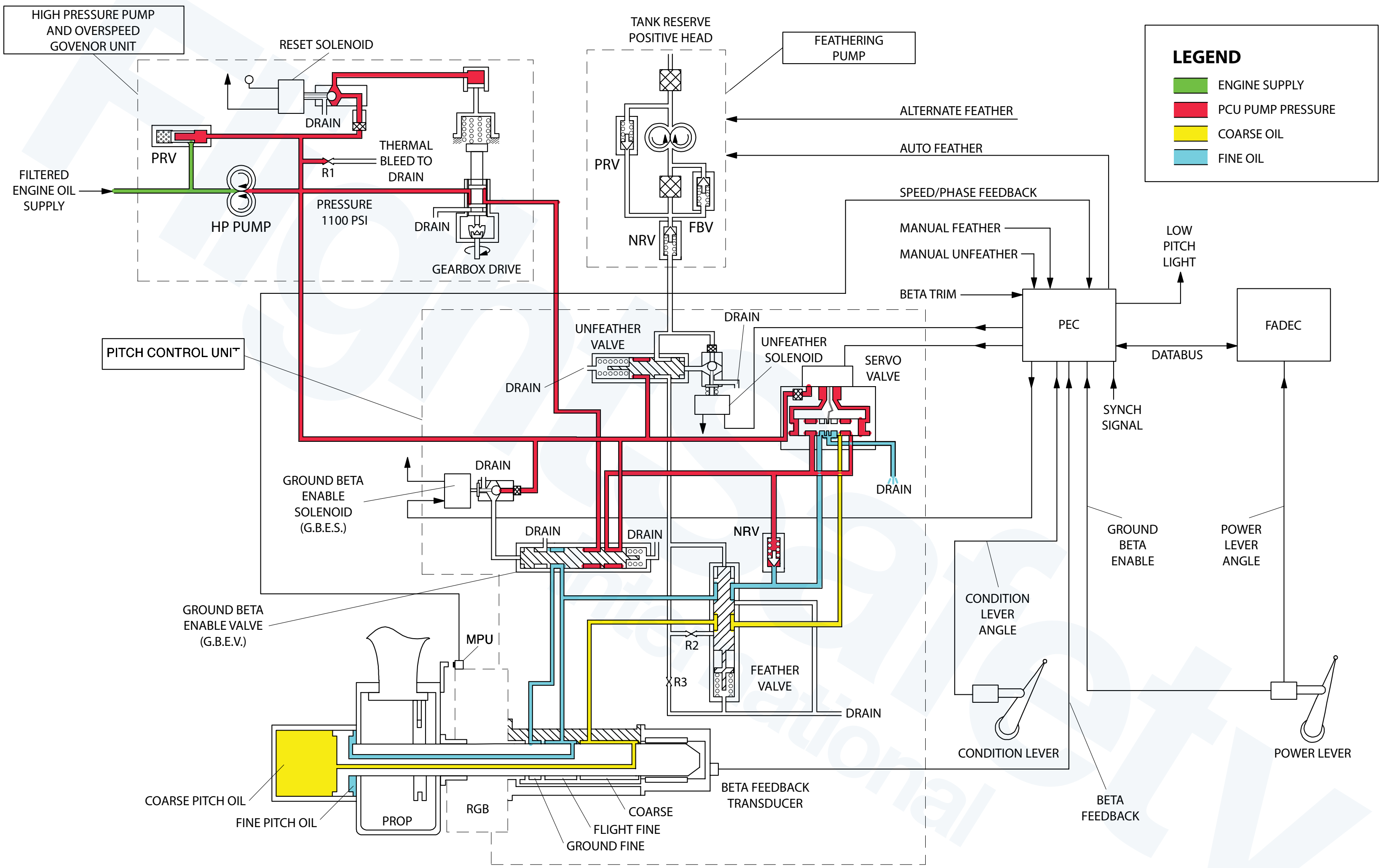

Figure 61-22. Normal Feather

Synchro-Phase Control

The purpose of the synchro-phase control is to control the propeller phase.

In this mode the system operates in closed loop propeller speed and propeller phase control. Synchro-phasing acts to reduce cabin noise by ensuring that the relative position, or phase difference, between the slave and master propellers is controlled.

The phase angle is calculated by timing the differences between master (#1) and slave propeller (#2) Dual Pulse Probe Assembly signals.

Normal Feather

Refer to Figure 61-22. Normal Feather.

The purpose of the normal feather mode is to feather the propeller on routine engine shutdowns.

Normal feather is set when the condition lever is moved to either the START/FEATHER or FUEL OFF detent. This signal commands the PEC to drive the servo-valve towards coarse pitch.

NOTES

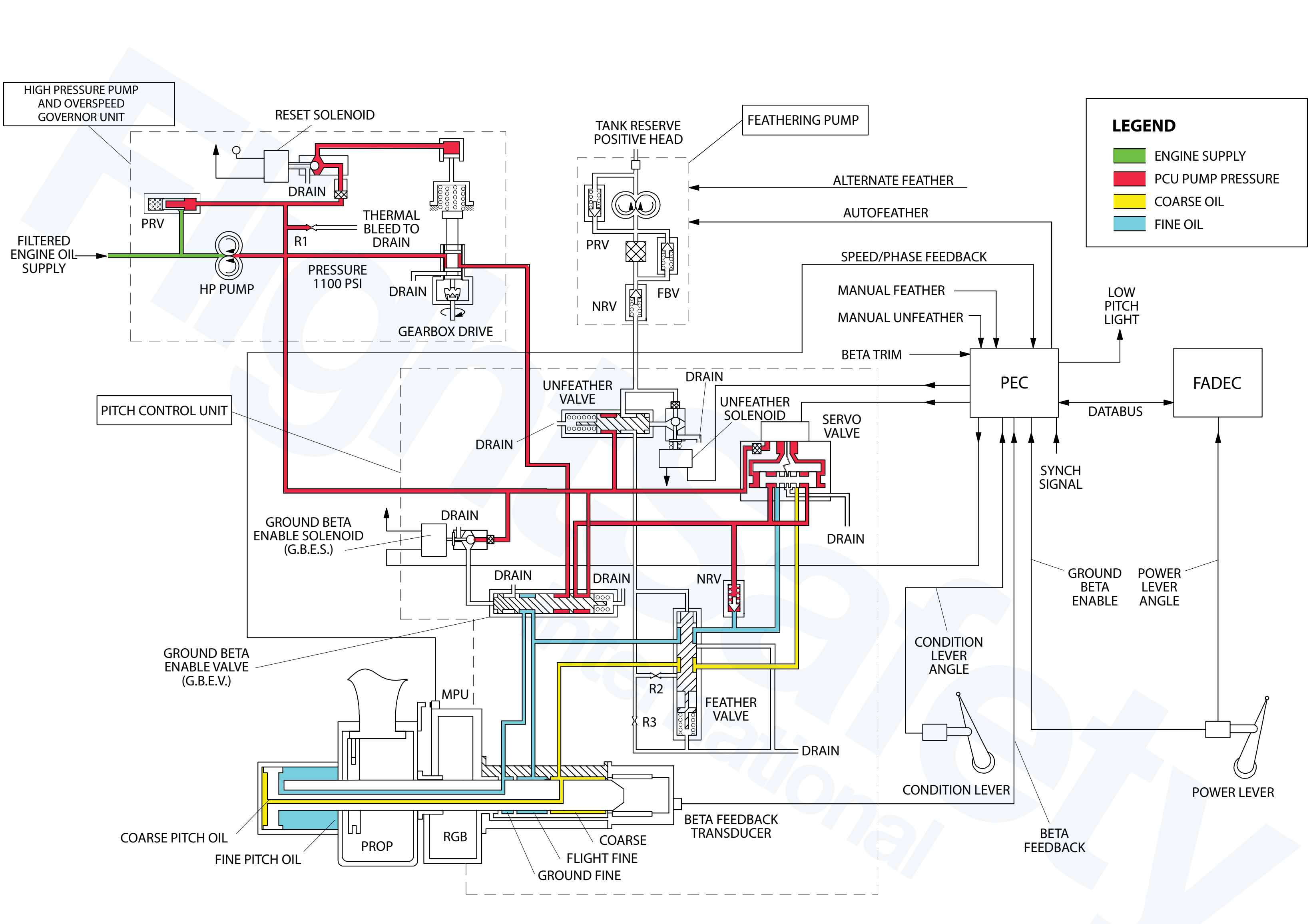

Figure 61-23. Maximum Reverse

Reverse Speed Control

Refer to Figure 61-23. Maximum Reverse.

The purpose of the reverse speed control is to control the propeller speed in reverse between 950 and 1030 RPM.

In this mode the system operates in closed loop propeller RPM control.

The FADEC schedules fuel based on a power schedule versus PLA, with a maximum limit of 1500 SHP.

NOTES

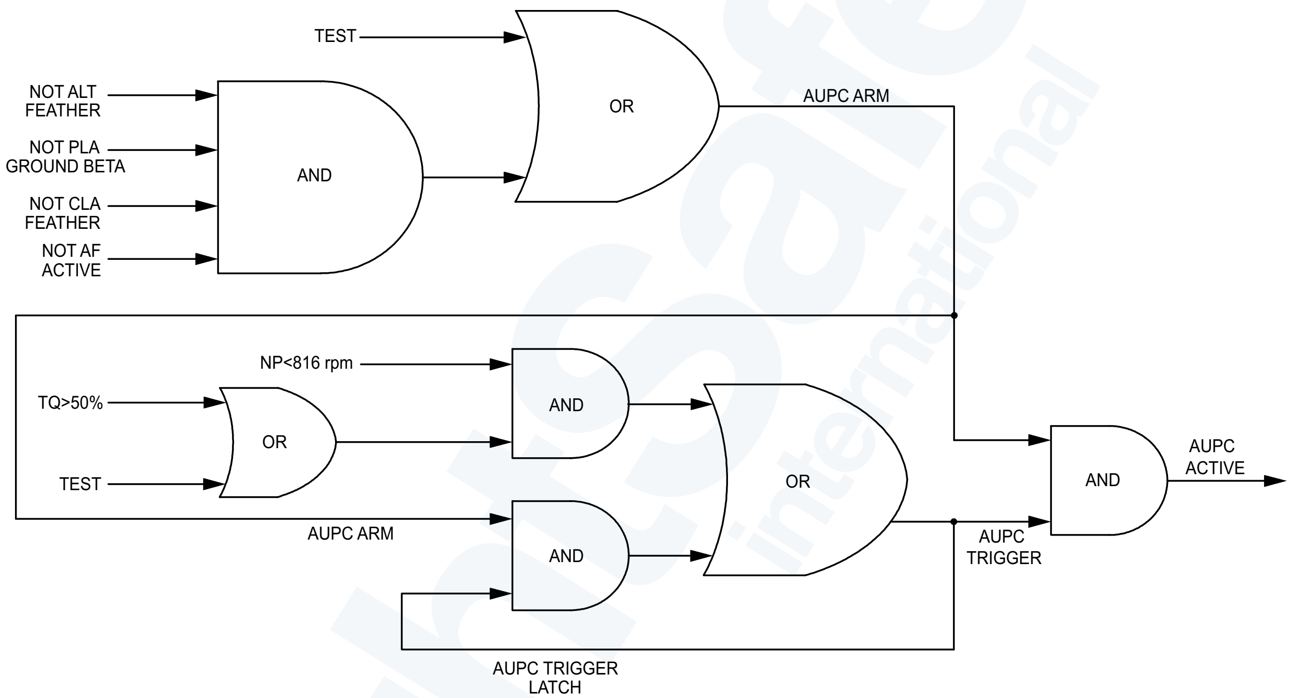

Figure 61-24. AUPC Logic

Automatic Underspeed Protection Circuit (AUPC)

Refer to Figure 61-24. AUPC Logic.

The purpose of AUPC is to protect the aircraft from lack of thrust caused by a common software problem in both PEC. This would result in a drive coarse signal and loss of thrust.

This function is implemented in hardware independent of the PEC control lane software.

If underspeed condition is detected on both N_p sensors, it will cause a drive fine signal to be generated and the N_p will increase.

The propeller speed increase will be arrested by the OSG. This function overrides the authority of the control lane software but is, overridden by the autofeather function.

The AUPC is armed when:

- PLA is above FI
- CLA above Start/Feather and
- Autofeather or manual feather is not demanded for longer than 0.5 seconds.

AUPC is activated by:

- Propeller speed is less than 80% and torque is over 50% for longer than 1 second.

AUPC is disarmed by any of the following:

- PLA moved to FI
- CLA moved to Start/Feather
- Autofeather or manual feather demanded.

The AUPC function is tested during Autofeather Test.

Automatic Take-Off Thrust Control System (ATTCS)

The purpose of the ATTCS is to give the necessary protection against engine failure during the critical part of the take-off roll.

The ATTCS causes an uptrim of the remote powerplant and an auto-feather on the failed powerplant.

Any erroneous feather of the local propeller will cause an uptrim of the remote powerplant.

The ATTCS has two sub-systems:

- Autofeather
- Uptrim.

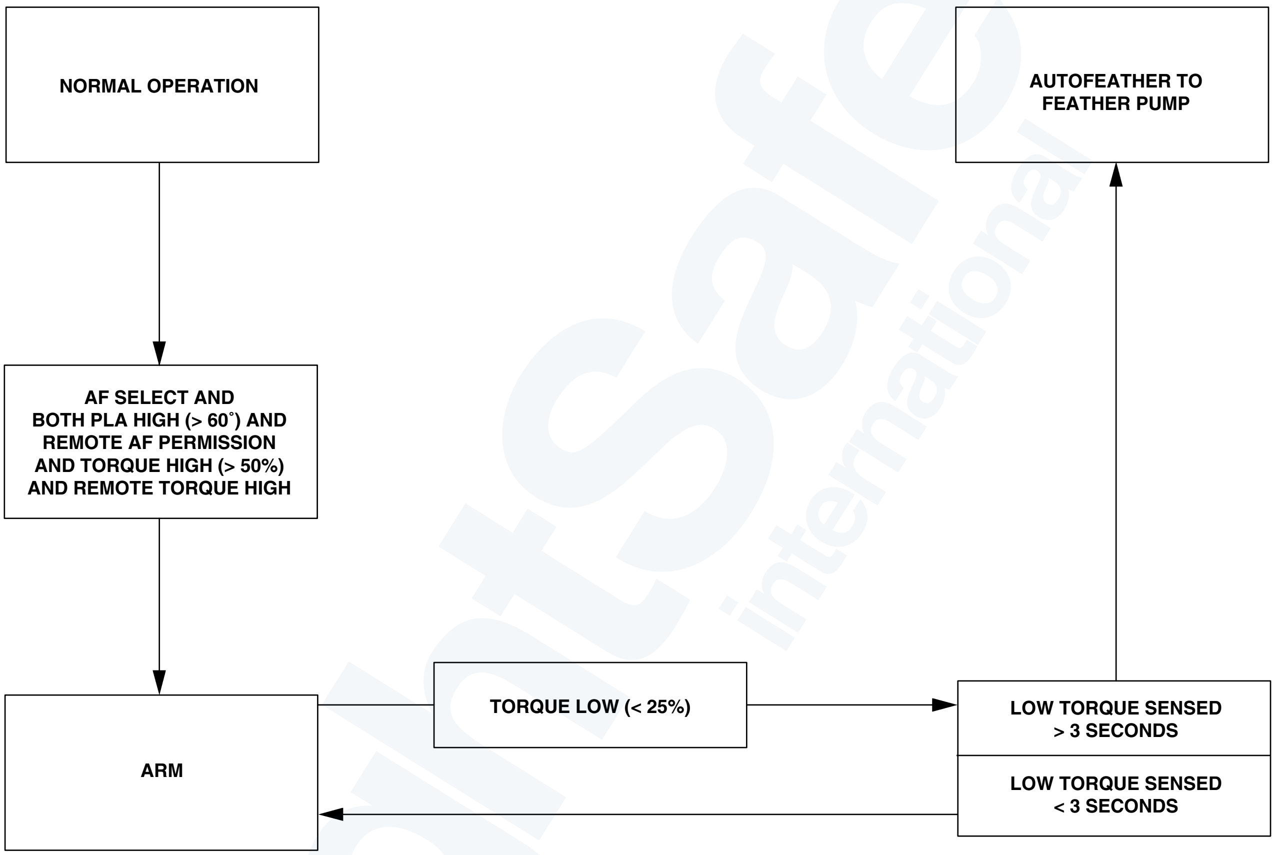

Figure 61-25. Autofeather State Transitions

Autofeather

NOTES

The purpose of the autofeather system is to:

- Reduce drag on the failed engine
- Increase power on the serviceable engine
- Prevent the two engines from going into autofeather simultaneously.

Refer to Figure 61-25. Autofeather State Transitions.

The autofeather function is implemented in PEC hardware. It includes cross-wing communication with the remote PEC.

Two torque signals from the N_{pt}/Q sensors on an engine are needed to show less than 25% torque for three seconds before the system autofeathers.

The autofeather function is armed by:

- PLA to above 60°
- Local and remote torques above 50%
- Pressing the AUTOFEATHER SELECT switch.

When both powerplant Autofeathers are armed, an A/F ARM indication is shown on the ED.

Autofeather is disarmed by either de-selecting the AUTOFEATHER SELECT switch, or moving either PLA below 60.

An autofeather will cause the PEC to:

- Output a servo-valve drive coarse signal
- Energize the auxiliary pump relay.

At this point the A/F ARM indication is removed from the ED.

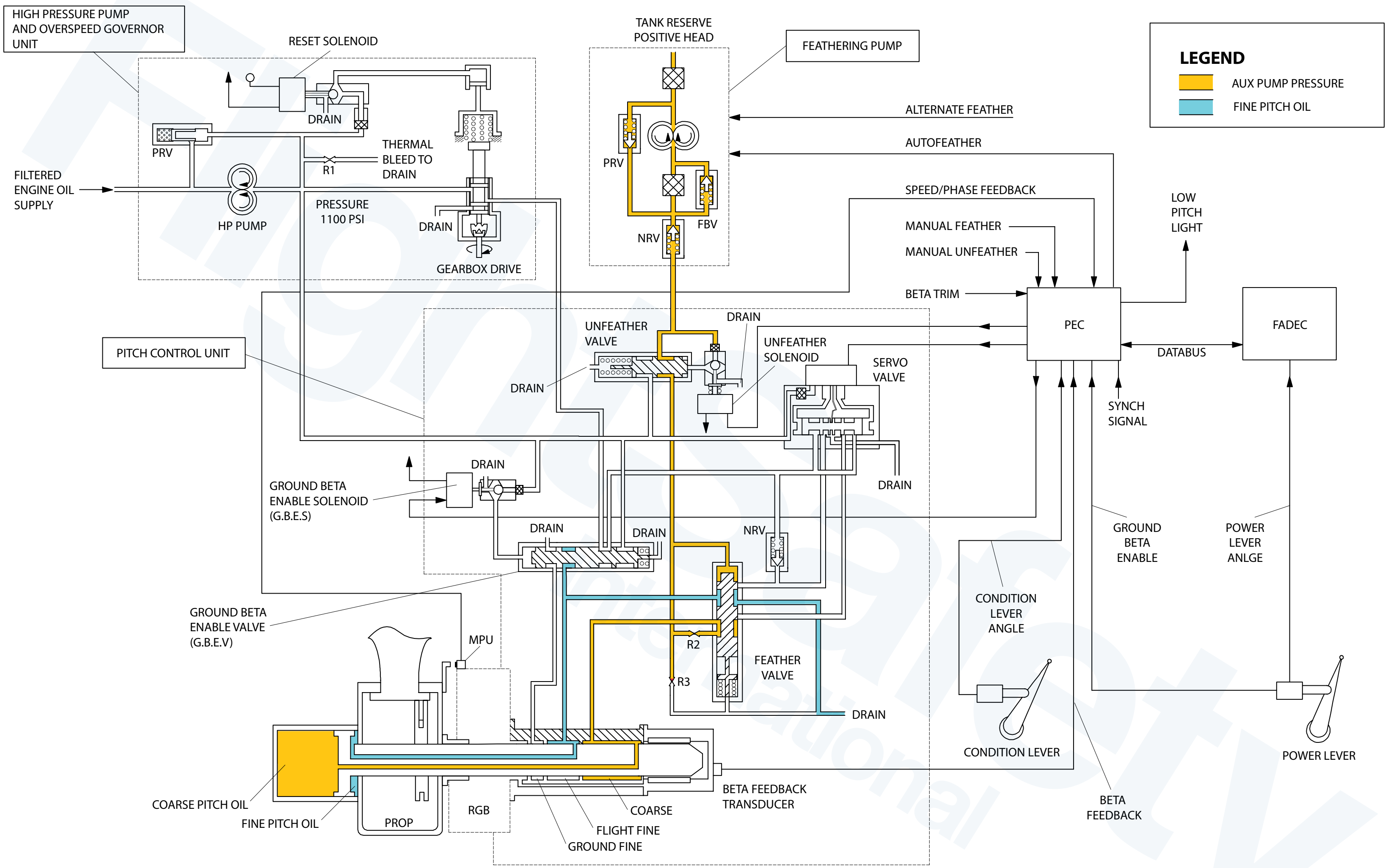

Figure 61-26. Autofeather Mode

Refer to Figure 61-26. Autofeather Mode.

When the propeller is auto-feathered the system can only be dis-armed by de-selecting the AUTOFEATHER SELECT switch.

A detected failure in the ATTCS results in the system inhibiting the A/F SELECT indication on the ED.

NOTES

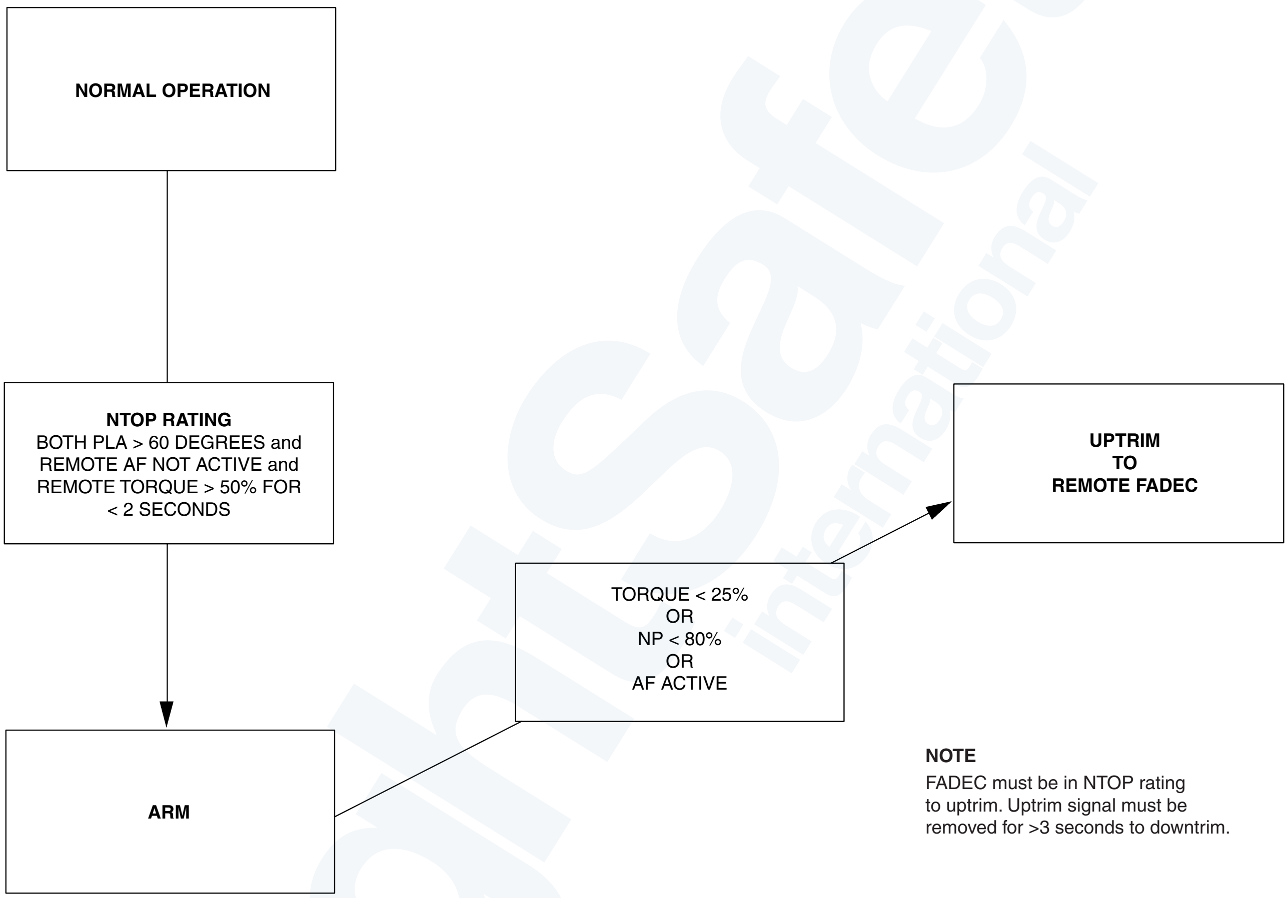

Figure 61-27. Uptrim For Low Np or Low Torque

Uptrim

Refer to Figure 61-27. Uptrim For Low Np or Low Torque.

The purpose of the Uptrim system is to uptrim the local engine by 10%, when signaled to do so by the remote PEC.

The uptrim function is implemented in the PEC hardware, with communication from the local PEC to the remote FADEC.

Uptrim is armed by moving both PLA above 60° and having remote torque high (Q) for > two seconds.

Uptrim is disarmed by moving either PLA below 60° or by autofeather of the remote powerplant.

When the system is armed an uptrim will occur if:

- Both torque signals drop below 25% or
- Both torque sensors show a speed of less than 800 N_p or
- Autofeather active.

An uptrim will cause the PEC to command the FADEC to change from the NTOP schedule, to the Maximum Take-Off Power schedule.

An UPTRIM message will appear on the ED.

NOTES

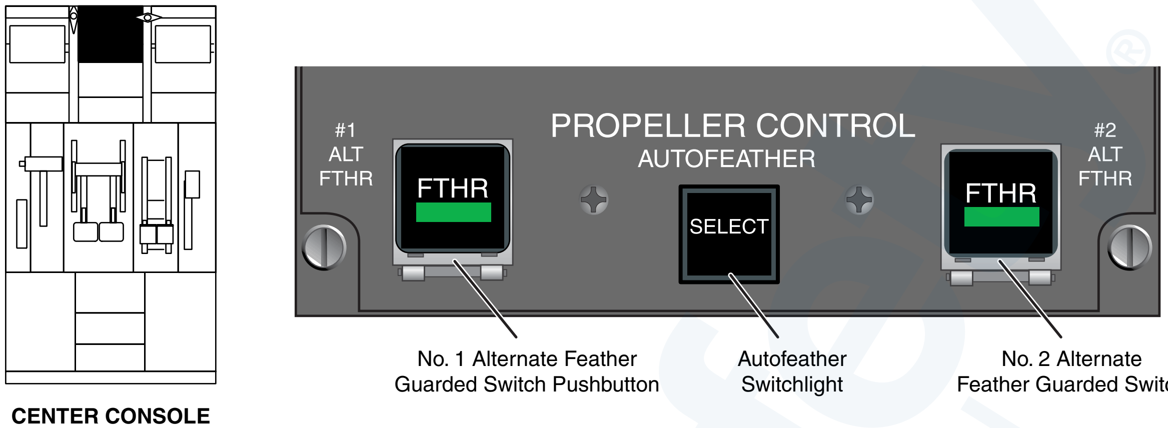

Figure 61-28. Propeller Control Panel

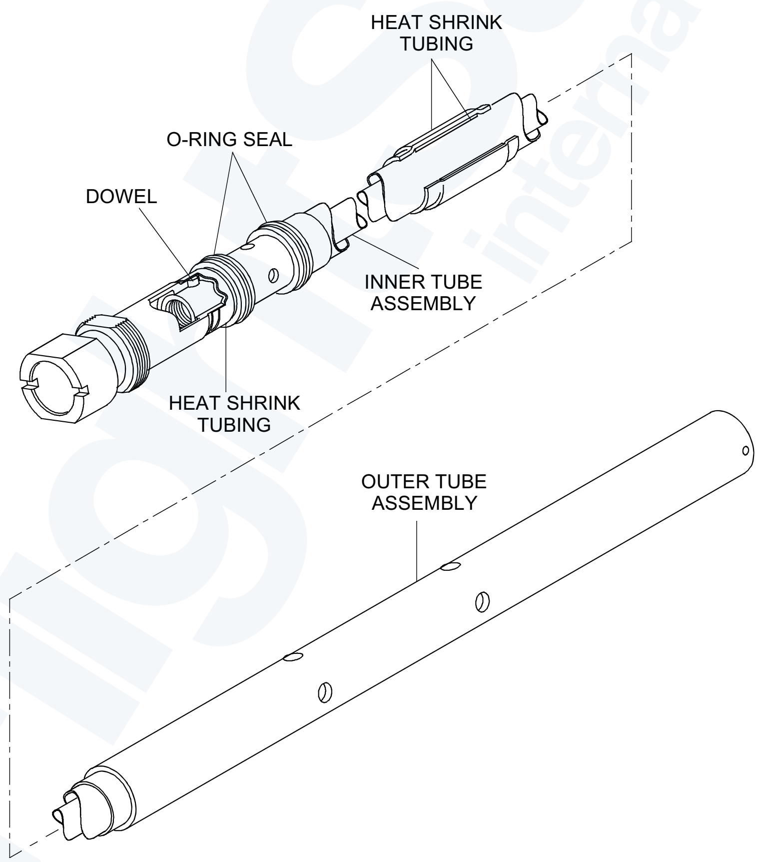

Figure 61-29. Beta Tube Assembly

Alternate Feather and Propeller Control panel

Refer to Figure 61-28. Propeller Control Panel.

The purpose of the Alternate Feather system is to give an alternate means to feather the propeller of a failed engine.

Alternate feather is set when the condition lever is moved to the START/FEATHER, or FUEL OFF position, operating the CLA < 40°, or the microswitch and the applicable ALT FTHR switchlight, on the PROPELLER CONTROL panel is pushed.

The auxiliary pump starts (30 second run time) and supplies pressure to enable the back-up feather valve and drive the propeller to coarse pitch.

The control panel is installed on the upper center console in the flight compartment.

COMPONENT DESCRIPTION

Beta Tube Assembly

Refer to Figure 61-29. Beta Tube Assembly.

The beta tubes are installed in the center of the propeller hub.

The assembly is attached to the front of the crosshead assembly by means of an adjustable screw thread and locking collar with a spline. The adjustable screw permits setting of the assembly within the Beta Feedback Tube to achieve full range of feedback.

The beta tubes are used to transfer fine and coarse pitch oil pressure from the PCU to the propeller pitch change mechanism. The tubes also monitor the blade pitch angle for beta control.

Dual Pulse Probe Assembly

Refer to Figure 61-30. Dual Pulse Probe Assembly (MPU).

The probe assembly is a dual channel device installed on the brush block bracket on the front of the reduction gearbox.

The speed and phase of the propeller is sensed by a dual pulse probe assembly and a set of seven targets on the deicing slip-ring. Six targets give speed signaling inputs and the seventh acts as a master reference for balancing purposes and synchro-phase.

TASK 61-20-06-820-801 describes how to measure and adjust the gap for the Dual Pulse Probe to the target screws.

Rotate the propeller and measure the gap at each target screw to the probe using non-magnetic feeler gauges.

Turn the propeller to position the target screw with the smallest gap directly below the probe.

Adjust as necessary to achieve the correct gap by adding/removing the shim(s) below the dual pulse probe attachment

Figure 61-30. Dual Pulse Probe Assembly (MPU)

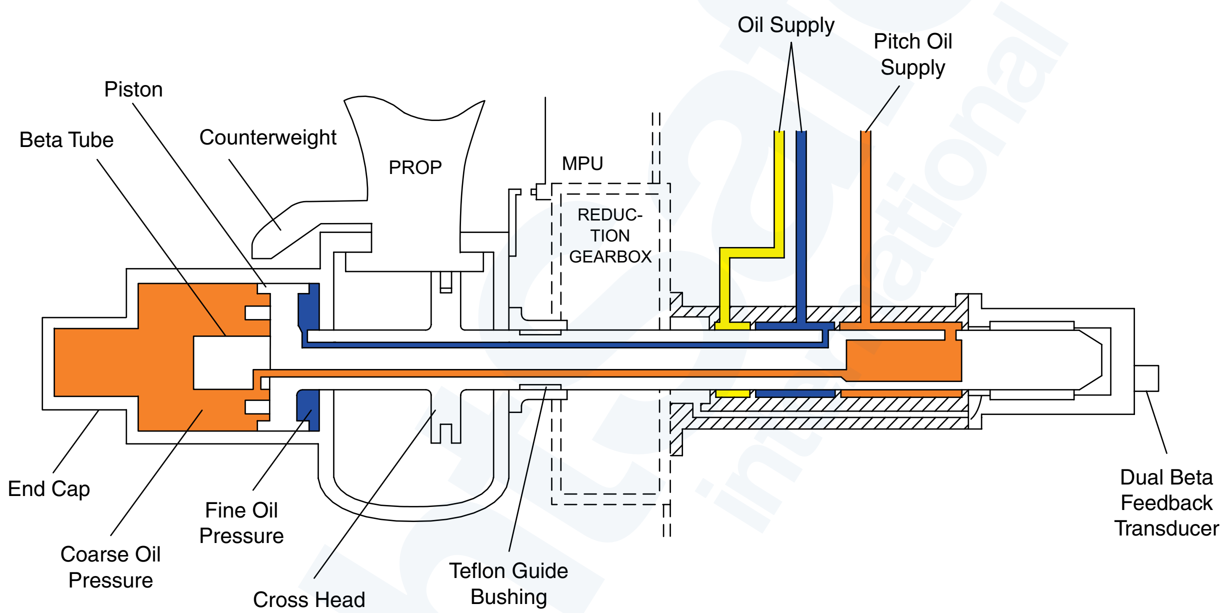

Figure 61-31. Beta Tube Assembly and Beta Feedback Transducer

Beta Feedback Transducer (BFT)

Refer to Figure 61-31. Beta Tube Assembly and Beta Feedback Transducer.

The Beta Feedback transducer is on the PCU, at the back of beta tube assembly.

The Beta Feedback transducer monitors the blade pitch angle for beta control.

NOTES

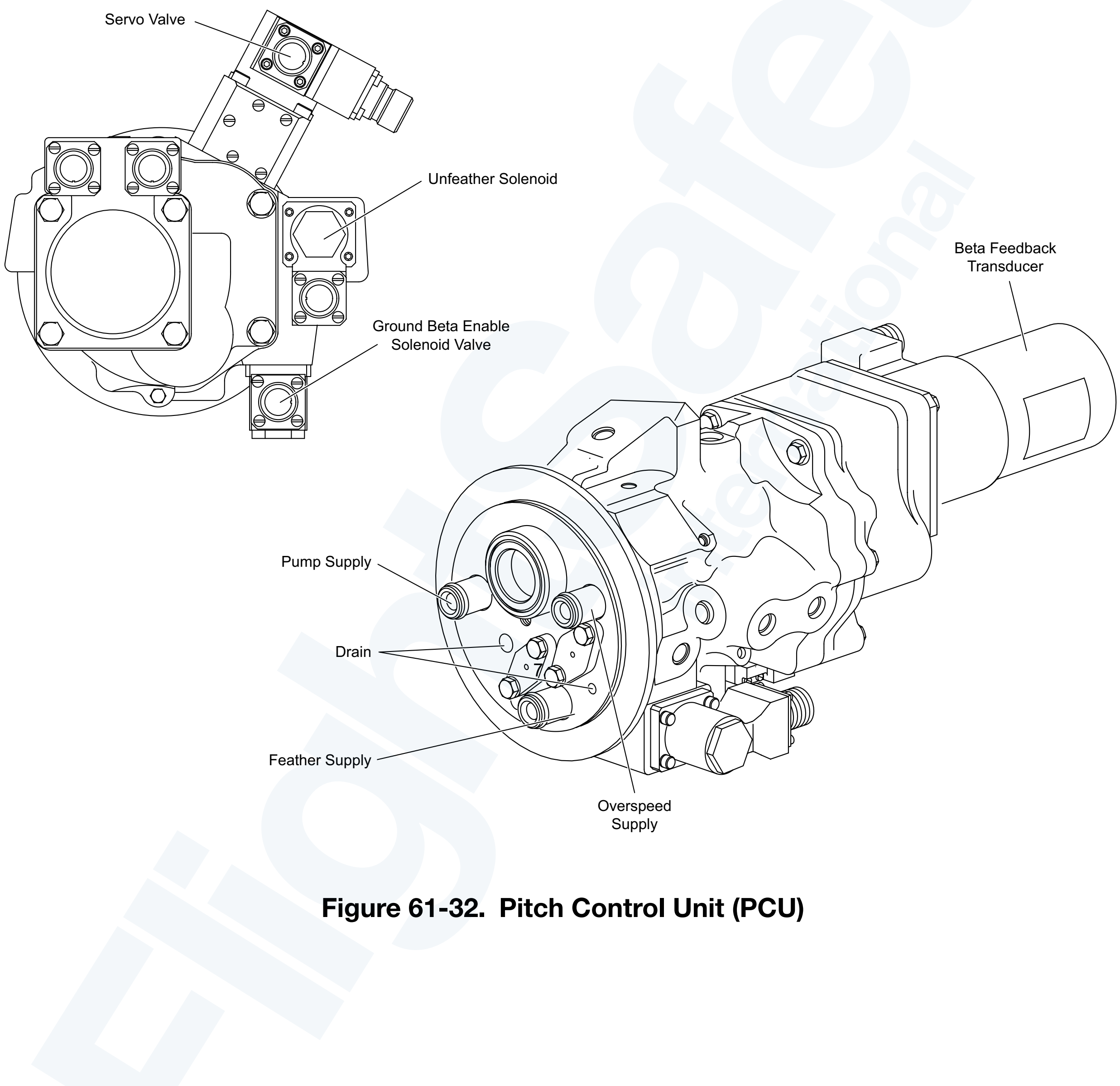

Figure 61-32. Pitch Control Unit (PCU)

Pitch Control Unit (PCU)

NOTES

Refer to:

- Figure 61-32. Pitch Control Unit (PCU).
- Figure 61-33. Pitch Control Unit.

The PCU is on the rear face of the reduction gearbox and is attached to a bolted on adaptor on the gearbox by a V-band clamp.

The PCU controls the flow of oil pressure to the fine and coarse pitch sides of the propeller pitch change mechanism. The following components comprise the PCU:

- Servo-valve
- Ground Beta Enable Solenoid valve
- Unfeather valve
- Back-up feather valve and
- Beta Feedback Transducer.

Pitch Control Unit Adapter

The adapter is installed between the reduction gearbox and the pitch control unit. It provides a mounting point for the PCU.

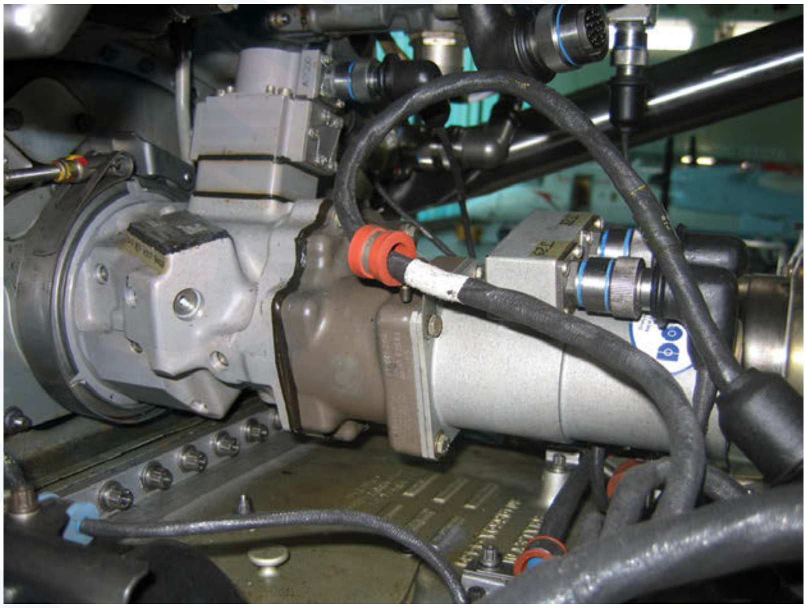

Figure 61-33. Pitch Control Unit

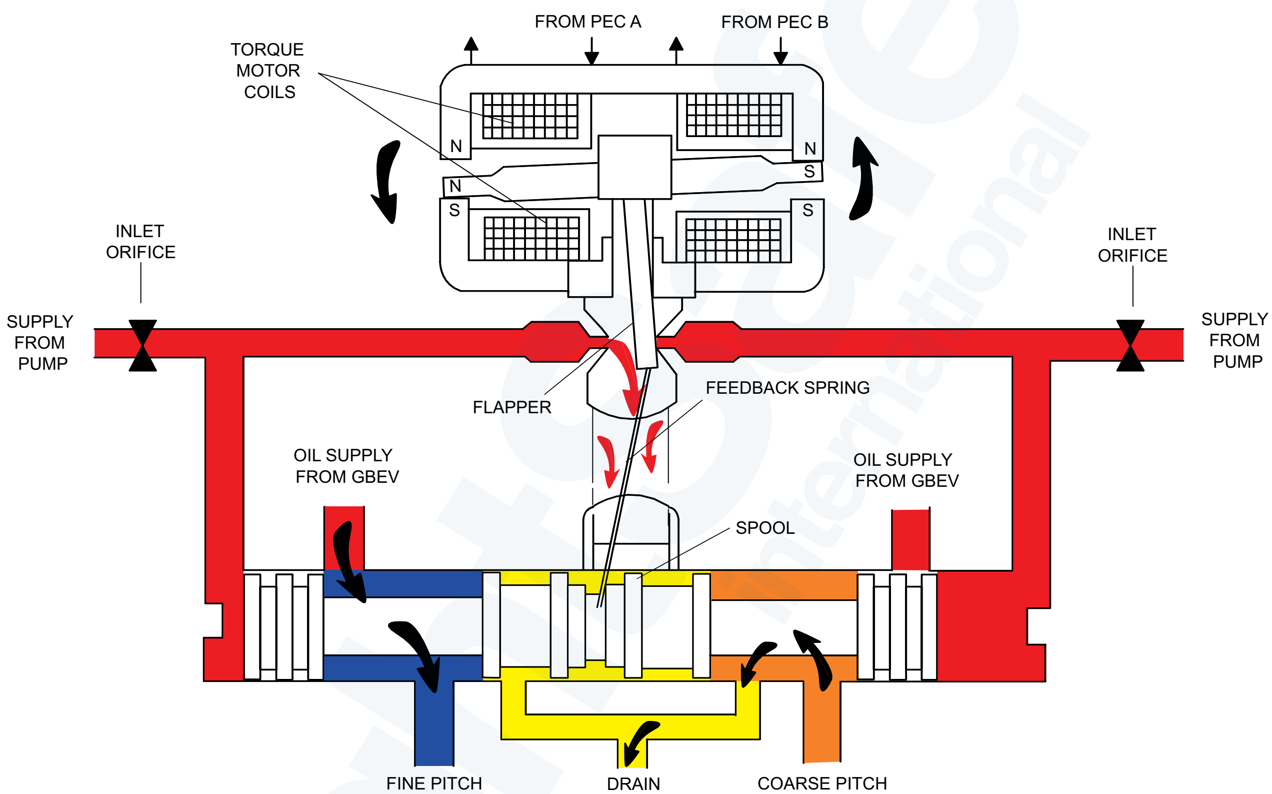

Figure 61-34. Servo-valve

Servo-valve

Refer to Figure 61-34. Servo-valve.

The servo-valve is a two stage nozzle flapper design used to control blade pitch in all control modes. It does this by controlling the flow of oil. The torque motor that drives on the first stage has two coils, with lines connected to each PEC lane.

Opening the valve schedules high pressure oil to one line, while venting the other line to drain.

In total absence of hydraulic supply to the servo-valve, or during loss of hydraulic supply to the second stage only, the propeller will be influenced by the blade forces.

A drive fine would be arrested by the OSG and a drive coarse, at higher power, would be arrested by the AUPC.

NOTES

Ground Beta Enable Solenoid Valve (GBEV)

Refer to Figure 61-35. Ground Beta Enable Solenoid Valve (GBEV).

The GBEV is installed in the PCU. The ground beta enable solenoid controls the valve. When energized, it vents the enable valve end chamber to drain, allowing spring pressure to move the valve from the flight position into the ground position.

In the flight position (with the solenoid de-energized), the ground fine chamber in the PCU is vented to drain. This action stops the propeller pitch from being driven below the flight fine stop.

The OSG is connected into the HP oil system with the valve in this position.

In the ground position (with the solenoid energized), the ground fine chamber is connected to the fine pitch oil line. This allows the propeller to be driven into the ground beta pitch range.

The OSG is isolated from the system in ground beta and the engine fuel control system gives overspeed protection with the valve in this position.

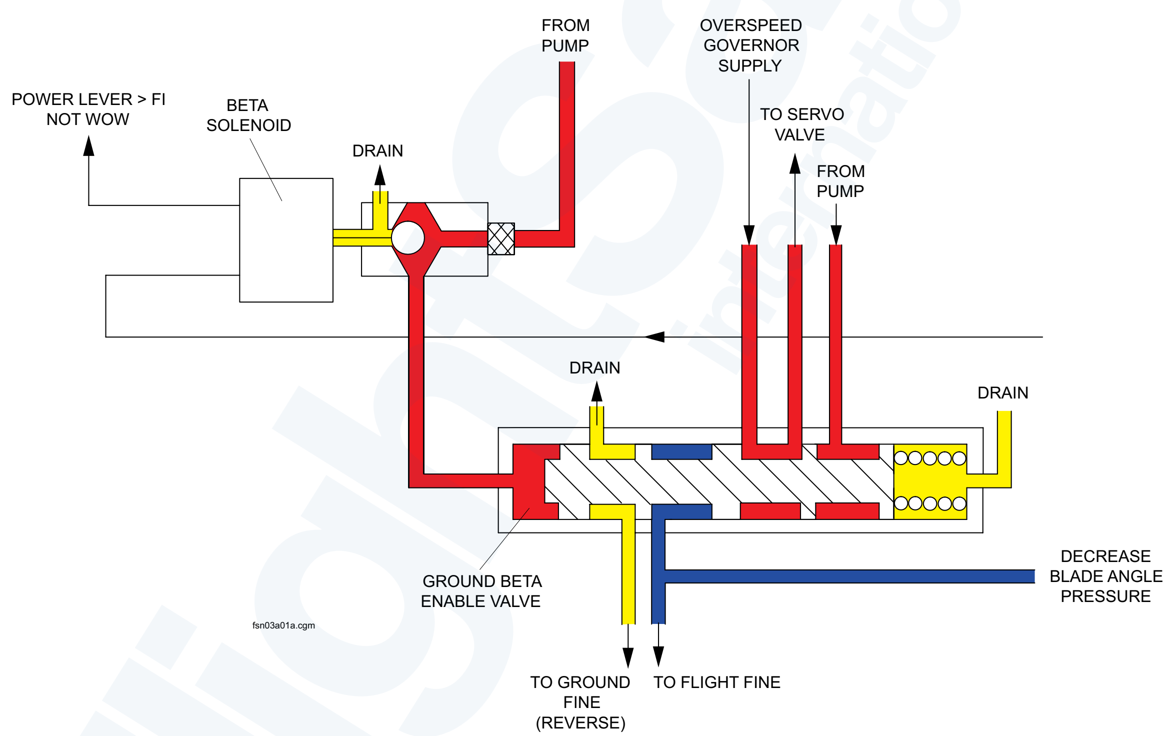

Figure 61-35. Ground Beta Enable Solenoid Valve (GBEV)

61 PROPELLER

Unfeather Valve and Solenoid

Refer to Figure 61-36. Unfeather Valve and Solenoid.

The purpose of the Unfeather Valve and Solenoid is to direct oil from the auxiliary pump to the PCU servo-valve.

The valve is enabled when the unfeather solenoid and auxiliary pump are energized.

The unfeather solenoid is activated by a maintenance switch, and is monitored by the PEC, although the PEC has no control over the valve.

NOTES

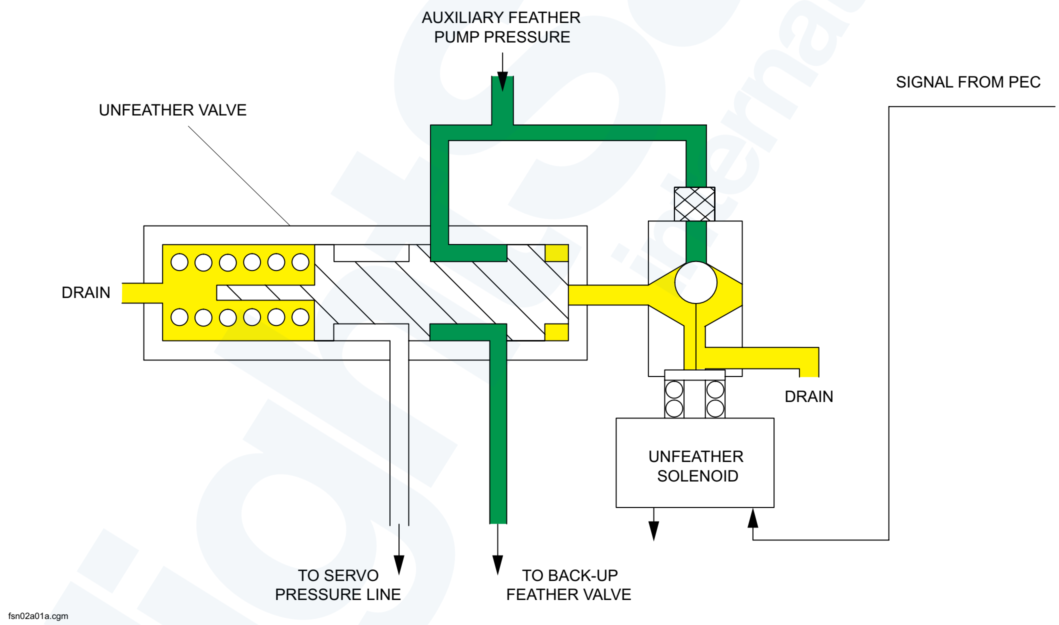

Figure 61-36. Unfeather Valve and Solenoid

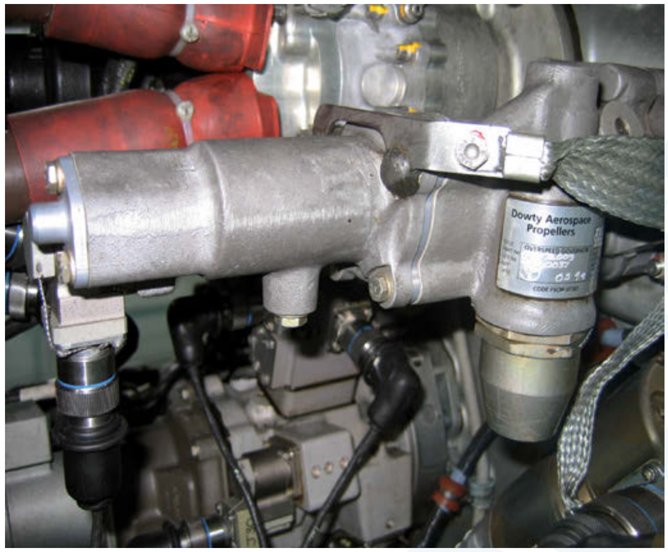

Figure 61-37. Overspeed Governor (OSG)

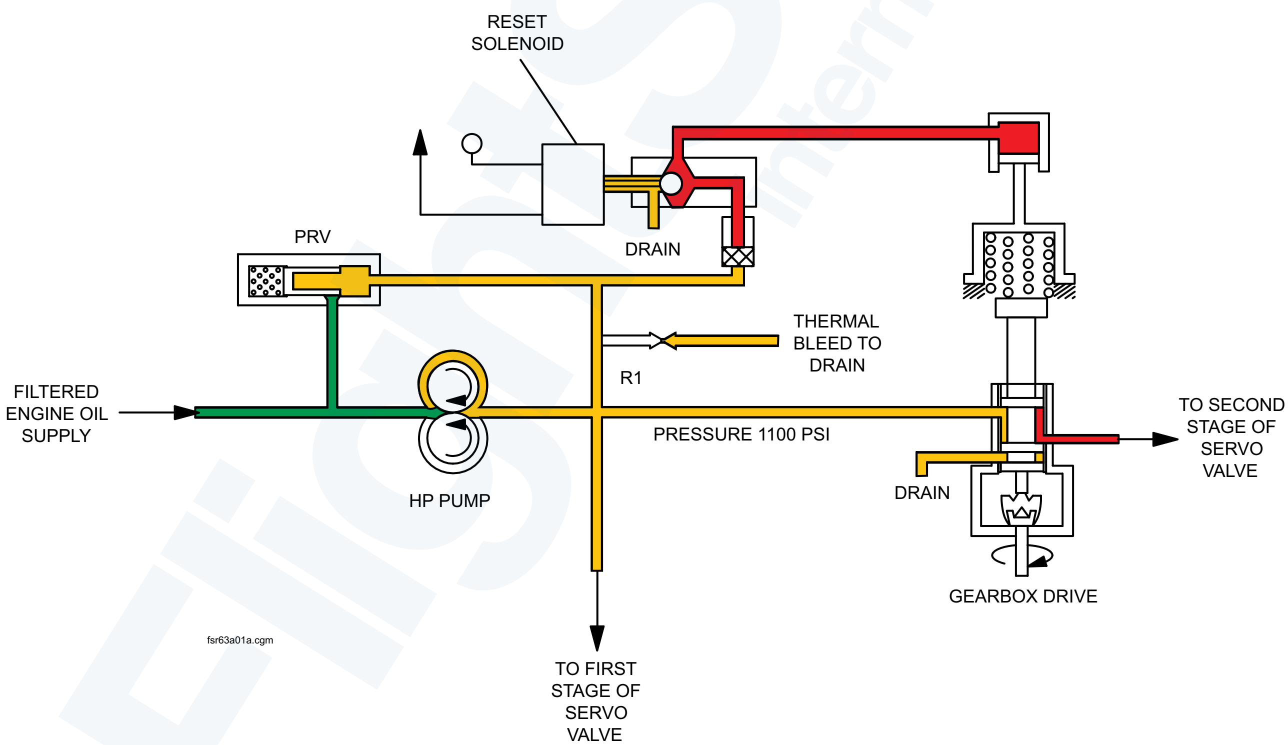

Figure 61-38. Propeller Overspeed Governor and Pump

Overspeed Governor (OSG)

NOTES

Refer to:

- Figure 61-37. Overspeed Governor (OSG).
- Figure 61-38. Propeller Overspeed Governor and Pump.

The OSG and pump are attached to the rear of the reduction gearbox by four studs and nuts. Correct installation is assured by a single dowel. A bonded seal plate provides sealing to the gearbox casing for the four oil ports.

The unit is a two piece aluminium constructed body, housing the gear pump case and governor/spool driven by the reduction gearbox. The governor/spool body houses the reset solenoid valve and a pressure relief valve which limits pump maximum output pressure to 1100 psi (758.4 kPa).

The OSG provides overspeed protection for the propeller system that is controlled by flyweights. The pump in the OSG increases the engine oil pressure for propeller actuation.

At approximately 104% N_P (1060 rpm) an overspeed condition occurs. The flyweights will move the spool against the spring venting the oil supply to the PCU to drain. This restricts the oil supply to the servo-valve allowing the propeller counterweights to coarsen the blade angle to slow the rotational speed.

The unit has a reset solenoid energized by the OSG test switch on the pilot's side console. This allows functioning the OSG at a lower value of approximately 860 N_P. An interlock to prevent operation in flight is provided by the WOW input to a low side switch in PEC.

Feathering Pump

Refer to Figure 61-39. Feathering Pump.

The 28 VDC gear type feathering pump is installed with a mounting pad on the rear of the reduction gearbox.

The feathering pump is energized during autofeather or alternate feather. It draws oil from a dedicated reservoir in the RGB. It can feather the propeller in the air or on the ground and is not dependant on the rotation of the RGB.

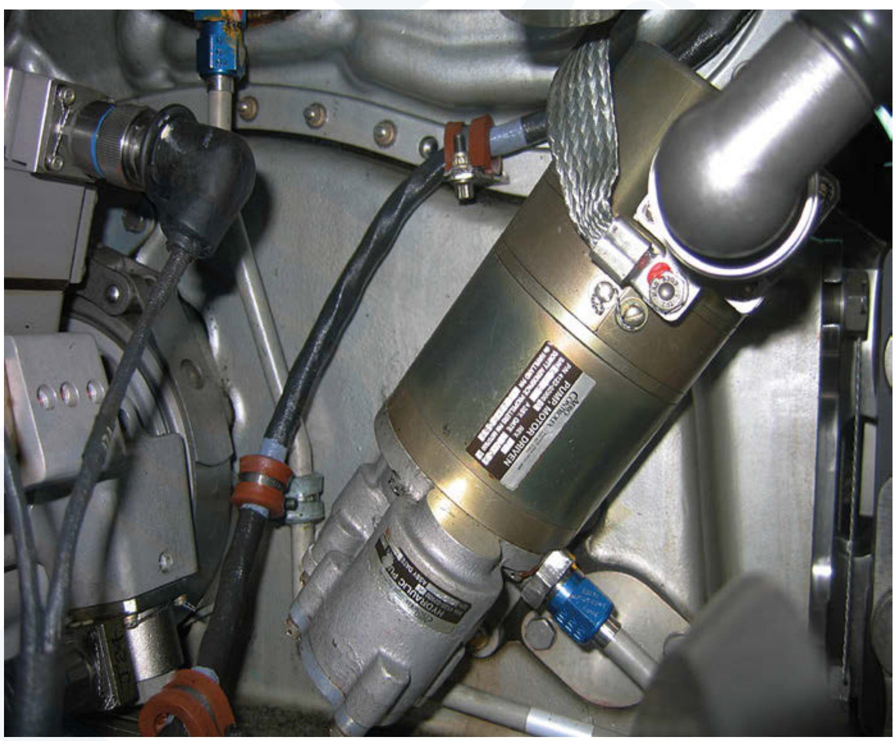

Figure 61-39. Feathering Pump

Propeller Electronic Control (PEC)

Refer to Figure 61-40. Propeller Electronic Control (PEC).

The PEC is installed under the spine cowl at the top forward nacelle, between the two hinged forward doors.

The PEC is a digital electronic control unit that incorporates two independent lanes for:

- Propeller speed governing
- Beta schedule control functions
- Synchrophasing control
- Uptrim function
- Autofeather function
- Automatic underspeed protection function.

The PEC controls the propeller pitch in relationship to the controls:

- The CLA inputs from dual RVDTs are provided as analog signals directly to PEC, with excitation provided by the PEC
- The PLA signals inputs from dual RVDTs are provided to the PEC by the FADEC on an RS422 digital data bus
- Propeller speed input is from a dual pulse probe assembly installed on the brush block bracket and sensing the target screws, on the periphery of the slip ring
- Engine control panel selection
- WOW
- PEC Channel Change (No Fault) with WOW and CLA to START and FEATHER Position.

The PCU provides:

- Governed constant speed operation
- Power lever controlled beta range (Flight idle to reverse)
- Manual feather
- Unfeather.

Propeller Ground-Range Annunciator

The propeller Ground Range lights are installed on the pilot's glareshield panel. They are controlled by the appropriate PEC at a blade angle of 1 degree below Flight Idle.

Figure 61-40. Propeller Electronic Control (PEC)

OPERATIONAL TEST OF THE PROPELLER FAULT CODE INDICATION (MRB #612000-204)

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

The maintenance procedure that follows is for the operational check of the propeller fault code indication.

There are two procedures given to do this task.

The first procedure uses fault code data stored in the FADEC and is displayed on the torque gauge.

The second procedure uses fault code data stored in the EMU which is accessed from the CDS and is displayed on the ARCDU.

Energize the aircraft electrical system.

Do the operational check of the propeller fault code indication as follows (Engine Display):

NOTE

This check must be completed with the two engines stopped and the propellers in the feather position.

1. Move the two condition lever angles (CLA’s) to the FUEL OFF position.
2. Move the two power lever angles (PLA’s) to the DISC position.
3. Set the MAINT DISC switch to the on position.
4. Make sure the adjacent amber CDS LED comes on.
5. Read the LRU codes from the digital indication on the torque and N_H gauges.
6. Make a record of the fault codes on the display.
7. Refer to the AMM PSM 1-84-2 and FIM PSM 1-84-23 for instructions and disposition of the fault codes shown.
8. Set the MAINT DISC switch to the off position.

Do the operational check for propeller fault code indications as follows (EMU):

1. Make sure the aircraft is in ground mode, (WOW).
2. Make sure the indicated airspeed, (IAS) is less than 50 knots.
3. Set the CDS GND MAINT switch to the on position.
4. Make sure the adjacent amber CDS LED comes on.
5. Set either ARCDU selector switch to the ON position
6. On the ARCDU, select the MAINT push-button.
7. Follow the menu and select the side key adjacent to the POWERPLANT FAULTS and look at the selection for “Engine 1 or Engine 2”
8. Use the NEXT or PREVIOUS keys to make an analysis of the fault codes.
9. If the display is white, there are no engine fault code indications.
10. If the display is amber, select the side key adjacent to Engine 1 or Engine 2 and record all the fault code indications.
11. Refer to the AMM PSM 1-84-2 and FIM PSM 1-84-23 for instructions and disposition of the fault codes shown.

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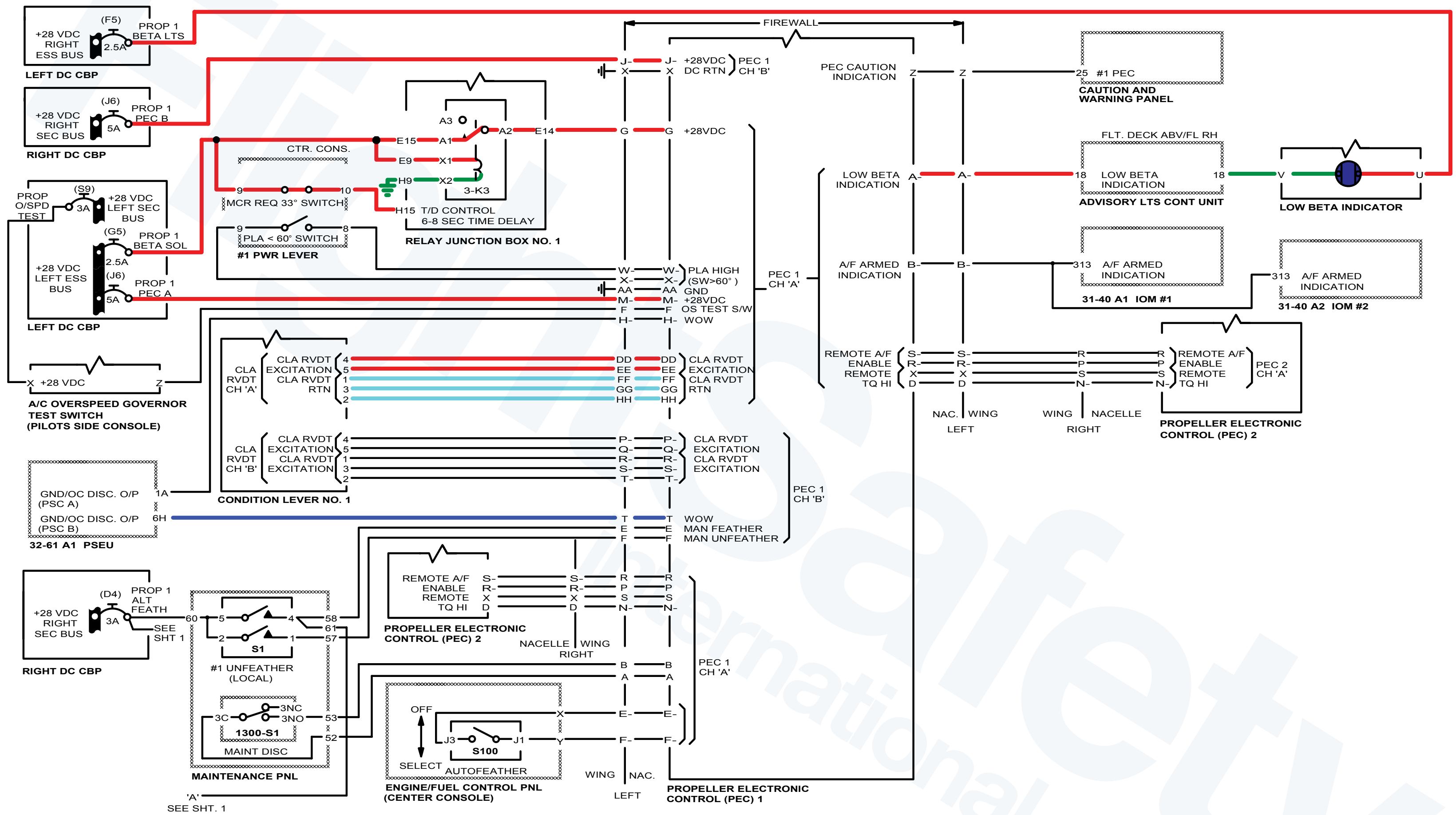

Figure 61-41. Normal Operation (Sheet 1 of 2)

OPERATION

Ground Start Mode

- Power Lever in Disc position
- Condition Lever in Fuel Off position.

Refer to:

- Figure 61-41. Normal Operation (Sheet 1 of 2).
- Figure 61-42. Normal Operation (Sheet 2 of 2).

When electrical power is supplied from the essential busses to PEC, the PEC will then supply the excitation current from pins DD and EE to the dual RVDT of the condition lever.

The excitation current enters the RVDT at pins 4 and 5. The RVDT return signal to PEC comes from pins 1, 2 and 3 of the RVDT. It then returns to pins FF, GG and HH of PEC.

With the PLA <33° signal at pin H15 on the 6 - 8 second time delay relay, the relay will energize.

- Power will pass from contact A1 to A2 and from there to pin G of PEC
- PEC also gets a WOW signal from the PSEU from pin T
- With these signals PEC will energize the ground beta enable solenoid by sending a discrete output on pin T to pin A on the GBES with a return signal out on pin B to PEC pin J.

Press the starter switch and when N_H is observed move CLA to Start and Feather Position. This will cause:

- Fuel to flow, controlled by FADEC
- Signal from CLA RVDT to PEC.

PEC will command the servo valve in the pitch control unit using pins N and U of PEC to pins A and B of the servo valve. The servo valve dual torque motor, controlled by the signal from PEC, will position the flapper valve and send high oil pressure, from the propeller pump to one end of the directional control valve. The valve will be positioned to supply coarse pitch oil to the transfer sleeve. From the transfer sleeve the oil will pass through the outer beta tube to the front of the Pitch Change Piston holding the crosshead assembly on the feathering stop.

Unfeather After Start

- PLA in Disc position
- CLA to MIN, 900 or MAX.

A signal from CLA RVDT is sent to PEC, and PEC commands the servo valve to supply fine pitch oil to the ground fine oil line from the ground beta enable valve to the transfer sleeve. From the transfer sleeve the oil will pass through the inner beta tube to the rear of the pitch change piston, driving the piston and pitch change shaft forward. This will decrease the propeller pitch to 0° pitch.

As the propeller pitch decreases through 10°, the low beta indicator (Prop ground range lights) will come ON. Power is from the Essential. Bus through the light to the advisory lights control unit that provides a ground when it receives a signal from pin A of PEC.

From the other side of the pitch change piston, coarse pitch oil will return to the RGB through the outer beta tube, transfer sleeve and the servo valve.

As the propeller pitch decreases towards 0°, N_P will increase to 660 N_P where it will be governed by FADEC limiting engine fuel in N_P Underspeed Governing mode.

During Taxi with PLA between Flight Idle and disc the propeller will remain in N_P underspeed governing.

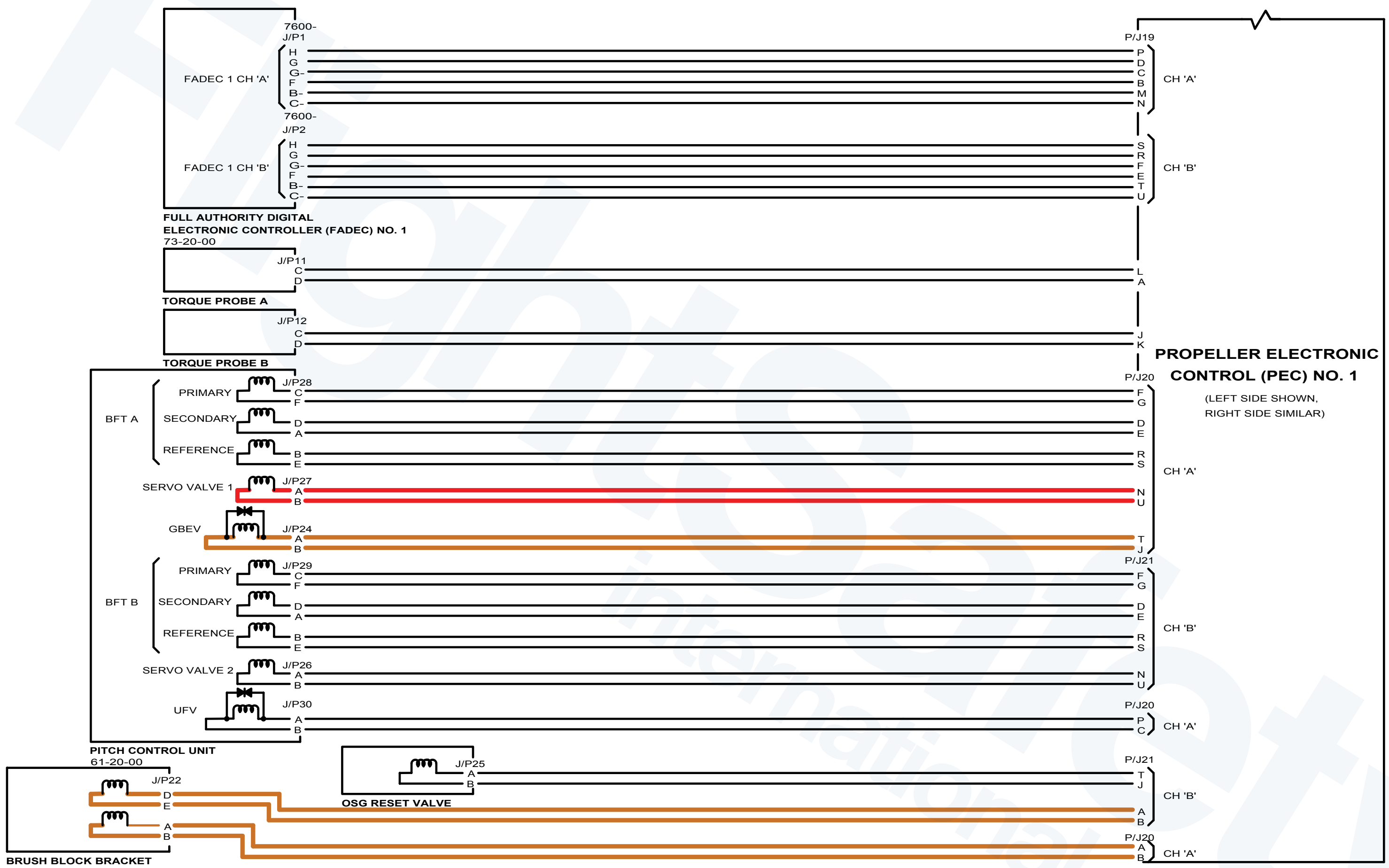

Figure 61-42. Normal Operation (Sheet 2 of 2)

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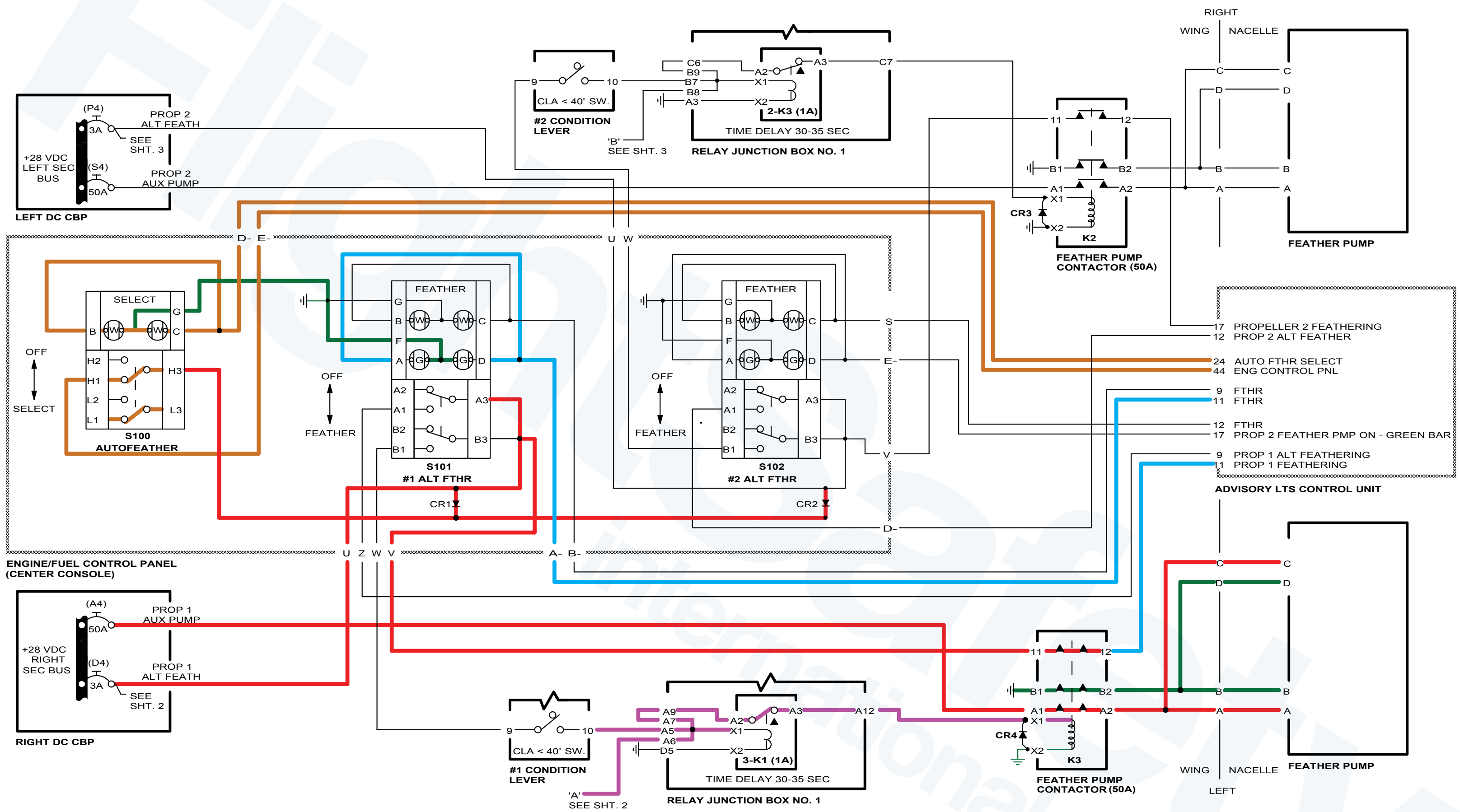

Figure 61-43. Autofeather (Sheet 1 of 2)

Increase Power for Take-Off

- PLA/Rated Power Detent
- CLA/Max.

As PLA is advanced above flight Idle towards the rated power detent, power is removed from pin H1 on time delay relay to de-energize the relay. The signal is removed from pin G of PEC and the ground beta enable solenoid is de-energized after a 6 - 8 second delay. Oil is drained from one end of the ground beta enable valve high oil pressure positions it to the inflight position.

PEC gets propeller speed signals from the dual pulse probe assy input on pins A and B of connectors P/J 20 and 21. PEC also gets reference N_p signals from FADEC on ARINC 429 bus.

As PLA is advanced, N_p will increase up to 1020 N_p. PEC will now command the servo valve to progressively increase propeller pitch to absorb the increasing engine power and torque will increase to match the torque bug (MTOP or NTOP).

After Take-Off

Pilot will retard CLA to 900 position. PEC will command increase propeller pitch.

Oil will be directed by the servo valve into the coarse pitch oil line, and through the transfer sleeve to the outer beta tube. The oil from the beta tube goes to the front of the pitch change piston increasing the pitch. The propeller will slow down to 900 N_p (MCL).

Constant Speed Mode

With signals from the dual pulse probe assy, and a reference signal from FADEC, PEC can control propeller speed to match CLA input.

PEC will command the servo valve in the PCU to direct oil to the coarse or fine pitch sides of the pitch change piston.

Doing this will constantly change propeller pitch to maintain the selected N_p.

Operation on the Beta Schedule

The beta feedback transducer sends pitch angle signals to PEC when pitch is near to 27° and below. Above 27° pitch, the transducer signal is saturated and therefore not useful.

With PLA >60° the minimum blade pitch allowed by the beta schedule is 27°. The beta schedule is one of PEC's control loops.

- As PLA is retarded < 60° PEC allows < 27° pitch angle as necessary
- As PLA is retarded to Flight Idle, PEC will allow pitch to decrease to 16.5°
- As PLA is retarded to 33° and below, PEC will allow pitch to decrease through disc to maximum reverse -11° to match PLA.

Approach

PEC will control propeller pitch on the beta schedule to match power lever angle.

CLA will be at Max 1020.

Landing and Reverse

When the aircraft lands PEC gets:

- <20 ft AGL from radio altimeters
- PLA <33°.

NOTE

WOW and AGL signals are in parallel.

With these signals PEC will energize the ground beta enable solenoid.

Spring pressure will now move the ground beta enable valve.

This action of the valve will allow fine pitch oil, when commanded by PEC, to enter the ground fine line and the pitch can now be decreased below 16.5° (F.I.).

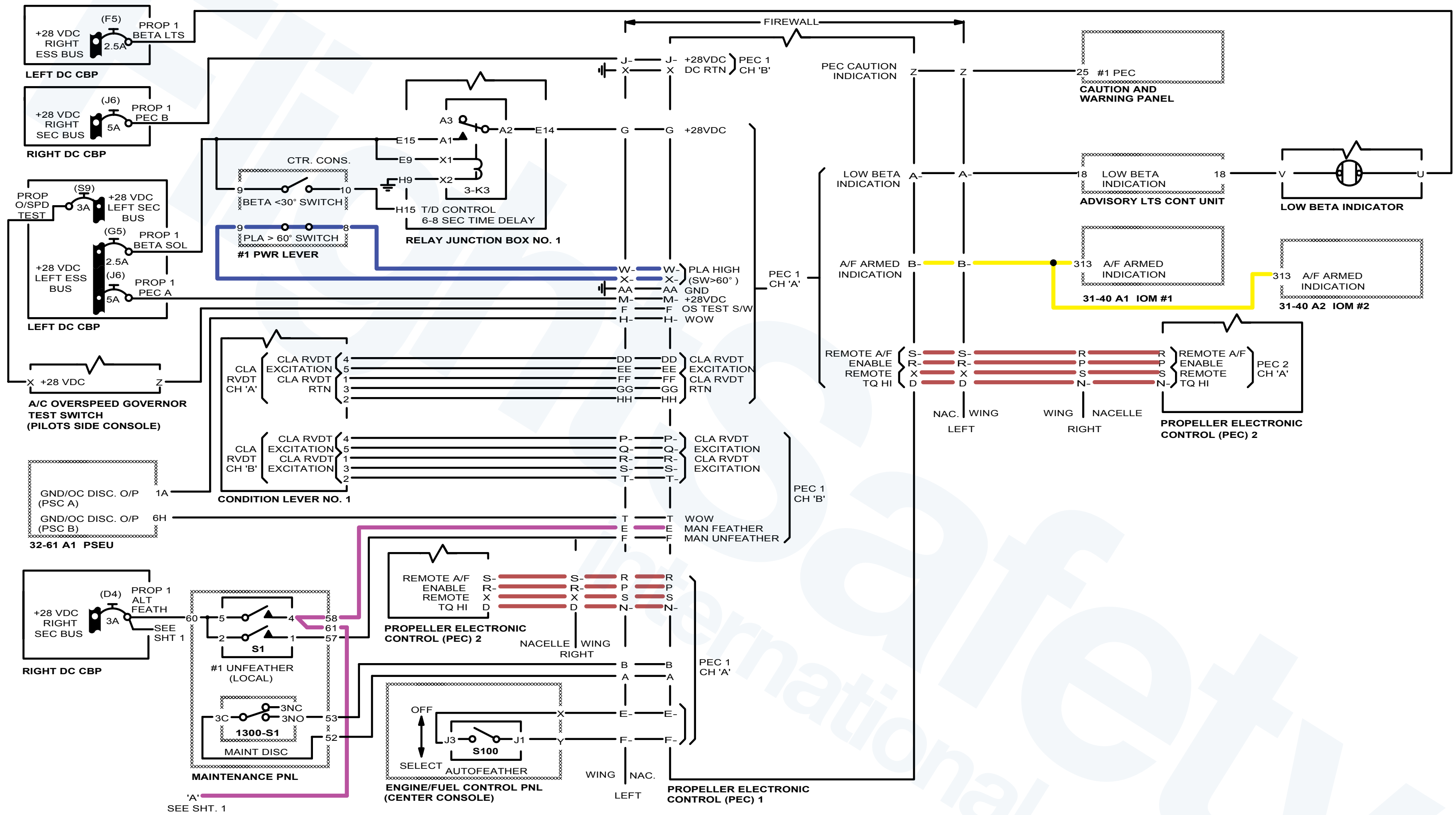

Figure 61-44. Autofeather (Sheet 2 of 2)

As the pitch decreases to <10°, PEC will command the Propeller Ground Range lights ON.

As the PLA is retarded towards disc and into reverse pitch mode, FADEC will increase power and PEC will control propeller pitch to absorb this power and maintain N_p between 950 and 1020 N_p dependent upon ambient conditions.

The ground beta enable valve when in ground mode isolates the propeller overspeed governor, so maximum N_p in reverse is limited by FADEC at 1020 by limiting engine fuel.

Autofeather System

Refer to Figure 61-43. Autofeather (Sheet 1 of 2).

Autofeather is selected and armed for Take-Off. The system is de-selected by aircrew after Take-Off.

Power is supplied from the secondary bus through diode CR1 to pin H3 on the Autofeather Select Switch.

When the select switch is pushed:

- Power flows from pin H3 to pin H1 and from there to Advisory lights control unit pin 44
- The unit then outputs power through pin 24 to pins B and C of the autofeather select switch and out through pin G to ground
- The autofeather SELECT ON switchlight illuminates.

NOTES

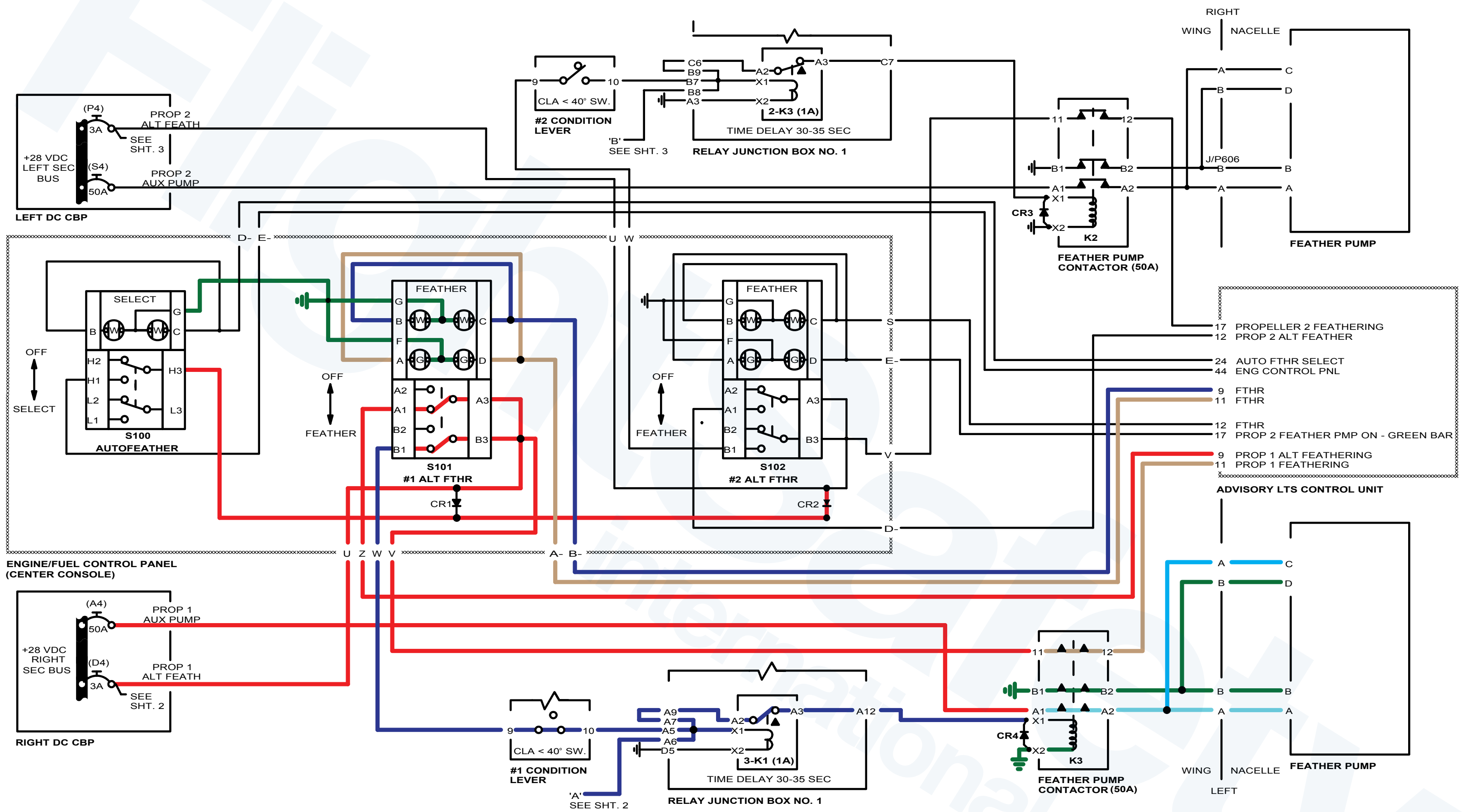

Figure 61-45. Alternate Feather

Refer to Figure 61-44. Autofeather (Sheet 2 of 2).

The autofeather system is ARMED when:

- There is power from the Essential bus through PLA >60° switch to pin W of PEC
- PEC receives local TQ Hi signal from FADEC on ARINC bus and remote TQ Hi from the remote PEC on pins X and D
- Autofeather permission from remote PEC on pins S and R
- There are no faults on the autofeather boards in the PEC's.

An output from pin B of the PEC's to pins 121 and 313 of the IOM's will bring on the A/F ARM message on the ED.

Autofeather Activation

Refer to:

- Figure 61-43. Autofeather (Sheet 1 of 2).
- Figure 61-44. Autofeather (Sheet 2 of 2).

If PEC now senses a low torque signal from BOTH N_{pt}/Q sensors on the failing engine for a period of 3 seconds, it will output power from pin E.

This power will go to:

- Pin 58 of the Maintenance Panel
- From pin 58 the power will be output on pin 61
- This power will now go to Time Delay Relay 3-K1 (or 2-K3)
- The power will enter on pin A6 and connect to A9
- From pin A9, power will go from contact A2 to A3
- From contact A3 through pin A12 to coil X1 to X2 of relay K3 Feather Pump Contactor.

This will energize the Alternate Feathering Pump.

- Also, power from the Sec bus will, through contacts 11 to 12 of relay K3 go to pin 11 of P/J1 of Advisory Light Control
- Power would then be output from pin 11 of P/J2 to pins D and A of switch S101 (or S102) to bring ON the green bar)
- The Alternate Feathering Pump using oil from the reserve oil tank in the RGB will send oil pressure to the Unfeather valve
- The de-energized Unfeather valve will direct the oil pressure to the top of the Feather Valve.
- This will move the feather valve against the action of its spring
- This action will open a port into and out of the feather valve through a restrictor R2
- The restrictor will control the rate of feather
- The oil will pass through the feather valve to the coarse pitch oil chamber of the transfer sleeve
- From the transfer sleeve the oil will flow into the inner beta tube to the front of the pitch change piston
- The pitch change piston will drive the blades to feather
- After 30 to 35 seconds the time delay relay will energize:
    - Contacts A2 to A3 will open
    - Power will be removed from K3 relay and the pump will stop operating
    - Green bar in alternate feather switch will go OFF, power removed from pin D.

Alternate Feather

Refer to Figure 61-45. Alternate Feather.

Power from the Secondary bus is applied to pin A3 and B3 on the ALTERNATE FEATHER switch on the PROPELLER CONTROL panel.

When Alternate Feather S101 selected power goes:

- From pin A3 to A1 and
- From there to pin 9 of P/J1 to pin 9 of P/J2, to bring ON the FTHR (white) light on switch S101 through pins B and C of the switch.

Power also goes:

- From pin B3 to pin B1 in alternate feather switch
- From pin B1 to pin 9 in CLA < 40° switch
- From 40° switch pin 10 to Time Delay Relay pin A5
- From pin A5 to contacts A2 and A3 of time delay relay
- Output from A3 energizes feather pump contactor K3 closing contacts A1 to A2 and B2 to B1
- This energizes the alternate feathering pump
- Contacts B2 to B1 provide the ground.
- Power also from pin B3 of switch S101 goes to contact 11 on Feather Pump contactor
- From contact 11 to contact12 to pin 11 of P/J 1 advisory lights control
- Output from pin 11 of P/J2 goes to pins A and D on switch S101 to bring ON the green bar (pump operating).

The pump using oil from the reserve oil tank in the RGB will send oil pressure to the Unfeather valve.

- The de-energized unfeather valve will direct the oil pressure to the top of the Feather Valve
- This will move the feather valve against the action of its spring
- This action will open a port into and out of the feather valve through a restrictor R2
- The restrictor will control the rate of feather
- The oil will pass through the feather valve to the coarse pitch oil chamber of the transfer sleeve
- From the transfer sleeve the oil will flow into the inner beta tube to the front of the pitch change piston
- The pitch change piston will drive the blades to feather
- After 30 to 35 seconds the time delay relay will energize:
  - Contacts A2 to A3 will open
  - Power will be removed from K3 relay and the pump will stop operating.
  - Green bar in alternate feather switch will go Off, power removed from pin D.

Automatic Underspeed Protection Circuit

Circuit is Armed if:

- NOT Manual feather
- NOT Autofeather
- NOT Alternate Feather
- NOT PLA Ground Beta.

If PEC detects high torque >50% AND low N_p< 815 N_p from both N_pt/Q sensors, the AUPC circuit will output a drive fine signal to the servo valve of the Pitch Change Unit.

This drive fine signal will override any other signal coming from PEC and the PEC Caution Light will be ON.

The fine pitch oil, directed by the servo valve, will go to the flight fine pitch port of the Ground Beta Enable valve, and from there to the transfer sleeve.

From the transfer sleeve the oil will flow through the outer beta tube to behind the Pitch Change Piston driving the pitch change mechanism to fine pitch.

The ground beta enable valve will prevent any fine pitch below 16°.

The propeller speed will be limited by the overspeed governor.

Overspeed Governor

(Propeller control system NOT in ground beta control)

The flyweights of the overspeed governor are driven by the RGB, and therefore, have a direct relationship to propeller speed.

The action of the flyweights on the governor sliding valve is opposed by governor springs and oil pressure.

When N_p is below 1060 the springs and oil pressure are holding the sliding valve in position, to allow oil to flow through the governor to the GBEV.

From the GBEV the oil flows to the second stage of the servo valve where it is then controlled by the servo valve, which is controlled by PEC to control N_p.

If the propeller overspeeds to 1060 N_p the flyweights are now providing enough force to overcome the springs and oil pressure.

- The flyweights move outwards which raises the sliding valve
- This action cuts of oil to the second stage of the servo valve
- No oil supply to the propeller pitch change mechanism results in the counterweight assemblies now becoming the controlling force on the blades
- Blade pitch will be increased to control the propeller at 1060 N_p
- If the pitch increase caused by the counterweights causes N_p to decrease below 1060 the sliding valve will re-open and N_p control will revert to normal.

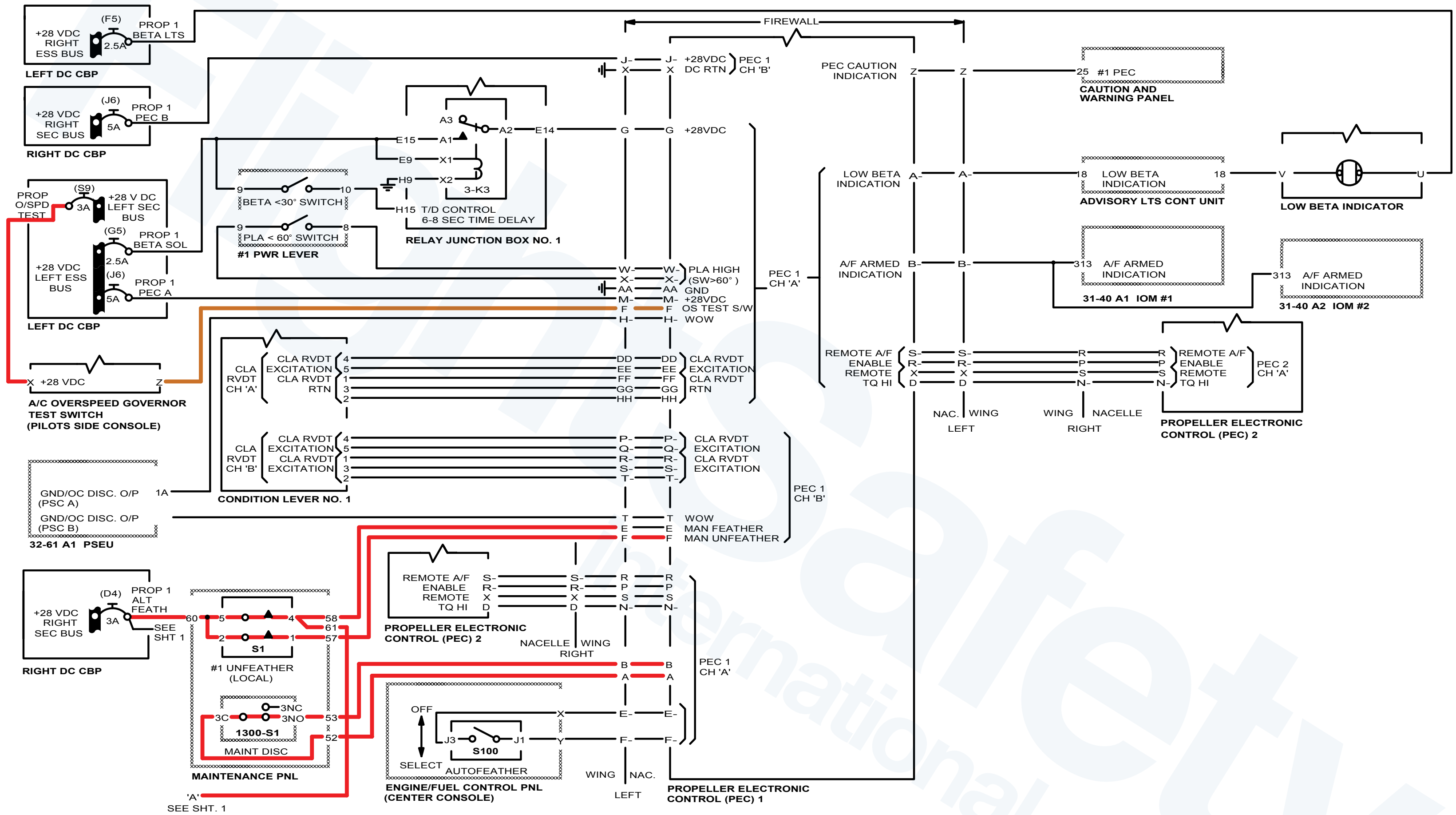

Figure 61-46. Propeller Overspeed Governor Test. (Sheet 1 of 2)

Overspeed Governor Test

Refer to Figure 61-46. Propeller Overspeed Governor Test. (Sheet 1 of 2).

Select overspeed governor test switch:

- Power from Sec bus to pin X of the switch (permanently)
- From pin X to pin Z
- From pin Z to pin F of PEC.

NOTES

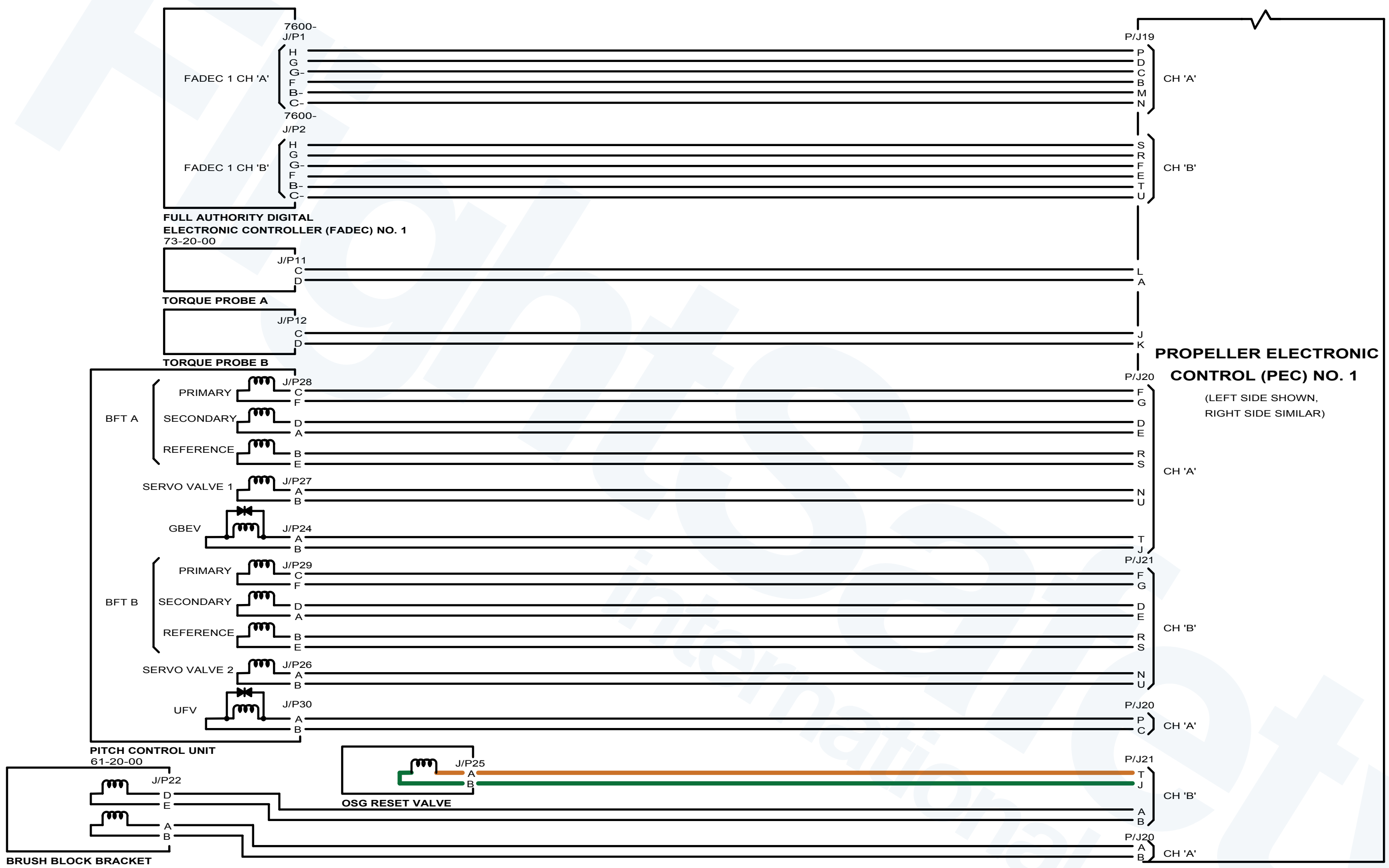

Figure 61-47. Propeller Overspeed Governor Test. (Sheet 2 of 2)

Refer to Figure 61-47. Propeller Overspeed Governor Test. (Sheet 2 of 2).

- Output from PEC P/J21 pin T to Overspeed Governor Reset solenoid
- Return signal from OSG reset solenoid to PEC pin J of P/J21.

The energized OSG reset solenoid allows the oil pressure to drain away from the top of the governor.

This means that the flyweights only have to overcome the force of the governor springs.

This action resets the governor down to approximately 860 N_p.

The action is then as described previously for the overspeed governor but at a lower N_p.

NOTES

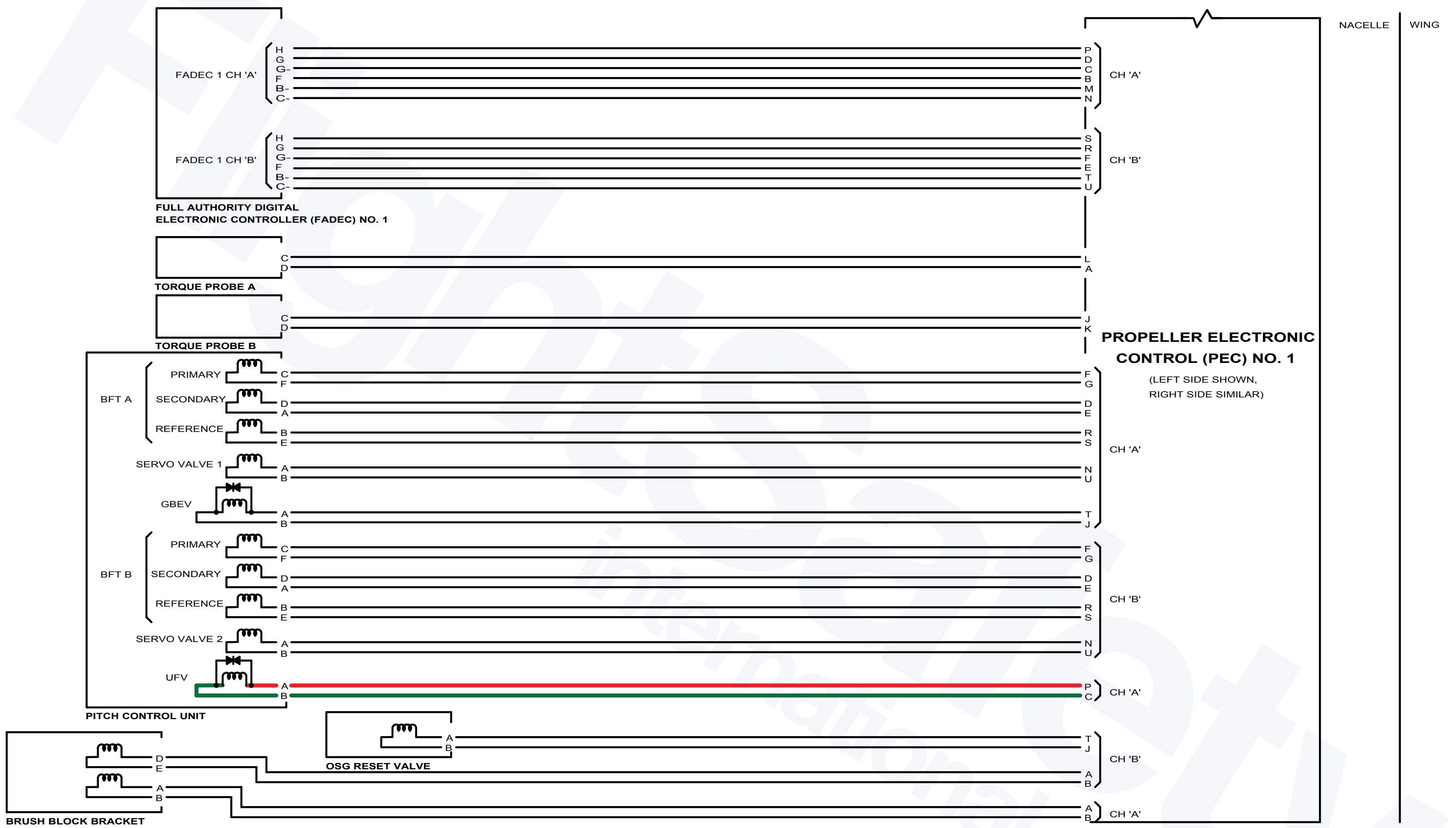

Figure 61-48. Maintenance Unfeather (Sheet 1 of 2)

Maintenance Unfeather

Refer to:

- Figure 61-48. Maintenance Unfeather (Sheet 1 of 2).
- Figure 61-49. Maintenance Unfeather (Sheet 2 of 2).

Prerequisites

- WOW DETECTED
- No propeller rotation
- Maintenance mode selected.

Power is always supplied from Sec bus to pins 5 and 2 of the Unfeather switch.

With all of the above and unfeather switch selected on the Maintenance Panel:

- Power goes from contact 5 to contact 4 of the switch and then to pin E of PEC
- Power also from contact 2 to contact 1 of the switch to pin F of PEC
- With these signals PEC will energize the solenoid of the Unfeather valve
- The power output from contact 4 also goes to pin 61 of the unfeather switch
- From pin 61 the power goes to the time delay relay pins A5 and A9
- From pin A9 to contacts A2 to A3 to coil X1 and X2 of relay K3
- This will connect power from the Sec bus to the pump. Power from Sec bus also through 11 and 12 of K3 relay to pin 11 advisory lights control
- Advisory lights control will output power on pin 11 P/J2 to contacts D and A on switch S101 (or S102)
- The green bar in the switch will illuminate because the pump is operating.

The oil output from the alternating feathering pump will go to the Unfeather Valve:

- The solenoid in the unfeather valve has been energized by PEC
- Oil through the solenoid ball valve to end of unfeather valve
- Unfeather valve moves against action of its spring
- This connects output port of valve to servo pressure line
- Servo valve commanded by PEC to fine pitch
- Oil pressure from pump directed by servo valve to fine pitch chamber of transfer sleeve
- From transfer sleeve, through outer beta tube to rear of pitch change piston
- This fine pitch oil can drive the propeller to Maximum Reverse.

When the Time Delay Relay times out, 30 - 35 seconds, the relay will energize opening contacts A2 to A3:

- This will remove power from coil X1 and X2 in K3 relay
- Relay contacts A1 to A2 will open
- Alternate feathering pump will stop
- Contacts 11 and 12 will open in K3 relay
- Advisory lights Control will remove power from green bar in switch S101 (or S102).

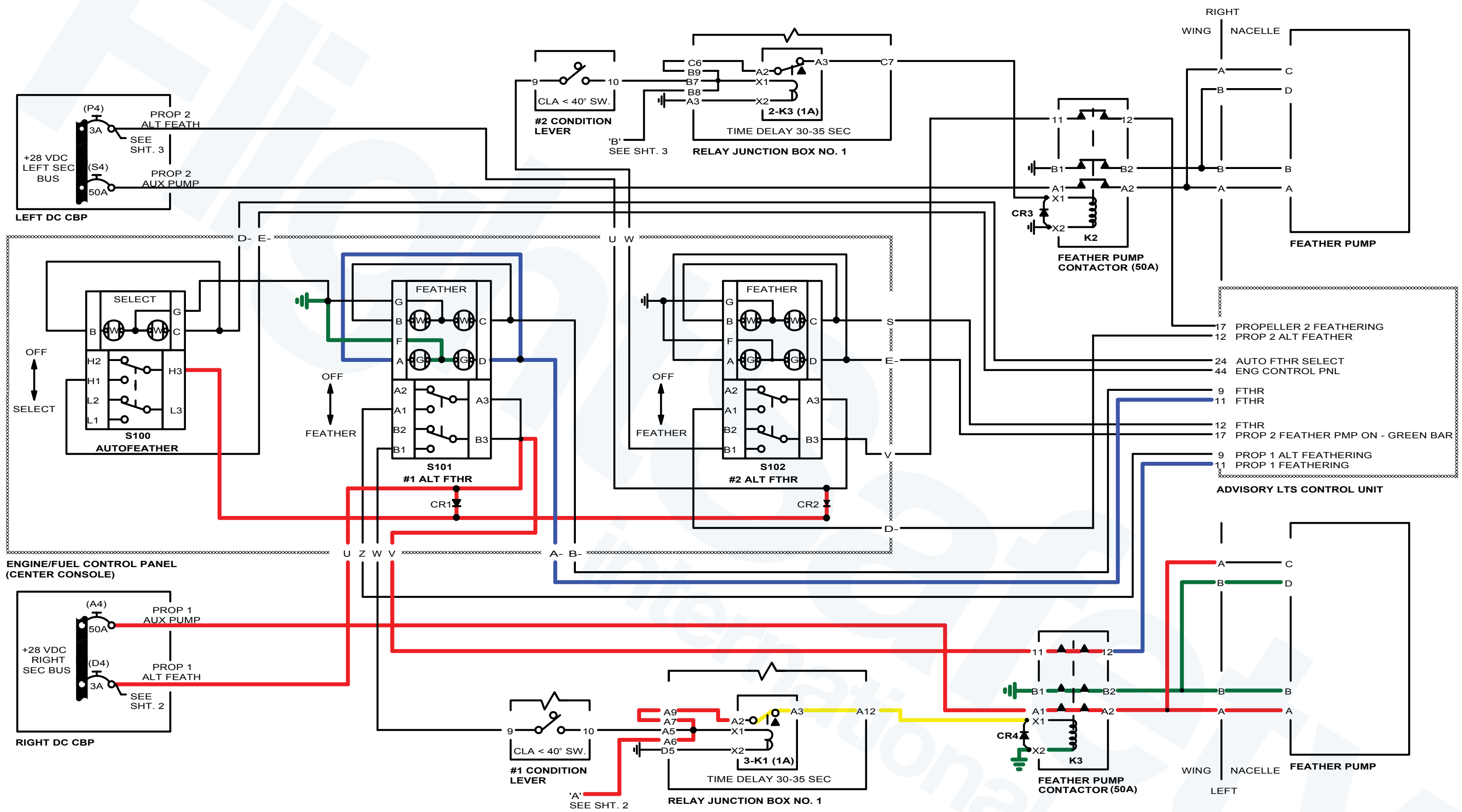

Figure 61-49. Maintenance Unfeather (Sheet 2 of 2)

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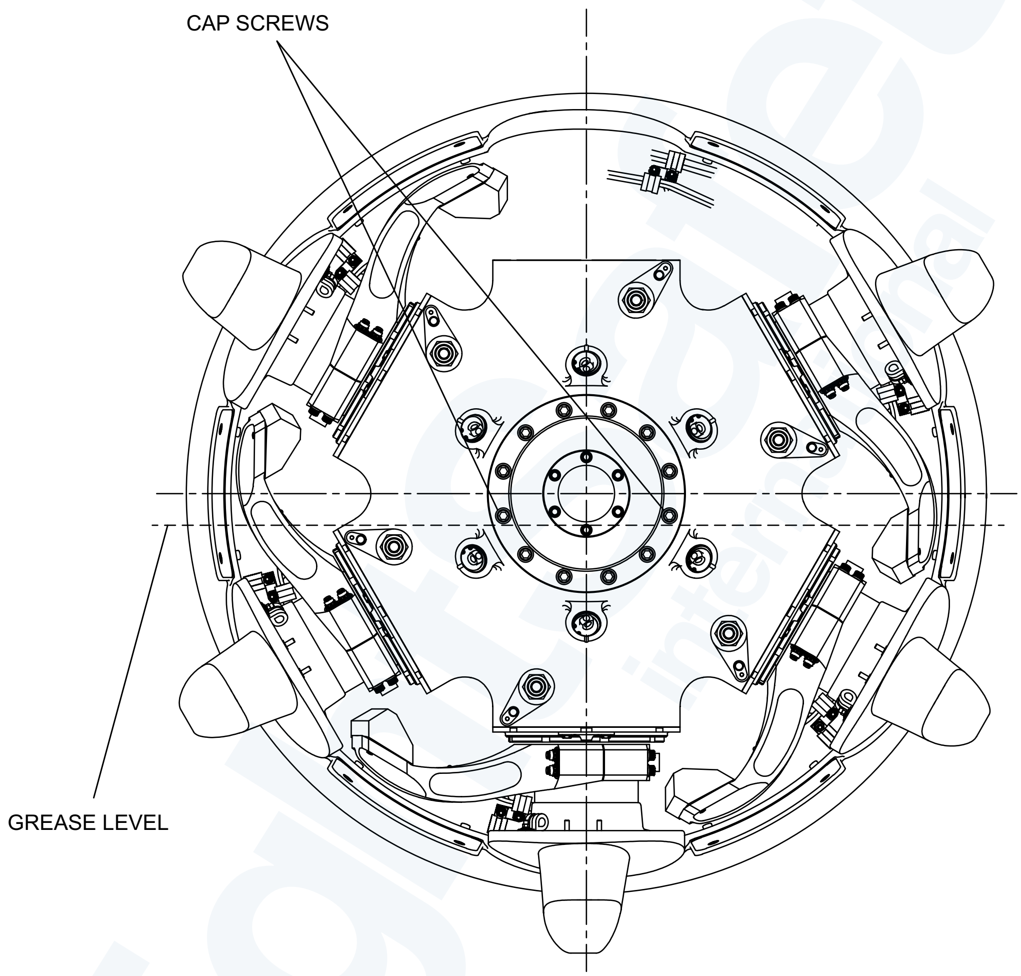


Figure 61-50. Propeller Hub Grease Level Check

61-00-00 APPENDIX

MAINTENANCE CONSIDERATION

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Safety Precautions

Do not install the blade bats in the blade cuff area. If you do this, you can cause damage to the propeller blades.

Replace the propeller attachment nuts if the running torque is less than 50 lbf in (5.65 Nm). If you do not do this, the attachment nuts will not be correctly locked.

Aircraft or System Limitations

TASK 05-13-00-990-802.

Time Limits, Propellers The time limited items are identified in Dash 8-400 Dowty Aerospace's Propeller Maintenance Manual Publication No. 1096, Component Maintenance Manual 61-00-00.

Mandatory Airworthiness Limitations:

- Life Limitations
- Mandatory Inspections.

Servicing

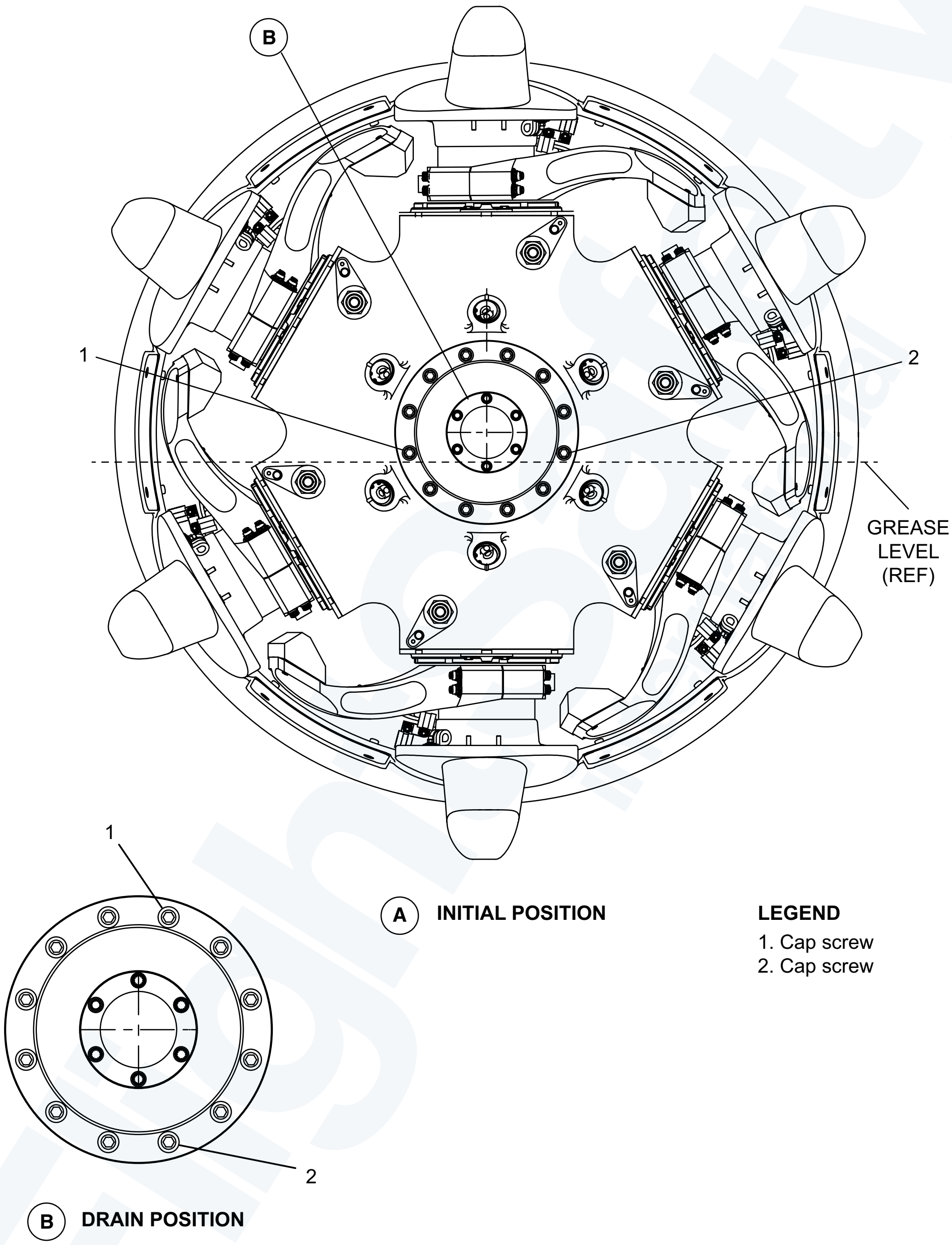
Propeller Grease Level Check

Refer to Figure 61-50. Propeller Hub Grease Level Check.

1. Remove the spinner
2. Turn the propeller so that the blade assembly at the highest position is vertical
3. Carefully remove one of the cap screws that attach the cylinder to see if grease comes out of the hole
4. If grease does come out of the hole, the grease level is satisfactory
5. Install the cap screws and torque to 34 to 36 lbf ft (46.1 to 48.8 Nm)
6. If grease does not come out of the hole, remove the other cap screw at the same level as the cap screw removed
7. Add grease to the hub assembly through one of the cap screw holes. Continue to add grease until the grease is level with or comes out of the other cap screw hole
8. Install the cap screws and torque to 34 to 36 lbf ft (46.1 to 48.8 Nm)
9. Install the spinner.

NOTE

If you find signs of excessive grease leakage and/or signs of loose blades (Refer to AMM05-61-00-210-805).



LEGEND

1. Cap screw
2. Cap screw

Figure 61-51. Add New Grease to Propeller

Replacing the Propeller Hub Grease

Refer to Figure 61-51. Add New Grease to Propeller.

1. Make sure that the temperature of the propeller hub is more than 15°C (60°F).
2. Turn the propeller so that the blade assembly at the highest position is vertical (initial position).
3. Carefully remove the two cap screws (1, 2) from the propeller cylinder as shown in the illustration.
4. Use your hand to turn the propeller until grease comes out of one of the cap screw holes.

NOTE

Use a container to collect the grease from the cap screw hole.

5. Turn the propeller until that hole is at the lowest position to the ground (drain position).
6. Let all of the grease drain from the propeller.
7. Add new grease to the propeller as follows:
   - Use your hand to turn the propeller so that the cap screw holes (1, 2) are in the initial position again (horizontal)
   - Add grease through one of the cap screw holes until the grease is level with, or comes out of the other cap screw hole.
8. Install the cap screws (1, 2) and torque to 34 to 36 lbf ft (46.1 to 48.8 Nm).
9. If necessary, clean the propeller of any grease spillage with a clean lint-free cloth.

Special Tooling

Refer to the DHC-8 Q400 Maintenance TASKS supplement for detailed task procedures.

- DAPT70-0021-00 Lifting Equipment
- DAPT61-0015-00 Torque Adapter
- DAPT60-0223-00 Outer Sleeve Clamps
- DAPT65-0079-00 Blade Sling
- DAPT60-0126-00 Blade Bat
- DAPT65-0093-00 Adapter
- DAPT65-0094-00 Hydraulic Pump
- DAPT60-0116-00 Ring Nut Wrench
- DAPT60-0140-00 Ball Extractor
- DAPT60-0347-00 Ball Assembly Tool
- DAPT60-0348-00 Ball Assembly Tool
- DAPT60-0115-00 Bearing Wrench
- DAPT60-0188-00 Bearing Wrench (Alternative to DAPT60-0115-00)
- DAPT65-0078-00 Installation Bullets
- DAPT65-0087-00 Spinner Removal Tool (Qty 2)
- DAPT60-0189-00 Beta Tube Wrench.

Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2.

- TASK 05-53-00-210-811 Engine Inspection after Propeller Sudden Stoppage
- TASK 05-53-00-210-812 Engine Inspection after Propeller Strike Causing Blade Structural Damage
- TASK 05-53-00-210-813 Engine Inspection after Propeller Strike Causing Minor Blade Damage.
- TASK 05-53-00-210-821 Propeller Inspection after a Bird Strike
- TASK 05-53-00-210-822 Propeller Inspection after a Lightning Strike
- TASK 05-53-00-210-823 Propeller Inspection after an Engine Fire
- TASK 05-53-00-210-824 Propeller Inspection after an Over-speed Condition
- TASK 05-53-00-210-825 Propeller Inspection after an Over-torque
- TASK 05-53-00-750-801 Engine Inspection after Propeller Lightning Strike.

OPERATIONAL TEST OF THE PROPELLER AUTOFEATHER AND UPTRIM SYSTEM (MRB #612000-201)

The maintenance procedure that follows is for the operational check of the propeller autofeather and uptrim system.

On aircraft without Modsum 4-113588, do an operational check of the propeller autofeather and uptrim system as follows:

NOTE

This check can be completed with the two engines started or with the two engines stopped.

If you do the check with the two engines started, do the pre-start checks and start the engines.

Make sure that the two condition lever angles (CLA's) are at START AND FEATHER (engine started) or at FUEL OFF (engine stopped).

Make sure the two power lever angles (PLA's) are at DISC.

On the propeller control panel, push the AUTOFEATHER SELECT switchlight and make sure that:

a. The SELECT advisory light comes on.
b. The 'A/F SELECT' and the 'A/F TEST IN PROGRESS' messages show on the engine display.

NOTE

During the test sequence the uptrim indication and ITT red radial increase should show on the engine display. 'NTOP' changes to 'MTOP' and the torque bug increase to the applicable uptrim set position. The 'A/F ARM' message will then replace the 'A/F SELECT' message on the engine display. This will occur approximately three seconds before the 'A/F SELECT' message shows again. The sequence will then occur again (this is to permit the test on the two power plant autofeather systems).

Make sure the 'A/F TEST PASSED' message shows on the engine display after the test sequence is completed.

On the propeller control panel, push the AUTOFEATHER SELECT switchlight and make sure that:

a. The SELECT advisory light goes out.
b. The 'A/F SELECT' and the 'A/F TEST PASSED' messages do not show on the engine display.

If the 'A/F TEST PASSED' message does not show during the check then the system is defective.

If you did the check with the engines started, shutdown the engines.

On aircraft with Modsum 4-113588, do an operational check of the propeller autofeather and uptrim system as follows:

Make sure that the two condition lever angles (CLA's) are at START AND FEATHER (engine started) or at FUEL OFF (engine stopped).

Make sure the two power lever angles (PLA's) are at DISC.

On the propeller control panel, push the AUTOFEATHER SELECT switchlight and make sure that the SELECT advisory light comes on.

a. Observe the engine display, the messages that follow must show:
   - 'A/F SELECT'.
   - 'A/F TEST IN PROGRESS'.

b. Observe the engine display, the messages that follow must show twice:
   - ‘UPTRIM’ shows.
   - ‘ITT and N_H’ red radials increase.
   - ‘NTOP’ changes to ‘MTOP’ and torque rating and torque bugs increase.
   - ‘A/F ARM’ shows.
   - ‘A/F SELECT’ shows.
   - ‘UPTRIM’ goes out of view.
   - ‘MTOP’ changes to ‘NTOP’ and torque rating and torque bugs decrease.

c. Observe the engine display. If the test is satisfactory, the ‘A/F TEST PASS’ message will show.

NOTE

If the autofeather test is aborted, ‘A/F TEST ABORT’ will show on the engine display. If the autofeather test fails, ‘A/F TEST FAILED’ will show on the engine display. If the ‘A/F TEST ABORT’ message shows, do the autofeather test again.

On the propeller control panel, push the AUTOFEATHER SELECT switchlight and make sure that:

a. The SELECT advisory light goes out.

b. The ‘A/F SELECT’ and the ‘A/F TEST PASSED’ messages do not show on the engine display.

If the ‘A/F TEST PASSED’ message does not show during the check then the system is defective.

If you did the check with the engines started, shutdown the engines.

If you did the check with external electrical power, de-energize the aircraft electrical system.

OPERATIONAL TEST OF THE PROPELLER ALTERNATE FEATHER (MRB #612000-202)

The maintenance procedure that follows is for the operational check of the propeller alternate feather.

Do an operational check of the propeller alternate feather as follows:

1. Make sure that the two engines are stopped and the two propellers are at feather.
2. Make sure that the condition lever angles (CLA’s) are at FUEL OFF.
3. Make sure the two power lever angles (PLA’s) are at DISC.
4. Set the MAINT DISC switch to on at the engine maintenance panel.
5. Set the UNFEATHER (manual) switch on, at the engine maintenance panel.
6. Make sure that the propeller moves to a low blade angle.
7. Set the UNFEATHER (manual) switch off, at the engine maintenance panel.
8. Set the CLA to MIN/850.
9. Push the ALT FTHR switchlight on, at the propeller control panel:
   a. Make sure the indication on the ALT FTHR switch stays off.
   b. Make sure the propeller stays unfeathered.
10. Set the condition lever angle to START AND FEATHER:
    a. Make sure the indication FTHR on the switchlight comes on.
    b. Make sure the Green Bar comes on for between 25 and 35 seconds.
    c. Make sure that the propeller fully feathers.
11. Move the CLA to FUEL OFF.
12. Refer to the FIM, PSM 1-84-23 for more data if any of the alternate feather checks are defective.

OPERATIONAL TEST OF THE PROPELLER OVERSPEED GOVERNOR (MRB #612000-203)

The maintenance procedure that follows is for the operational check of the propeller overspeed governor.

1. Do the pre-start checks and start the engines
2. Make sure that the two condition lever angles (CLA's) are at MAX/1020.
3. Make sure the two power lever angles (PLA's) are at FLIGHT IDLE.
4. Set and hold the PROP O'SPEED GOVERNOR switch to TEST on the pilot's side console:
    a. Make sure that the OSG TEST IN PROG message is shown on the engine display after approximately 3 seconds for the two power plants.
5. Slowly move the PLA's until the OSG TEST PASS message is shown on the engine display for the two power plants.
6. Move the two PLA's to FLIGHT IDLE:
    a. Make sure the propeller speed decreases to 660 rpm.
7. Release the PROP O'SPEED GOVERNOR switch.
8. If the check is unsatisfactory, do the procedure in steps (2) to (7) again one more time only.
9. Refer to the FIM, PSM 1-84-23 for more data for any of the propeller overspeed governor checks, as necessary.

OPERATIONAL TEST OF THE PROPELLER FAULT CODE INDICATION (MRB #612000-204)

The maintenance procedure that follows is for the operational check of the propeller fault code indication.

T

here are two procedures given to do this task.

The first procedure uses fault code data stored in the FADEC and is displayed on the torque gauge.

The second procedure uses fault code data stored in the EMU which is accessed from the CDS and is displayed on the ARCDU.

Energize the aircraft electrical system

Do the operational check of the propeller fault code indication as follows (Engine Display):

NOTE

This check must be completed with the two engines stopped and the propellers in the feather position.

1. Move the two condition lever angles (CLA's) to the FUEL OFF position.
2. Move the two power lever angles (PLA's) to the DISC position.
3. Set the MAINT DISC switch to the on position.
4. Make sure the adjacent amber CDS LED comes on.
5. Read the LRU codes from the digital indication on the torque gage.
6. Make a record of the fault codes on the display.
7. Refer to the AMM PSM 1-84-2 and FIM PSM 1-84-23 for instructions and disposition of the fault codes shown.
8. Set the MAINT DISC switch to the off position.

Do the operational check for propeller fault code indications as follows (EMU):

1. Make sure the aircraft is in ground mode, (WOW).
2. Make sure the indicated airspeed, (IAS) is less than 50 knots.
3. Set the CDS GND MAINT switch to the on position.

4. Make sure the adjacent amber CDS LED comes on.
5. Set either ARCDU selector switch to the ON position
6. On the ARCDU, select the MAINT push-button.
7. Follow the menu and select the side key adjacent to the POWERPLANT FAULTS and look at the selection for "Engine 1 or Engine 2"
8. Use the NEXT or PREVIOUS keys to make an analysis of the fault codes.
9. If the display is white, there are no engine fault code indications.
10. If the display is amber, select the side key adjacent to Engine 1 or Engine 2 and record all the fault code indications.
11. Refer to the AMM PSM 1-84-2 and FIM PSM 1-84-23 for instructions and disposition of the fault codes shown.

OPERATIONAL CHECK OF THE PROPELLER AUTOFEATHER SYSTEM IN MAINTENANCE MODE (CMR# 612000-106)

The maintenance procedure that follows is for the operational check of the propeller autofeather system in maintenance mode.

On aircraft without Modsum 4-113558, do an operational check of the propeller autofeather system in maintenance mode as follows:

1. Make sure that the two condition lever angles (CLA's) are at 'FUEL OFF'.
2. Make sure the two power lever angles (PLA's) are at 'DISC'.
3. Set the 'MAINT DISC' switch to on.
4. Push the AUTOFEATHER SELECT switchlight and make sure that the SELECT advisory light comes on.
5. Make sure that you get the indications that follow:
   a. The 'A/F SELECT' and the 'A/F TEST IN PROGRESS' messages show on the engine display.

   NOTE

   During the test sequence, the uptrim indication must show on the engine display and the torque bug increase to the applicable uptrim set position.

   b. The 'A/F ARM' message will then replace the 'A/F SELECT' message on the engine display.
   c. After approximately three seconds the 'A/F SELECT' message shows again.
   d. On the propeller control panel, the ALT FTHR advisory light (on the ALT FTHR switchlight) comes on.
   e. The LH indicator will come on first and then the RH indicator.
   f. The sequence will then occur again (this is to permit the test on the two power plant autofeather systems).
6. Make sure the 'A/F TEST PASSED' message shows on the engine display after the test sequence is completed.
7. Push the AUTOFEATHER SELECT switchlight and make sure that:
   a. The SELECT advisory light goes out.
   b. The 'A/F SELECT' and the 'A/F TEST PASSED' messages do not show on the engine display.
8. Set the 'MAINT DISC' switch to off.
9. If the 'A/F TEST PASSED' message does not show during the check then the system is defective. Refer to the FIM TASK 61-20-00-710-804.

On aircraft with Modsum 4-113558:

1. Make sure that the two condition lever angles (CLA's) are at 'FUEL OFF'.
2. Make sure that the two power lever angles (PLA's) are at 'DISC'.
3. Set the 'MAINT DISC' switch to on.
4. Push the AUTOFEATHER SELECT switchlight and make sure that the SELECT advisory light comes on.
   a. Observe the Foolowing messages on the engine display:
      - 'A/F SELECT'
      - 'A/F TEST IN PROGRESS'.
   b. On the propeller control panel, make sure that the ALT FTHR advisory light (on the ALT FTHR switchlight) comes on. NOTE: The #1 ALT FTHR advisory light will come on during the first cycle, and the #2 ALT FTHR advisory light will come on during the second cycle. (c) Observe the engine display. If the test is satisfactory, the 'A/F TEST PASS' message will show. NOTE: If the autofeather test is aborted, 'A/F TEST ABORT' will show on the engine display. If the autofeather test fails, 'A/F TEST FAILED' will show on the engine display. If the 'A/F TEST ABORT'messages shows, do the AUTOFEATHER TEST again.
5. On the propeller control panel, push the AUTOFEATHER SELECT switchlight and make sure that:
   a. The SELECT advisory light goes out.
   b. The 'A/F SELECT' and the 'A/F TEST PASSED' messages do not show on the engine display.
6. On the ENGINE MAINTENANCE section of the Central Maintenance Panel (above the wardrobe), set the 'MAINT DISC' switch to off.
7. If the 'A/F TEST PASSED' message does not show during the check then the system is defective. Refer to the FIM TASK 61-20-00-710-804.

OPERATIONAL TEST OF THE PROPELLER REDUCED N_P FUNCTION

The maintenance procedure that follows is for the return to service check after troubleshooting the reduced N_p function.

Set the WOW system for Air Mode

Do the pre-start checks and start the engines

Do a functional check of the propeller reduced N_p as follows:

1. Set the two condition lever angles (CLA's) at 850.
2. Move the two power lever angles (PLA's) until the two propellers speed govern at 850.
3. Move the two power lever angles (PLA's) again to give an increase of approximately 3% Tq to make sure that the propellers are on the speed governing schedule.
4. Push the RDC N_p switch for approximately 1 second and make sure that the RDC N_p advisory is displayed on the ED.
5. In less than 10 seconds, move the CLA to MAX 1020.
6. Make sure that the ED rating changes from MCR to NTOP.
7. Make sure the N_p stays at 850.
8. Move the two power lever angles (PLA's) again to give an increase of approximately 3% Tq and make sure that the N_p stays at 850.
9. Push the RDC N_p switch for approximately 1 second and make sure that the N_p has increased. Make sure that the RDC N_p advisory has gone off on the ED.

Set the WOW system to Ground Mode.

FUNCTIONAL TEST OF THE PROPELLER TIME TO UNFEATHER (MRB #612000-205)

Do a functional check of the propeller time to unfeather as follows:

1. Do the pre-start checks and start the engines
2. Make sure that the condition lever angle (CLA) is at START AND FEATHER:
   a. Make sure the power lever angle (PLA) is at DISC.
3. Set the condition lever angle to MIN/850:
   a. Make sure that the time to unfeather the propeller is less than 30 seconds.

NOTE

To calculate the time to unfeather: record the time from when the CLA is moved to MIN/850 until the PROPELLER GROUND RANGE light on the glareshield comes on.

   b. If the propeller unfeathers in more than 30 seconds, refer to the FIM (Refer to FIM 61-20-00-810-803.

NOTE

SPECIAL TOOLS & TEST EQUIPMENT

- DAPT70-0021-00 Lifting Equipment
- DAPT61-0015-00 Torque Adapter
- DAPT65-0078-00 Installation Bullets
- DAPT60-0422-00 Compression Tool
- DAPT60-0192-00 Bush Extraction Tool
- DAPT60-0357-00 Liner Turning Tool
- DAPT65-0108-00 Puller
- Commercially available Mega ohmmeter, 500 volts
- DAPT60-0223-00 Outer Sleeve Clamps
- DAPT65-0079-00 Blade Sling
- DAPT60-0126-00 Blade Bat
- DAPT65-0093-00 Adapter
- DAPT65-0094-00 Hydraulic Pump
- DAPT60-0116-00 Ring Nut Wrench
- DAPT60-0140-00 Ball Extractor
- DAPT60-0115-00 Bearing Wrench
- DAPT60-0188-00 Bearing Wrench (Alternative to DAPT60-0115-00)
- DAPT60-0400-00 Ring Nut Wrench (Alternative to DAPT60-0116)
- DAPT60-0346-00/DAPT60-0347-00 Ball Assembly Tool
- DAPT65-0087-00 Spinner Removal Tool (Qty 2)
- DAPT61-0014-00 Torque Adapter
- GSB1000028 Pin, MLG Door Ground Lock
- GSB2700008-16D Target - Actuator (Steel)
- GSB2700008-32 Target - De-actuator (Copper)
- GSB3411011 Test Set - Pitot/Static
- DAPT60-0189-00 Beta Tube Wrench
- Commercially available Four terminal low resistance test equipment
- Commercially available Resistance test meter
- Commercially available Insulation test meter
- DAPT02-0065-00 Bullet, Seal
- PWC57095 Wrench
- Commercially Available Computer, Laptop

- 400-PC-TE-0430 Test Cable, Lane A
- 400-PC-TE-0440 Test Cable, Lane B
- 005-SW-EC-001-01-01.EXE Program Software

61-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 61-10-00-210-801: General Visual Inspection of the Propeller Blades (MRB #611000-203).
- AMM 61-10-00-000-801: Removal of the Propeller.
- AMM 61-10-00-400-801: Installation of the Propeller.
- AMM 61-10-01-000-801: Removal of the Blade Assembly and Bearing.
- AMM 61-10-01-400-801: Installation of the Blade Assembly and Bearing.
- AMM 61-10-01-400-802: Propeller Blade Re-Torque.
- AMM 61-10-06-000-801: Removal of the Hub, Actuator and Backplate Assembly.
- AMM 61-10-06-400-801: Installation of the Hub, Actuator and Backplate Assembly.
- AMM 61-10-11-000-801: Removal of the Spinner.
- AMM 61-10-11-400-801: Installation of the Spinner.
- AMM 61-10-00-820-801: Restoration of the Propeller Hub (MRB #611000-201).
- AMM 61-10-00-820-802: Restoration of the Propeller Blades and Bearing Assemblies (MRB #611000-202).
- AMM 61-10-00-640-801: Grease Level Check in the Propeller.
- AMM 61-10-00-640-802: Replacement of the Propeller Grease.
- AMM 61-10-00-680-801: Service the Propeller by Replacing the Propeller Hub Grease (MRB#612000-208).
- AMM 61-10-00-350-802: Replacement of the Propeller Hub, Actuator and Backplate Assembly, Inboard Bearing Liner.
- AMM 61-10-00-720-801: Functional Test for the Balancing of the Propeller.
- AMM 61-20-01-000-801: Removal of the Beta Tubes.
- AMM 61-20-01-400-801: Installation of the Beta Tubes.
- AMM 61-20-06-820-801: Adjust the Dual Pulse Probe Assembly.
- FIM 61-20-00-810-802: Propeller #1 (#2), Failure to achieve 100% Np - Fault Isolation.
- FIM 61-20-00-810-803: Propeller #1 (#2), Slow to unfeather - Fault Isolation.
- FIM 61-20-00-810-804: Propeller #1 (#2), Fails to unfeather - Fault Isolation.
- FIM 61-20-00-810-805: Propeller #1 (#2), Autofeather test fails - Fault Isolation.
- FIM 61-20-00-810-807: Propeller #1, Uncommanded Propeller Feather - Fault Isolation.

- AMM 61-20-00-710-801: Operational Test of the Propeller Autofeather and Uptrim System (MRB #612000-201).
- AMM 61-20-00-710-802: Operational Test of the Propeller Alternate Feather (MRB #612000-202).
- AMM 61-20-00-710-804: Operational Test of the Propeller Fault Code Indication (MRB #612000-204).
- AMM 61-20-00-710-805: Operational Check of the Propeller Autofeather System in Maintenance Mode (CMR# 612000-106).
- AMM 61-20-00-710-803: Operational Test of the Propeller Overspeed Governor (MRB #612000-203).
- AMM 61-20-41-710-801: Operational Test of the Propeller Control Panel.
- AMM 61-20-36-070-801: PEC Fault Code Clear.
- AMM 61-20-36-070-802: Propeller Electronic Control Unit (PEC) Reset.